Project 5: Mission Control

Documentation set, October 9, 2026

This PDF is built from the files in the Mission Control folder. Links to other folders appear as plain text.

Contents

Part 1: Overview
Part 2: Requirements
Part 3: Design document
Part 4: Operations
Part 5: Model card
Part 6: Security and privacy
Part 7: Architecture decision records
Part 8: Data contracts
Part 9: API specification
Part 10: Changelog

Part 1: Overview

`Mission Control/README.md`

Project 5: Mission Control. Document set

Project 5, Mission Control | Document: index (README) | Set: rev 4, Oct 9, 2026

Status: Draft for review by John. Owner: John. Last updated: Oct 9, 2026.

Mission Control is a searchable and playable archive of Apollo 11 Mission Control audio. The set below is the complete project 5 documentation. The platform that the project runs on is documented in the platform design doc and the platform requirements; this set links to it instead of repeating it.

Documents in this set

Document	Type	Status	Where
Requirements	Before coding	Draft for review	requirements.md
Design doc (17 sections, including the glossary and the open questions)	Before coding	Draft for review; open questions in section 16	design-doc.md
Architecture decision records	Before coding	28 records, 0001 to 0028; each marked accepted, default or open	adrs/
Data contracts (ODCS v3.0.2)	Before coding	23 contracts, draft	contracts/
API specification (OpenAPI 3.0.3)	Before coding	Draft; limits are open (OQ-12)	openapi.yaml
Operations	Written while building	Draft; runbooks written after real failures	operations.md
Model card	Written while building (the project uses AI)	Draft; thresholds open (OQ-16)	model-card.md
Security and privacy (threat model, personal-data inventory, licence and attribution register)	Written while building	Draft; licences partly verified	security-and-privacy.md
Decision log (answered questions and open items, split out of the shared logs on Oct 9, 2026)	Kept current	Draft	decisions.md
Changelog	Kept current	Rev 4 entry added	changelog.md

Not yet written

- Reference generated from code. There is no code yet; it is generated when the first code exists.

Open questions

Open questions are in design-doc.md, section 16. Two are answered and two are partly answered (OQ-10 and OQ-17). The rest need a spike, a measurement, an account, or a decision from John. Each has a GitHub issue to be created.

Platform

Project 5 depends on these platform documents:

- Platform design doc
- Platform requirements
- Platform ADRs

Labels

Every document in this set has a label line directly under its title: the project, the document type and the set revision. The PDF of this set is in this folder: `full-documentation.pdf` (revision 4). Earlier PDFs were removed as old versions on Oct 9, 2026.

Part 2: Requirements

Mission Control/requirements.md

Project 5: Mission Control. Requirements

Project 5, Mission Control | Document: requirements | Set: rev 4, Oct 9, 2026

Status: Draft for review by John. Last updated: Oct 9, 2026. Style: arc42 section 1 (introduction and goals) and section 10 (quality requirements and scenarios). Related: design-doc.md, adrs/, contracts/, openapi.yaml. Platform documents: ../Platform/ (placeholder path; the platform documents do not exist yet).

Source rule: every value here is a recorded decision (ADR cited) or a fact from the Build Plans, Build Plan Decisions or Documentation Plan documents. Anything not determined is an open question (OQ-n), listed in design-doc.md section 16.

Citation key: BP = Build Plans for the Five Projects; BPD = Build Plan Decisions; DP = Documentation Plan. ADR = architecture decision record (adrs/). OQ-n = open question n in design-doc.md section 16. CI = continuous integration, the automatic tests GitHub runs on every change. DAG = one scheduled Airflow workflow. WER = word error rate. Session box = the AWS server started for work sessions and the nightly run; "with the session box off" means only always-on parts (S3, CloudFront, Lambda, DynamoDB) are serving. Landing hour = the hour of the landing covered in Phase A; landing window = the wider window covered in Phase B (exact limits OQ-6). Temporary 7-day area = the platform clean-up rule under which temporary files are deleted after 7 days. OGG = a compressed audio format (the version downloaded); FLAC = a lossless audio format about 5 times larger. Faithfulness = whether every claim in an answer is supported by the cited segments. LLM judge = a language model that scores answers, calibrated against hand-graded answers. Red-team = deliberately hostile test prompts. OpenAPI = the standard format for describing the HTTP API. Table, component and parameter names in these documents are Design (proposed) unless an ADR or the plan names them. Other terms are in design-doc.md section 17 (Glossary).

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1. Introduction and goals (arc42 section 1)

1.1 What the system is

A searchable, playable archive of Apollo 11 Mission Control. Every spoken line is transcribed and tagged with its channel (the console position) and Ground Elapsed Time (GET, the mission clock). A timeline shows who was talking at any moment, and "Ask Mission Control" answers questions with the exact clips. A public API exposes the data. (BP, Project 5)

Source material: Apollo 11 Mission Control audio from roughly 170 thirty-channel tapes (no hours figure is used until the catalog is measured, OQ-22), digitized by UT Dallas and NASA and published on archive.org and the Explore Apollo site. NASA's public air-to-ground transcript, with GET timestamps, is the reference for timing and accuracy. Terms must be confirmed before clips are published (ADR 0019).

1.2 Why it exists

John's reason: the earlier PDF search API was boring; this adds unstructured audio at scale with AI and a memorable demo (BPD, "Projects and scope"; ADR 0001).

1.3 Users and their needs

The plan does not define personas. The three groups below are derived from the features the plan describes and are for review.

Group	Who	What they need	Features that serve them
Visitor	Anyone opening the public site	Find a spoken moment by searching, hear it at the right GET, see who was talking when, ask a question and get clips as evidence, know how far to trust a line	F-A5, F-B7, F-B9, F-B10
API developer	Someone using the public API or MCP server	A documented, versioned API under /v1 with keys, predictable limits, clear errors, pagination	F-A5 (keys on request), F-B8

Group	Who	What they need	Features that serve them
Owner and reviewer	John	Costs that cannot surprise, accuracy numbers that can be checked, reproducible results, a decision record for the model choice	F-B3, F-B4, F-B5, F-B6, all quality requirements

1.4 Quality goals (top five)

1. Trustworthy: every answer and line traces to a clip and a source file offset; uncertainty is shown, never hidden (QR-6, QR-7, QR-8, QR-9).
2. Always available: the public site and search work with the session box off (QR-1).
3. Cost-safe: no paid item can run up a surprise bill (QR-3 to QR-5).
4. Secure by design: the public API cannot be abused (QR-10 to QR-13).
5. Measured: accuracy, retrieval and cost numbers are published (QR-14 to QR-18).

1.5 Stakeholders

Stakeholder	Interest
John (owner, sole approver)	Builds it, approves each document before coding of that part begins (DP, "How the documents get drafted")
Visitors and API users	Use the site and API
NASA, UT Dallas, archive.org	Source providers; credited; no endorsement implied (ADR 0019)

1.6 Constraints

Constraint	Value	Source
Monthly budget	Under $50, browser only	BPD
Project 5 Phase B month	About $32 to $48 all in (8 to 12 GPU hours, $2.20 to $3.60; $1 to $3 of batch model calls; model API capped at $10)	BP, Monthly cost
Project 5 Phase C month	About $35 to $48 all in (15 to 25 GPU hours, $4 to $7.50; $2 to $5 batch model calls; model API capped at $5)	BP, Monthly cost
AI spend limit	$15 a month, cheapest model only for now	BPD
GPU	g4dn.xlarge spot only (one NVIDIA T4); waits for spot	BP
Disk	Box disk under 40 GB in Phase B processing	BP 5B.1, ADR 0016
Time	All timestamps in UTC, converted on the page only; GET stored separately	BP rule 15 (store every timestamp in UTC and convert only on the page), ADR 0010
Licenses	NASA works generally not copyrighted; credit NASA; no endorsement; Databricks Free Edition is non-commercial	BP, Risks; ADR 0019
Tools	Spark, Airflow, Iceberg with Glue, faster-whisper, NeMo, LanceDB, FastAPI on Lambda, API Gateway, React and TypeScript, DynamoDB, MLflow on Databricks, Great Expectations	BP, One tool per job
Tools not used in this project	dbt, Snowflake, Kafka (per the "Technology by project" table, these cells are empty for Project 5)	BP

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2. Features by phase

Phase A is live and small, Phase B adds depth, Phase C is scale and starts only after every project has reached Phase B. (BP, "How the plan works")

Phase A: live

Scope: the landing hour on the two clearest channels, air-to-ground and the flight director loop (ADR 0002).

ID	Feature	Build step
F-A1	Catalog of archive items with channel map and recorded terms	5A.1
F-A2	Audio prep: download landing hour (OGG), convert to 16 kHz mono, speech detection	5A.2
F-A3	Transcription with faster-whisper large-v3 on spot GPU, word timestamps and confidence, checkpoint per file	5A.3
F-A4	Mission clock: tape time to GET, offset per tape	5A.4
F-A5	Search and timeline: LanceDB index, Opus clips, two-lane GET timeline, search box; API keys on request with rate limit and daily quota	5A.5

Phase B: depth

ID	Feature	Build step
F-B1	All 30 channels for the landing window, one tape at a time	5B.1
F-B2	Speaker separation (pyannote), jargon fixes, batch correction of low-confidence segments	5B.2
F-B3	Accuracy: word error rate per channel type, tracked in MLflow	5B.3
F-B4	Transcription model selection: Parakeet versus Whisper large-v3	5B.4
F-B5	Transcription cost tuning on the GPU	5B.5
F-B6	Retrieval evaluation: 200 questions, recall at 10, MRR, time error	5B.6
F-B7	Event extraction with timeline markers	5B.7
F-B8	Public API under /v1 with self-service keys, docs, load test, MCP server	5B.8
F-B9	Ask Mission Control agent with citations, refusals, evals	5B.9
F-B10	Quality checks with Great Expectations; grey low-confidence lines	5B.10

Phase C: scale (second round)

ID	Feature
F-C1	The full mission on the five busiest channels, sized from Phase B's measured cost per audio hour
F-C2	Fine-tune a small Whisper model on a few spot-GPU hours using NASA's aligned air-to-ground transcript and corrected minutes as labels; compare word error rate with base and large models on a held-out hour
F-C3	A replay mode that plays all channels from any GET
F-C4	Noise reduction tested on the noisiest back-room channels, kept only where it lowers word error rate

Phase C acceptance criteria: the plan states none per step, and John decided (Oct 9, 2026) that the Phase C done-when checks are written when Phase C is designed. They are recorded here as missing. The plan gate for each Phase C is that the design note is updated with measured numbers.

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3. Functional requirements

Each functional requirement names one actor ("the system", or a named component) and one action, and is checked by the acceptance criterion of its feature in section 5 or by the quality scenario it cites.

Each requirement traces to a feature. "Must" requirements come from the Build Plans text.

Phase A

ID	Requirement	Feature
FR-A1	The system must pull archive item metadata (tape, channel, start time, duration, format, checksum) into Iceberg with an Airflow DAG.	F-A1
FR-A2	The system must hold a channel map from tape track to console position.	F-A1
FR-A3	The system must record the source terms in the README.	F-A1
FR-A4	The system must keep a manifest of in-scope files with hashes (ADR 0011).	F-A1
FR-A5	The system must download the OGG files of the landing hour for the two channels (ADR 0005).	F-A2
FR-A6	The system must convert audio to 16 kHz mono with ffmpeg in Spark and run Silero speech detection so only speech is transcribed (ADR 0021).	F-A2
FR-A7	The system must record the speech share per channel.	F-A2
FR-A8	Every speech segment must trace back to a file and an offset.	F-A2
FR-A9	The system must transcribe with faster-whisper large-v3 on a g4dn.xlarge spot box, or a smaller model on CPU if the GPU quota is not yet granted (ADR 0003).	F-A3
FR-A10	Transcription must save word timestamps and confidence, with a checkpoint per file.	F-A3
FR-A11	The system must record audio hours per GPU hour in the design note.	F-A3
FR-A12	The system must map tape time to GET by decoding the time code on track 1 or, if it cannot be decoded, by anchoring known events in NASA's air-to-ground transcript (the 1202 alarm, the landing call), and store the offset per tape (ADR 0009).	F-A4
FR-A13	The system must chunk the transcript, embed on CPU and write a LanceDB index to S3 (ADR 0006). In Phase A a chunk is a run of continuous speech with the speaker blank; the index is rebuilt by speaker turn after 5B.2 (Decided by John, Oct 9, 2026; pause length and chunk sizes OQ-8).	F-A5
FR-A14	The website route (unkeyed, used by the public page) must apply per-visitor limits and a daily cap to every request (ADR 0022). The ask setting on the website route is open (OQ-13) and ask is off there until it is decided. Checked by a limit test that gets a 429 beyond the limit.	F-A5
FR-A15	The system must cut speech into Opus clips served through CloudFront (ADR 0018).	F-A5
FR-A16	The React page must show two lanes on a GET timeline and a search box.	F-A5
FR-A17	The public page must work with the session box off.	F-A5
FR-A18	All pipelines must satisfy the platform engineering standards (immutable raw, event time and source key on every record, idempotent tasks, write-audit-publish, contracts). See ../Platform/ and design doc section 12.	all
FR-A19	The publish step (design doc C14) must run a short set of hand-written Python checks (row counts, nulls, schema, time ranges) and stop publishing when one fails (ADR 0023). Checked by a unit test file for these checks, to be written with the checks.	F-A5
FR-A20	The api_key_service (design doc C8) must store each key request from the request form (a web form where a visitor asks for a key) in DynamoDB from Phase A (ADR 0020). Checked by submitting a test request and reading the row.	F-A5
FR-A21	Audio prep (design doc C2) must write downloaded audio and 16 kHz intermediates only to the temporary 7-day area (ADR 0011). Checked by listing where audio prep writes.	F-A2
FR-A22	The web_app (design doc C6) must, when the last nightly run is late or failed, keep showing the old data with a visible late notice and the time of the last good run (ADR 0025). Checked by QR-28.	F-A5
FR-A23	The developer API (REST type) must require a key on every request and apply that key's rate limit and daily quota; keys are issued on request in Phase A (ADR 0007, ADR 0022). Checked by contract tests.	F-A5

Phase B

ID	Requirement	Feature
FR-B1	The system must extend to every channel for the landing window (the amount of raw audio is measured from the catalog first, OQ-22), transcribing speech only, one tape at a time with the disk under 40 GB (ADR 0016).	F-B1
FR-B2	5B.4 and 5B.5 must be run on the landing hour first; 5B.1 then uses the chosen model and settings. This is a prerequisite of 5B.1 (see its acceptance criterion).	F-B1
FR-B3	The system must record throughput and cost per audio hour in the design note.	F-B1
FR-B4	The system must separate speakers with pyannote on mixed channels (air-to-ground: CAPCOM and the crew).	F-B2
FR-B5	The system must fix jargon with glossary rules, then run a batch model call for low-confidence segments only, in structured output listing each change; the original text is always kept (ADR 0013).	F-B2
FR-B6	A hand check of 100 corrections must report how many were right.	F-B2
FR-B7	The system must compute word error rate on air-to-ground against NASA's transcript and on 30 hand-corrected minutes of back-room audio, tracking every model and setting as MLflow runs on Databricks (ADR 0017).	F-B3
FR-B8	The system must publish word error rate per channel type.	F-B3
FR-B9	The system must transcribe the same landing hour with parakeet-tdt-0.6b-v2 (NeMo, English, word timestamps, fp16) and with faster-whisper large-v3 on the same T4 and compare word error rate on air-to-ground and back-room audio, audio hours per GPU hour and cost per audio hour (ADR 0004).	F-B4
FR-B10	If Parakeet fails on the T4, the comparison must use a smaller Whisper model instead.	F-B4
FR-B11	The system must record the model decision in a decision record.	F-B4
FR-B12	For the chosen model, the system must run the same hour at fp16 and int8 and at several batch sizes, record GPU utilization and memory with nvidia-smi or DCGM every few seconds, throughput and cost per audio hour, find the bottleneck (decoding, audio loading or the GPU), fix the largest one, and measure again, tracking every run in MLflow.	F-B5
FR-B13	The system must evaluate retrieval with 200 test questions tied to known moments (150 from the transcript and mission timeline, 50 written by hand), measuring recall at 10, MRR and the time error of the top hit, comparing chunk sizes and hybrid versus vector-only search.	F-B6
FR-B14	CI must fail if recall at 10 drops below its threshold.	F-B6
FR-B15	A batch model call must extract events (alarms, go/no-go polls, system problems, decisions, handovers) with type, GET, channel, speakers, confidence and the exact transcript span; events without a valid span are rejected (ADR 0014).	F-B7
FR-B16	The timeline must show events as markers.	F-B7
FR-B17	The system must publish precision and recall per event type against the mission timeline and one hand-labeled hour.	F-B7
FR-B18	The system must serve FastAPI on Lambda behind API Gateway's REST type under /v1 with search, segments, timeline, clips, events and ask (ADR 0007).	F-B8
FR-B19	The api_key_service (design doc C8) must let a visitor obtain a key without the owner in Phase B (self-service), apply per-key rate limits and daily quotas, and make revocation take effect instantly (ADR 0007). Checked by contract tests.	F-B8
FR-B20	The Lambda function must have a reserved-concurrency cap (ADR 0008).	F-B8
FR-B21	The ask endpoint must require a key and have its own daily cap under the model spend limit.	F-B8
FR-B22	The API must validate request size and format.	F-B8
FR-B23	The API must cache repeated searches.	F-B8

ID	Requirement	Feature
FR-B24	An alarm on spikes in rejected requests must disable a key that keeps hitting its limits.	F-B8
FR-B25	The developer API must paginate list results with cursors. Checked by contract tests.	F-B8
FR-B26	No AWS WAF (firewall rule) is added until a trigger condition for abuse is met; the trigger condition is open (OQ-30).	F-B8
FR-B33	The project must publish OpenAPI docs on a live docs page (hosting OQ-13).	F-B8
FR-B34	The project must have contract tests against openapi.yaml that pass in CI.	F-B8
FR-B35	The project must have a k6 load test that reports latency at a stated request rate.	F-B8
FR-B36	The project must provide an MCP server that exposes the API as tools (design doc C15).	F-B8
FR-B27	An agent must answer only from retrieved segments, cite channel, speaker and GET with a playable clip for every claim, and refuse when the tapes do not cover a question (ADR 0012).	F-B9
FR-B28	Evals must cover faithfulness (LLM judge calibrated on 50 hand-graded answers), exact checks that cited times fall inside the cited clips, refusal tests, and red-team prompts planted in transcript text.	F-B9
FR-B29	Great Expectations (replacing the Phase A Python checks, ADR 0023) must flag likely transcription errors (repeated phrases, text where no speech was detected), GET running backward within a channel, and missing clips.	F-B10
FR-B30	Low-confidence lines must be shown in grey and marked unverified, and the README must publish word error rate per channel.	F-B10
FR-B31	Results and models must leave Databricks by manual export (download, commit to GitHub); a spike tests automation (ADR 0017).	F-B3
FR-B32	The nightly run for Project 5 must do publish and quality checks only; transcription stays in session-time batches (ADR 0024).	F-B10

Phase C (intent only; done-when checks missing by decision until Phase C is designed)

ID	Requirement	Feature
FR-C1	Process the full mission on the five busiest channels, sized from Phase B's measured cost per audio hour.	F-C1
FR-C2	Fine-tune a small Whisper model and compare word error rate with the base and large models on a held-out hour.	F-C2
FR-C3	Provide a replay mode that plays all channels from any GET.	F-C3
FR-C4	Test noise reduction on the noisiest back-room channels and keep it only where it lowers word error rate.	F-C4

4. Quality requirements and scenarios (arc42 section 10)

Quality tree attributes: availability, cost-safety, security, correctness and trustworthiness, performance, maintainability, privacy, observability.

Numeric thresholds that the plan does not state are shown as "threshold: open (OQ-n)". Each such requirement still exists; the value is set when the open question is answered and before the part is built.

ID	Attribute	Scenario (stimulus, response, measure)	How verified	Source
QR-1	Availability	A visitor searches while the session box is off. The page and search answer from S3, CloudFront and Lambda.	Manual check on the public link with the session box off; same as the 5A.5 acceptance check	BP 5A.5, "Public sites are static files plus Lambda"
QR-2	Cold-start honesty	The first request after a period of inactivity is slow. The page says: "The first request after a period of inactivity may take up to 10 seconds while the service initializes."	Page text check; cold-start latency measured (platform step 1B.9)	BP 0.4

ID	Attribute	Scenario (stimulus, response, measure)	How verified	Source
QR-3	Cost: total	A normal month costs under $50; a Phase B month about $32 to $48 all in; a Phase C month about $35 to $48.	Cost per run recorded in the operations document (to be written while building); $40 AWS budget backup	BPD; BP, Monthly cost
QR-4	Cost: GPU	A GPU run cannot exceed its maximum GPU hours; GPU runs on spot only; job waits for spot. Value: open (OQ-12).	Terraform guard and job limit test; run record	BP, Hard limits; ADR 0015
QR-5	Cost: AI model	Spend stays within $15 a month (Phase B month cap $10, Phase C month cap $5); ask has a daily cap and a token limit per answer (values open, OQ-12).	Spend limit set in the API account before the first call; call log (model, prompt version, tokens, latency, cost) loaded into llm_calls; test that the cap refuses	BP; BPD; ADR 0015
QR-6	Traceability	Every speech segment traces back to a source file and offset.	Data test: no segment without file_id and offset (Great Expectations / unit test)	BP 5A.2
QR-7	Honesty of text	A corrected segment keeps its original text.	Data test: every corrected row has a non-null original; hand check of 100 corrections	BP 5B.2; ADR 0013
QR-8	Grounding of answers	Every claim in an Ask answer cites a clip, and each cited time falls inside the cited clip. Faithfulness judged by an LLM calibrated on 50 hand-graded answers. Thresholds: open (OQ-16).	Eval suite in CI	BP 5B.9; ADR 0012
QR-9	Refusal	When the tapes do not cover a question, the agent refuses.	Refusal tests in eval suite	BP 5B.9
QR-10	Security: quotas	An over-quota key gets a 429.	Contract tests	BP 5B.8
QR-11	Security: revocation	A revoked key is refused immediately.	Contract tests	BP 5B.8
QR-12	Security: concurrency	Traffic beyond the reserved-concurrency cap is rejected rather than billed. Cap value: open (OQ-12).	k6 load test	BP 5B.8; ADR 0008
QR-13	Security: prompt injection	Instructions planted in transcript text are not followed.	Red-team prompts planted in transcript text, in CI	BP 5B.9; platform rule 13 (least privilege, and source text is data, never instructions)
QR-14	Accuracy: transcription	Word error rate per channel type is published, measured against NASA's transcript (air-to-ground) and 30 hand-corrected minutes (back-room). Target: open (OQ-16).	MLflow runs; README table	BP 5B.3, 5B.10
QR-15	Accuracy: model choice	The transcription model is chosen from a README table of accuracy, speed and cost per channel type.	Decision record linked to the table	BP 5B.4; ADR 0004
QR-16	Retrieval quality	On 200 questions: recall at 10, MRR and time error of the top hit are reported; CI fails if recall at 10 drops below its threshold (open, OQ-16).	MLflow and CI	BP 5B.6
QR-17	Extraction quality	Precision and recall per event type are published; events without a valid span are rejected.	Eval against mission timeline and one hand-labeled hour; schema test	BP 5B.7; ADR 0014
QR-18	Performance: API	The load test reports latency at a stated request rate (rate: open, OQ-16).	k6 load test	BP 5B.8

ID	Attribute	Scenario (stimulus, response, measure)	How verified	Source
QR-19	Data integrity: clock	Anchor events line up across the two channels within a stated tolerance (tolerance: open, OQ-5). GET never runs backward within a channel.	Anchor agreement check in CI; Great Expectations check	BP 5A.4, 5B.10
QR-20	Data integrity: completeness	No clip is missing for a published segment.	Great Expectations check	BP 5B.10
QR-21	Resource: disk	Processing keeps the disk of the box doing the work under 40 GB (which box: OQ-11).	Disk monitor during 5B.1; job stops if exceeded (the stop rule is Design (proposed))	BP 5B.1; ADR 0016
QR-22	Reliability	Every job is safe to rerun; a spot interruption (two-minute warning) loses at most the work since the last per-file checkpoint.	Rerun test with sample files in CI	BP "How the box is run", 5A.3
QR-23	Maintainability: tests	Pipelines and checks are tested with small sample files in CI. No staged production failures.	CI	BP "How the plan works"
QR-24	Observability	Every DAG writes freshness and row counts to a metrics table; a public status page shows each pipeline's last good run; GPU utilization and memory are recorded every few seconds in tuning runs.	Status page; MLflow	BP 0.5, 5B.5
QR-25	Privacy	Email addresses or contact details of key requesters stay in DynamoDB and appear elsewhere only as a keyed-hash code. Retention period: to be set in the security and privacy document.	Data test on lake copy	BPD; ADR 0020
QR-26	Time handling	All wall-clock timestamps are UTC; GET stored separately; local time only on the page.	Schema tests on contract columns	BP rule 15; ADR 0010
QR-28	Lateness honesty	A nightly run is late or failed. The page shows the old data with a visible late notice and the last good run time; nothing is hidden. Freshness target: open (OQ-21).	Page test with a simulated late status.json	ADR 0025
QR-27	Provenance	NASA is credited on pages and in API responses; no endorsement is implied; source terms recorded in the README.	Page and README review	ADR 0019

5. Acceptance criteria

These are the "Done when" checks from the build plan, word for word in meaning. A feature is complete only when its criterion holds on the public link or in CI as stated.

Phase A

Feature	Step	Acceptance criterion
F-A1	5A.1	The manifest covers the in-scope files with hashes, and the terms note is in the README.
F-A2	5A.2	The speech share per channel is recorded, and every segment traces back to a file and offset.
F-A3	5A.3	The hour is transcribed, and audio hours per GPU hour are in the design note.
F-A4	5A.4	Anchor events line up across the two channels within a stated tolerance.
F-A5	5A.5	On the public link, a search for "computer alarm" plays the 1202 clip at the right GET with the session box off (through the unkeyed website route).

Phase A gate (BP, Build order and checkpoints): the landing hour is public first.

Phase B

Feature	Step	Acceptance criterion
F-B1	5B.1	Prerequisite: 5B.4 and 5B.5 have been run on the landing hour and the transcription model and settings are chosen in a decision record (the plan lists 5B.1 first but requires this order). Then: throughput and cost per audio hour are in the design note.
F-B2	5B.2	Each corrected segment keeps its original, and a hand check of 100 corrections reports how many were right.
F-B3	5B.3	Word error rate per channel type is published.
F-B4	5B.4	A comparison table in the README gives accuracy, speed and cost for both models per channel type, and the transcription model for 5B.1 is chosen from it in a decision record.
F-B5	5B.5	The README shows utilization, throughput and cost for each setting, the bottleneck found, and the improvement after the fix.
F-B6	5B.6	Scores are in MLflow and CI fails if recall at 10 drops below its threshold.
F-B7	5B.7	Precision and recall per event type against the mission timeline and one hand-labeled hour are published.
F-B8	5B.8	The docs page is live, contract tests pass, the load test reports latency at a stated request rate, an over-quota key gets a 429, a revoked key is refused, and traffic beyond the concurrency cap is rejected rather than billed.
F-B9	5B.9	The scores are in the README and CI fails below their thresholds.
F-B10	5B.10	Check tests pass in CI, low-confidence lines show in grey on the page, and accuracy per channel is in the README.

Phase B gate: the Whisper-versus-Parakeet comparison, the GPU tuning study, all 30 channels, the API and Ask Mission Control, with accuracy and retrieval scores published.

First-run checks (confirm an assumption before building on it; each has a fallback)

Phase	Check	Fallback	Open question
A	archive.org download speed is measured	none stated	OQ-2
A	Time code on track 1 can be decoded and the time zone in the file names is confirmed	Anchor events from NASA's transcript (ADR 0009)	OQ-1
A	OGG and FLAC versions of the same clip give the same accuracy	none stated; ADR 0005 is revisited	OQ-2
A	Whisper's speed on the T4 is measured on 10 minutes of audio	none stated	OQ-3
A	The GPU quota is approved	Smaller model on CPU	OQ-4
B	Parakeet transcribes a 1-minute clip on the T4 using fp16	5B.4 compares Whisper with a smaller Whisper model	OQ-3
B	The pyannote speaker-separation model is unlocked after accepting its terms on Hugging Face	none stated	OQ-28

Cross-cutting acceptance

Check	Criterion
Teardown	terraform destroy leaves nothing billing (platform gate; applies to this project's resources)
README	The README states exactly what is deployed (platform pattern from 1A.6 applied to this project)
Contracts	Every source and published table has a data contract; the schema is diffed before loading (BP rule 5 (give each source a contract and diff the schema before loading))

Part 3: Design document

Mission Control/design-doc.md

Design doc: Project 5, Mission Control

Project 5, Mission Control | Document: design doc | Set: rev 4, Oct 9, 2026

Template: Google design doc, arc42 sections, C4 model (Documentation Plan). Seventeen sections in the template order.

Conventions used in this document - "Decision" means a recorded decision with its ADR cited. "Fact" means confirmed in the Build Plans, Build Plan Decisions or Documentation Plan documents. "Design (proposed)" marks a choice made only in this document so that an engineer has something to build from; it is not a recorded decision and John confirms or changes it in review. Names of functions, classes, tables and columns are Design (proposed) working names unless an ADR or the plan names them. - Open questions are OQ-n in section 16. - Abbreviations: BP is the Build Plans for the Five Projects, BPD is the Build Plan Decisions and DP is the Documentation Plan (the three source documents). ADR is an architecture decision record in adrs/. Terms such as GET, tape time, session box, landing hour and the two doors are defined in section 17 (Glossary). - Terms used early (full list in section 17): Parquet: A columnar file format used for published tables; DuckDB: A small SQL engine that runs inside Lambda or on the box to query published files; FastAPI: The Python web framework used for the API; FLAC: A lossless audio format; about 5 times larger than OGG here; OGG: A compressed audio format; the version downloaded for processing (ADR 0005); Faithfulness: Whether every claim in an answer is supported by the cited segments; Keyed hash: A one-way scrambled code computed with a secret key (kept in Parameter Store), used instead of an email address or contact detail outside DynamoDB; Partition: One slice of a table (for example one pull or one tape) that a job overwrites as a unit; Red-team: Deliberately hostile test prompts, here including instructions planted in transcript text; LLM judge: A language model that scores answers, calibrated against hand-graded answers; Object Lock: An S3 setting that blocks deletion or overwrite of live raw files for 30 days; OpenAPI: The standard format for describing the HTTP API (openapi.yaml); Run of continuous speech: A stretch of speech with no break; the Phase A chunk. The pause length that ends a run is open (OQ-8).; Speaker turn: One speaker talking until another speaker takes over; known only after speaker separation (5B.2). Chunks follow speaker turns from then on.. Also: dbt, Kafka and Snowflake are platform tools that Project 5 does not use; OIDC is the login method GitHub Actions uses to reach AWS. - Proposed names: every function, class, table, column, component and API parameter name in this document is Design (proposed) unless an ADR or the plan names it. - Names: the page, the site and the web app all mean the React web app (component C6). The website route and the developer API are the two doors of the API (ADR 0022). The plan's term "public API" means the developer API. - Platform documents do not exist yet. Links to ../Platform/ are placeholder paths.

---

1. Status, owner, reviewers, last updated

Field	Value
Status	Draft, awaiting John's review. Coding of this project does not start until he approves.
Owner	John
Reviewers	John (sole approver)
Last updated	Oct 9, 2026
Related documents	requirements.md, adrs/, contracts/, openapi.yaml, platform design: ../Platform/design-doc.md
Build order position	Third project after the platform skeleton: Project 1, Project 3, then Project 5

2. Summary

Mission Control is a searchable, playable archive of Apollo 11 Mission Control audio. Audio from roughly 170 thirty-channel tapes (total size measured from the catalog first, OQ-22) is transcribed on a spot GPU, every spoken line is tagged with its channel and Ground Elapsed Time, and the result is exposed through a GET timeline with two or more lanes, full-text and vector search that plays the exact clip, an "Ask Mission Control" agent that answers only from retrieved segments with cited clips, and a keyed public API under /v1. Phase A puts the landing hour on the two clearest channels live; Phase B adds all 30 channels for the landing window, speakers, accuracy, the Whisper versus Parakeet

comparison, GPU tuning, retrieval evaluation, event extraction, the public API and the Ask agent; Phase C scales to the full mission on the five busiest channels. The outcome is a public demo plus published accuracy, retrieval and cost numbers, within the under-$50-a-month budget.

3. Context, scope and constraints

3.1 Background facts (confirmed)

- Source: archive.org "Apollo 11 MOCR ACR Collection": one channel per item, in FLAC, MP3 and OGG; uploaded by NASA with no license field. The audio was digitized by UT Dallas and NASA and is also published on the Explore Apollo site (collectSPACE). Roughly 170 thirty-channel tapes. No hours figure is used in these documents until the catalog is measured (Decided by John, Oct 9, 2026, OQ-22).
- NASA's public air-to-ground transcript, with GET timestamps, is the reference for timing and accuracy.
- The tape database lists start and end clock times without a time zone; landing-day tapes cover both readings, so the time zone needs the audio's time code or anchor events (OQ-1).
- NASA works are generally not copyrighted. Credit NASA and do not imply endorsement. Confirm the terms before publishing clips (ADR 0019).
- NVIDIA T4: appears in the Parakeet model's test hardware, but its architecture is not on the supported list; the T4 cannot use bf16 (ADR 0004).

3.2 System boundary

Inside: catalog, audio prep, transcription, mission clock, search index, clips, timeline web app, public API, key service, Ask agent, evaluation, quality checks, MCP server, this project's Terraform. Outside: the platform (scheduler, Spark, Airflow, Iceberg and Glue base, S3 buckets, CloudFront site, Parameter Store, serving template, status and incident pipeline, CI) which is described in ../Platform/design-doc.md, and the external services in section 6.

3.3 Constraints

Constraint	Value	Source
Budget	Under $50 a month; browser only	BPD
Phase B month	About $32 to $48 all in; 8 to 12 GPU hours; $1 to $3 of batch model calls; model API capped at $10	BP
Phase C month	About $35 to $48 all in; 15 to 25 GPU hours; $2 to $5 of batch model calls; model API capped at $5	BP
AI spend	$15 a month, cheapest model only for now	BPD
GPU	g4dn.xlarge spot only; waits for spot; spot quota of 4 vCPUs for G instances requested in step 0.1	BP, ADR 0003
Time zones	All wall-clock timestamps UTC; local only on the page	rule 15 (store every timestamp in UTC and convert only on the page), ADR 0010
Disk	Under 40 GB in Phase B	ADR 0016
Always-on parts	Lambda and Fargate write only to S3, DynamoDB, CloudWatch and X-Ray	rule 16 (anything running 24/7 writes only to always-on storage)
Licensing	Databricks Free Edition is non-commercial; check before charging for any project	BP assumptions
API doors	Two doors: unkeyed website route (per-visitor limits, daily cap, ask off until OQ-13 decides its setting) and keyed REST developer API (keys on request in Phase A, opened fully in Phase B); numbers open	ADR 0022, John, Oct 9, 2026
Late data	Show old data with a visible late notice and last good run time; never hide	ADR 0025, John, Oct 9, 2026
Nightly run	Publish and quality checks only; transcription in session-time batches	ADR 0024, John, Oct 9, 2026
Not used	dbt, Snowflake, Kafka (empty cells for Project 5 in the technology table). The empty cells are read as "not used"; confirm in review.	BP

3.4 Privacy rules that apply

- Key requesters' contact details stay in DynamoDB; elsewhere only a keyed-hash code (ADR 0020).
- Keys live in Parameter Store only; no key in code or GitHub (platform).
- Transcript text is data, never instructions (rule 13 (least privilege, and source text is data, never instructions)).
- The audio is historical public archive material; no personal data of visitors is collected by the pages described here, other than what the key request collects (OQ-14).

4. Goals and non-goals

Goals

1. Transcribe the Apollo 11 audio at the lowest cost per audio hour that meets measured accuracy, and publish the numbers.
2. Let anyone search and play the exact moment, with the session box off.
3. Show who was talking when on a GET timeline.
4. Answer questions only from evidence, with playable clips, or refuse.
5. Offer a secure public API that cannot be abused or run up a bill.
6. Choose the transcription model by measured accuracy and cost.

Non-goals

- No live streaming and no Kafka for this project (not in its technology column).
- No dbt models and no Snowflake marts (not in its technology column).
- No noise reduction before Phase C, and only where it lowers word error rate.
- No full-mission coverage before Phase C; Phase B covers the landing window only.
- No AWS WAF unless abuse appears (ADR 0007).
- No stronger model than the cheapest until Phase C evals justify it (BPD).
- No staged production failures (BPD).
- No commercial use without a license check (BP assumptions).

5. Requirements

Full text: requirements.md.

Summary of functional requirements - Phase A (FR-A1 to FR-A22; chunks are runs of continuous speech until 5B.2, ADR 0006): catalog with manifest and channel map; OGG download, 16 kHz mono, Silero speech detection; faster-whisper large-v3 on spot GPU with checkpoints; mission clock offset per tape; LanceDB index, Opus clips, two-lane GET timeline, search; keyed API access on request. - Phase B (FR-B1 to FR-B30): all 30 channels one tape at a time; pyannote, glossary and batch corrections; WER; Parakeet versus Whisper; tuning; retrieval eval; event extraction; public API; Ask agent with evals; Great Expectations checks and grey low-confidence lines. - Phase C (FR-C1 to FR-C4; done-when checks missing by decision until Phase C is designed): full mission on five busiest channels; Whisper fine-tune; replay mode; noise reduction test.

Summary of quality requirements (28 scenarios, QR-1 to QR-28) - Availability with the session box off (QR-1); cold-start wording (QR-2). - Cost: under $50 a month, GPU hours cap, $15 AI limit (QR-3 to QR-5). - Trust: traceability, original text kept, grounded answers, refusals, injection resistance (QR-6 to QR-9, QR-13). - Security: 429 over quota, instant revocation, concurrency cap (QR-10 to QR-12). - Accuracy and retrieval: WER, model choice, recall at 10, event precision and recall, API latency (QR-14 to QR-18). - Integrity and reliability: anchor agreement and GET monotonic, completeness, disk, rerun safety, tests, observability, privacy, UTC, provenance (QR-19 to QR-27).

6. Architecture

6.1 System context (C4 level 1)

```
flowchart TB
    visitor(["`Visitor
searches, plays, asks`"]):::person
    dev(["`API developer
```

```
keyed /v1 and MCP`"]):::person
    owner(["`Owner (John)
approves, reviews`"]):::person
    mc["`Mission Control
transcribes, indexes and serves
Apollo 11 Mission Control audio`"]:::cont
    archive["`archive.org
OGG, FLAC, MP3, metadata`"]:::ext
    nasa["`NASA air-to-ground
transcript, GET reference`"]:::ext
    modelapi["`Model API
cheapest model only`"]:::ext
    dbx["`Databricks Free Edition
MLflow tracking`"]:::ext
    hf["`Hugging Face
pyannote model`"]:::ext
    platform["`Platform
scheduler, Spark, Airflow,
Iceberg, Glue, CI, status`"]:::ext
    visitor -->|"searches, plays, asks"| mc
    dev -->|"calls /v1 with key"| mc
    owner -->|"approves, reviews"| mc
    mc -->|"downloads"| archive
    mc -->|"reads"| nasa
    mc -->|"corrections, events, ask"| modelapi
    mc -->|"logs runs"| dbx
    mc -->|"downloads model"| hf
    mc -->|"runs on"| platform

    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef gpu fill:#fce7f3,stroke:#be185d,color:#500724
```

6.2 Containers (C4 level 2)

```
flowchart LR
    visitor(["Visitor"]):::person
    dev(["API developer"]):::person
    subgraph mc["Mission Control"]
        direction LR
        web["`Web app
React, TypeScript`"]:::cont
        mcp["`MCP server
Python`"]:::cont
        cdn["`Site and clips
CloudFront, S3`"]:::store
        site_api["`Website route
unkeyed, per-visitor limits,
daily cap`"]:::cont
        api["`Developer API
FastAPI on Lambda,
API Gateway REST, keyed`"]:::cont
        idx[("`Search index
LanceDB in S3`")]:::store
        ddb[("`Visitor data
DynamoDB`")]:::store
        lake[("`Lake tables
Iceberg on S3, Glue`")]:::store
        batch["`Batch pipelines
Airflow, Spark on k3s`"]:::cont
        gpu["`GPU transcription
g4dn.xlarge spot,
faster-whisper, NeMo`"]:::gpu
    end
    archive["archive.org"]:::ext
    modelapi["Model API"]:::ext
    dbx["Databricks MLflow"]:::ext
    aws["`Parameter Store,
X-Ray`"]:::ext
    visitor -->|uses| web
    dev -->|"calls with key"| api
    dev -->|"uses tools"| mcp
    web -->|"loads site, clips"| cdn
    web -->|"search, timeline (ask off until OQ-13 is decided)"| site_api
    site_api -->|"same code, same index"| idx
```

```
    dev -->|"requests key (form)"| ddb
    mcp -->|calls| api
    api -->|queries| idx
    api -->|"published Parquet"| lake
    api -->|"key status"| ddb
    api -->|ask| modelapi
    site_api -->|"ask, if on"| modelapi
    api -->|"reads keys, traces"| aws
    batch -->|downloads| archive
    batch -->|"reads, writes"| lake
    batch -->|writes| idx
    batch -->|"publishes clips"| cdn
    batch -->|"starts job"| gpu
    gpu -->|"words, segments"| lake
    batch -->|"logs runs (export manual)"| dbx
    batch -->|"batch calls"| modelapi

    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef gpu fill:#fce7f3,stroke:#be185d,color:#500724
```

Note: the website route and the developer API share the same Lambda code and the same reserved-concurrency cap (ADR 0022). The diagram shows "API reads published Parquet" because the platform serving template runs DuckDB over published Parquet in Lambda (../Platform/); whether this project's API uses it for segments, timeline and events is Design (proposed) and OQ-13 and OQ-25 affect the exact split between LanceDB and Parquet reads.

6.3 Components (C4 level 3): pipeline and API containers

Components, part 1: the Phase A pipeline path.

```
flowchart LR
    subgraph batch["Batch pipelines (Airflow and Spark)"]
        c1["`C1 catalog_ingest
metadata, channel map,
manifest`"]:::cont
        c2["`C2 audio_prep
16 kHz mono,
speech segments`"]:::cont
        c4["`C4 mission_clock
tape time to GET,
offsets`"]:::cont
        c5["`C5 index_and_clips
chunks, LanceDB,
Opus clips`"]:::cont
    end
    subgraph gpu["GPU job"]
        c3["`C3 gpu_transcribe
words, confidence,
checkpoints`"]:::gpu
    end
    c1 -->|manifest| c2
    c2 -->|"`speech segments
via S3`"| c3
    c3 -->|"`transcript with
tape offsets`"| c4
    c4 -->|"`GET-aligned
transcript`"| c5

    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef gpu fill:#fce7f3,stroke:#be185d,color:#500724
```

Components, part 2: Phase B components and the API container.

```
flowchart LR
    c3["C3 gpu_transcribe"]:::gpu
    c4["C4 mission_clock"]:::cont
    c5["C5 index_and_clips"]:::cont
    subgraph batch["Batch pipelines (Phase B)"]
        c9["`C9 speaker_cleanup
speakers, glossary,
```

```
corrections`"]:::cont
        c11["`C11 retrieval_eval
recall at 10, MRR`"]:::cont
        c12["`C12 event_extraction
events with spans`"]:::cont
        c14["`C14 quality_checks
Great Expectations`"]:::cont
    end
    subgraph apis["Public API (Lambda)"]
        c7["`C7 api_service
FastAPI routers`"]:::cont
        c8["`C8 api_key_service
request, issue, revoke`"]:::cont
        c13["`C13 ask_agent
grounded answer
or refusal`"]:::cont
    end
    c3 -->|words| c9
    c9 -->|"`corrected text,
speakers`"| c5
    c5 -->|index| c11
    c9 -->|segments| c12
    c4 -->|offsets| c14
    c5 -->|"index, clips"| c7
    c7 -->|"ask route"| c13
    c7 -->|"key checks"| c8

    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef gpu fill:#fce7f3,stroke:#be185d,color:#500724
```

6.4 Data flow

Solid arrows are the pipeline path (audio prep, GPU transcription, mission clock, Phase B cleanup, indexing); dotted arrows are the NASA reference transcript and the authored evaluation sets feeding timing checks and evaluations.

```
flowchart TB
    subgraph src["Sources"]
        s1["archive_item_metadata"]:::ext
        s2["apollo_audio_files"]:::ext
        s3["nasa_air_to_ground_transcript"]:::ext
        s4["`retrieval_eval_questions,
ask_eval_cases`"]:::ext
    end
    r1["`Raw: OGG audio kept as a manifest (audio_manifest)`"]:::store
    subgraph lakeT["Lake tables (Iceberg)"]
        l1["`tape_catalog,
channel_map`"]:::store
        l2["speech_segments"]:::store
        l3["`transcript_words,
transcript_segments,
transcription_runs`"]:::store
        l4["`tape_clock_offsets,
anchor_events`"]:::store
        l5["`speaker_turns,
transcript_corrections,
events, quality_flags,
wer_results`"]:::store
    end
    subgraph pub["Published tables"]
        p1["`search_chunks,
search_index (LanceDB)`"]:::store
        p2["clips (Opus)"]:::store
    end
    subgraph srv["Serving"]
        v1["Public API /v1"]:::cont
        v2["Web app and site"]:::cont
    end
    s1 --> l1
    s2 --> r1
    r1 --> l2
    l2 --> l3
    l3 --> l4
    s3 -.-> l4
```

```
    l3 --> l5
    l4 --> p1
    l4 --> p2
    l5 --> p1
    s4 -.-> l5
    p1 --> v1
    p2 --> v2
    l5 --> v1
    v1 --> v2

    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef gpu fill:#fce7f3,stroke:#be185d,color:#500724
```

7. Component breakdown

Phase column: A, B or C is the phase in which the component is first built.

ID	Component	Phase	Build step
C1	catalog_ingest	A	5A.1
C2	audio_prep	A	5A.2
C3	gpu_transcribe	A	5A.3
C4	mission_clock	A	5A.4
C5	index_and_clips	A	5A.5
C6	web_app	A	5A.5
C7	api_service	A (keyed access), B	5A.5, 5B.8
C8	api_key_service	A, B	5A.5, 5B.8
C9	speaker_cleanup	B	5B.2
C10	accuracy_and_tuning	B	5B.3 to 5B.5
C11	retrieval_eval	B	5B.6
C12	event_extraction	B	5B.7
C13	ask_agent	B	5B.9
C14	quality_checks	B	5B.10
C15	mcp_server	B	5B.8
C16	infrastructure	A, B	all

C1 catalog_ingest (5A.1)

- Job: pull archive item metadata into Iceberg, build the channel map from tape track to console position, build the manifest, record the terms.
- Inputs: archive.org item metadata for the "Apollo 11 MOCR ACR Collection" (contract: archive_item_metadata); an authored channel-map source (OQ-6).
- Outputs: tape_catalog, channel_map, audio_manifest tables; the terms note in the README.
- Main functions and classes: `fetch_item_metadata(identifier)`, `list_collection_items()`, `parse_item(item_json)` returning `CatalogRow`, `build_channel_map(rows)`, `build_manifest(rows)`, `diff_schema(contract, observed)` (platform rule 5 (give each source a contract and diff the schema before loading)), `write_partition(table, rows, pull_date)` (overwrite one partition per pull, rule 3 (make tasks idempotent: overwrite a partition, never blind-append)), DAG `p5_catalog`. Classes: `CatalogRow`, `ManifestRow`.
- Behavior notes: stops on any non-200 response (pattern from Project 1 step 1A.1, Design (proposed) for this project); stores source URL, fetch time and hash with each raw response (rule 1 (keep raw files immutable with source URL, fetch time and hash)).

C2 audio_prep (5A.2)

- Job: download the OGG files for the landing hour on the two channels, convert to 16 kHz mono with ffmpeg in Spark, detect speech with Silero.
- Inputs: audio_manifest rows in scope; archive.org audio.
- Outputs: 16 kHz mono working files in the temporary 7-day area (Decided by John, Oct 9, 2026, ADR 0011); speech_segments; the speech share per channel (written to the pipeline metrics table, ../Platform/).
- Main functions and classes: `download_ogg(manifest_row)`, `verify_hash(path, manifest_row)`, `convert_to_16k_mono(path)`, `detect_speech(wav_path, params)` using Silero, `write_speech_segments(segments)`, `compute_speech_share(channel_id)`. Class: `SpeechSegment`.
- Checks: every segment has file_id and start and end offsets (QR-6).

C3 gpu_transcribe (5A.3, model per ADR 0004 in Phase B)

- Job: transcribe speech segments with faster-whisper large-v3 on a g4dn.xlarge spot box (smaller model on CPU if the GPU quota is not granted); save word timestamps and confidence; checkpoint per file.
- Inputs: speech_segments and the 16 kHz mono files via S3.
- Outputs: transcript_words, transcript_segments, transcription_runs (includes audio hours per GPU hour).
- Main functions and classes: `start_gpu_run(max_gpu_hours)`, `transcribe_file(wav_path, segments, model_config)`, `load_checkpoint(file_id)`, `save_checkpoint(file_id, progress)`, `handle_spot_interruption()` (two-minute warning, finish the current write, exit; rerun resumes), `write_words(rows)`, `record_run_metrics(run)`. Classes: `ModelConfig` (model name, precision, batch size, device), `TranscriptionJob`, `Checkpoint`.
- Failure handling: section 8.2. The orchestration of the GPU box (how Airflow starts it, how audio arrives, disk size) is OQ-11.

C4 mission_clock (5A.4)

Function-level detail in section 7.1 below.

C5 index_and_clips (5A.5)

- Job: chunk transcripts, embed on CPU, write the LanceDB index to S3, cut speech into Opus clips, publish clips through CloudFront, publish the timeline data the page needs.
- Inputs: transcript_segments with GET (after C4); 16 kHz mono or original audio for clipping (OQ-18).
- Outputs: search_chunks, search_index (LanceDB), clips table, Opus files in S3, published timeline JSON.
- Main functions and classes: `chunk_transcript(segments, policy)`, `embed_chunks(chunks, embedding_model)`, `write_lance_index(chunks_with_vectors, s3_path)`, `cut_clip(source, start_offset_s, end_offset_s)` producing Opus via ffmpeg, `publish_clips(clips)`, `build_timeline_json(channel_ids)`. Classes: `ChunkPolicy` (`continuous_speech` in Phase A, `speaker_turn` after 5B.2; size parameters compared in 5B.6), `Chunk`, `Clip`.
- NOTE (Decided by John, Oct 9, 2026, ADR 0006): the plan's 5A.5 says to chunk by speaker turn, but pyannote only arrives in 5B.2. In Phase A a chunk is a run of continuous speech with the speaker left blank (`search_chunks.speaker` is null); the index is rebuilt by speaker turn after 5B.2. The pause length that ends a run and the chunk sizes compared in 5B.6 are open (OQ-8).
- Open: embedding model and dimension (OQ-7), clip parameters (OQ-18).

C6 web_app (5A.5, extended in B)

- Job: React and TypeScript page from the platform site template: two lanes on a GET timeline, search box, clip player; later event markers, Ask panel, grey low-confidence lines; the cold-start wording from the serving template.
- Inputs: published timeline JSON and clips from CloudFront; search from the unkeyed website route (ask only if enabled there, OQ-13).
- Outputs: the public page; preview deploy per pull request (platform).
- Main components (React): `TimelinePage`, `GetTimeline` (lanes), `Lane`, `SearchBox`, `ResultList`, `ClipPlayer`, `EventMarkers` (B), `AskPanel` (B), `TranscriptLine` with `LowConfidence` styling (B), `CreditFooter`. Helper functions: `formatGet(seconds)`, `getToPixel(seconds, viewport)`, `playClip(clip)`.
- Late data: the page reads status.json and, when a run is late or failed, keeps showing the old data with a visible late notice and the time of the last good run in local time (ADR 0025); the freshness target is open (OQ-21).

- The page uses the unkeyed website route, not a key (ADR 0022); what identifies a visitor for per-visitor limits is open (OQ-13).

C7 api_service (5A.5 website route and keyed access; 5B.8 full)

- Job: one FastAPI code base serving two doors (ADR 0022): the unkeyed website route (per-visitor limits, daily cap; ask off until OQ-13 decides its setting) and the keyed developer API on Lambda behind API Gateway REST type, under /v1: search, segments, timeline, clips, events, ask. Validation, cursor pagination, caching of repeated searches, structured errors.
- Inputs: HTTP requests with an API key header; LanceDB index; published tables; DynamoDB key status.
- Outputs: JSON responses per openapi.yaml; X-Ray traces; metrics for alarms.
- Main functions and classes: `app = FastAPI()`, routers `search_router`, `segments_router`, `timeline_router`, `clips_router`, `events_router`, `ask_router`; `validate_request(request)` (size and format), `paginate(rows, cursor, limit)` and `encode_cursor` / `decode_cursor`, `cache_get(key)` / `cache_put(key, value)` (mechanism OQ-23), `search_index(query, mode, filters)`, `error_response(code, message)`, Lambda entry `handler` (via an ASGI adapter; the adapter library is not decided and is OQ-25). Classes: `ApiError`, `Page`, response models matching the OpenAPI schemas.
- Limits: reserved-concurrency cap; per-key rate limit and daily quota in the API Gateway usage plan (values OQ-12).

C8 api_key_service (5A.5, 5B.8)

- Job: request, issue, revoke and automatically disable keys; Phase A on request through a request form (a web form where a visitor asks for a key) that stores requests in DynamoDB (ADR 0020), Phase B self-service.
- Inputs: key requests from the form (fields OQ-14); API Gateway rejected-request metrics.
- Outputs: api_key_requests rows in DynamoDB; API Gateway API keys attached to usage plans; alarms.
- Main functions and classes: `request_key(contact, purpose)`, `issue_key(request_id)`, `revoke_key(key_id, reason)`, `disable_abusive_key(key_id)` (triggered by the alarm on spikes in rejected requests), `hash_contact(value)` (keyed hash; key in Parameter Store). Class: `KeyRequest`.

C9 speaker_cleanup (5B.2)

- Job: separate speakers with pyannote on mixed channels (air-to-ground: CAPCOM and the crew); fix jargon with glossary rules; batch model call for low-confidence segments only with structured output listing each change; keep the original.
- Inputs: transcript_words and transcript_segments; audio; glossary rules (content OQ-26).
- Outputs: speaker_turns, transcript_corrections; updated chunks.
- Main functions and classes: `run_diarization(wav_path)`, `assign_speakers(words, turns)`, `apply_glossary(text, rules)`, `select_low_confidence(segments, threshold)` (threshold OQ-16), `correct_batch(segments, prompt_version)` returning `CorrectionResult`, `record_correction(original, corrected, changes)`, `sample_for_hand_check(corrections, n)` (n = 100). Classes: `GlossaryRule`, `CorrectionResult`, `SpeakerTurn`.

C10 accuracy_and_tuning (5B.3, 5B.4, 5B.5)

- Job: word error rate; model comparison; tuning study.
- Inputs: transcripts; NASA's air-to-ground transcript; 30 hand-corrected minutes of back-room audio.
- Outputs: wer_results, transcription_runs; MLflow runs; README tables.
- Main functions and classes: `normalize_text(text)`, `compute_wer(reference, hypothesis)`, `align_to_reference(segments, nasa_lines)`, `run_benchmark(model_config, audio_set)`, `sample_gpu_metrics(interval)` (nvidia-smi or DCGM), `find_bottleneck(metrics)` (decoding, audio loading or GPU), `log_mlflow_run(run)`. Classes: `BenchmarkResult`, `GpuSample`.
- Settings compared: fp16 and int8, several batch sizes (values for batch sizes are chosen when 5B.5 runs).

C11 retrieval_eval (5B.6)

- Job: 200 test questions tied to known moments (150 from the transcript and mission timeline, 50 by hand); recall at 10, MRR, time error of the top hit; compare chunk sizes and hybrid versus vector-only.
- Inputs: retrieval_eval_questions; the index; the mission timeline (OQ-17).

- Outputs: MLflow runs; CI gate on recall at 10.
- Main functions and classes: `build_question_set()`, `recall_at_k(results, expected, k)`, `mrr(results, expected)`, `time_error(top_hit, expected)`, `run_retrieval_eval(index, questions, chunk_policy, mode)`. Classes: `EvalQuestion`, `EvalResult`.

C12 event_extraction (5B.7)

- Job: batch model call extracts events (alarms, go/no-go polls, system problems, decisions, handovers) with type, GET, channel, speakers, confidence and exact span; reject events without a valid span.
- Inputs: transcript_segments, speaker_turns.
- Outputs: events table; timeline markers.
- Main functions and classes: `extract_events_batch(segments, prompt_version)`, `validate_span(event, segments)` (span text must exist verbatim in the cited segments; Design (proposed)), `write_events(events)`, `score_events(events, reference)`. Classes: `Event`, `EventType` (alarm, go_no_go_poll, system_problem, decision, handover).

C13 ask_agent (5B.9)

- Job: answer only from retrieved segments with cited clips, refuse when not covered.
- Inputs: a question; retrieval tool over search_index; clips.
- Outputs: an answer with claims, each with channel, speaker, GET and a clip, or a refusal; logged call; X-Ray trace.
- Main functions and classes: `retrieve_segments(question, k)` (the single tool the agent may use; Design (proposed)), `compose_answer(question, segments)` (structured output), `verify_citations(answer, clips)` (each cited time inside its cited clip), `refuse(reason)`, `log_call(call)` to the call log in S3. Classes: `Claim`, `Citation`, `AskResult`.
- Evals: faithfulness LLM judge calibrated on 50 hand-graded answers; exact citation checks; refusal tests; red-team prompts planted in transcript text. Thresholds: OQ-16.

C14 quality_checks (5B.10)

- Job: Phase A: a short set of hand-written Python checks (row counts, nulls, schema, time ranges) as the publish gate (ADR 0023). Phase B: Great Expectations suites replace them and add likely transcription errors, GET running backward, missing clips; publish gate (write, audit, publish).
- Inputs: staging tables.
- Outputs: quality_flags; pass or fail of the publish step.
- Main functions: `check_repeated_phrases(segments)`, `check_text_without_speech(segments, speech_segments)`, `check_get_monotonic(segments)`, `check_missing_clips(segments, clips)`, `run_publish_gate(table)`. Phase A functions: `check_row_counts`, `check_not_null`, `check_schema` (against the contract), `check_time_ranges`.

C15 mcp_server (5B.8)

- Job: expose search, segments, timeline, clips, events and ask as MCP tools, calling the public API with a key.
- Main functions: one tool per endpoint: `tool_search`, `tool_segments`, `tool_timeline`, `tool_clips`, `tool_events`, `tool_ask`. Where it runs is not decided (OQ-13).

C16 infrastructure (Terraform)

- Job: this project's resources as code, cheap to destroy: GPU launch configuration (spot only, 4-hour or per-run limits), S3 prefixes, LanceDB path, CloudFront behaviors for clips, API Gateway REST API with usage plans and keys, Lambda with reserved concurrency, DynamoDB table with point-in-time recovery, alarms (rejected-request spike, storage size), IAM roles per pipeline with least privilege, Parameter Store entries. Reuses the platform modules (../Platform/).

7.1 Function-level detail: transcript time alignment (5A.4), component C4 mission_clock

Purpose: convert positions on the tape ("tape time") to GET, with one offset per tape, using the time code on track 1 or, if it cannot be decoded, anchor events from NASA's air-to-ground transcript (ADR 0009). Wall-clock tape times are stored in UTC once the zone is confirmed (ADR 0010). The format of the time code and the time zone are unknown until the first-run check (OQ-1), so `decode_timecode` is specified by interface only; its implementation is chosen after the check.

Definition (Design (proposed)): the offset of a tape is the value to add to the number of seconds since the tape's start to obtain GET. For an anchor match, offset = NASA GET of the anchor minus the tape time of the matched segment. The definition of "tape start" for items of the same tape is fixed in review (OQ-1, OQ-6).

Classes

Class	Purpose	Fields	Notes
`ClockMethod` (enum)	Which method produced an offset	`TIMECODE`, `ANCHOR`	Stored with the offset
`TranscriptLine`	One line of NASA's air-to-ground transcript	`get_s`, `speaker`, `text`	From `parse_nasa_transcript`
`AnchorEvent`	A known event used for anchoring	`anchor_id`, `label`, `nasa_get_s`, `match_phrases`	Plan names the 1202 alarm and the landing call
`AnchorMatch`	Where an anchor was found on a tape and channel	`anchor_id`, `channel_id`, `file_id`, `tape_time_s`, `segment_id`, `score`	
`OffsetFit`	Result of a fit	`offset_s`, `residual_s`, `n_points`, `method`	`residual_s` is the fit error
`TapeClockOffset`	The stored result for a tape	`tape_id`, `offset_s`, `method`, `fit`, `anchor_ids`, `status`, `run_id`	One row per tape
`AgreementReport`	Result of the cross-channel check	`passed`, `max_diff_s`, `tolerance_s`, `per_anchor`	

Functions

Function or class	Purpose	Takes	Returns
`parse_nasa_transcript(raw_transcript)`	Turn NASA's air-to-ground transcript into rows with GET and text	The raw transcript file (format OQ-17)	List of `TranscriptLine`
`parse_get(get_text)`	Convert a GET string as printed in the transcript to seconds	GET text	Seconds as a number
`format_get(get_s)`	Convert seconds to display text for the page and citations	Seconds	GET text
`load_anchor_definitions()`	Read the anchor events (1202 alarm, landing call) with their NASA GET	Anchor configuration file under version control	List of `AnchorEvent`
`resolve_tape_time_base(catalog_row, tz_name)`	Convert a tape's published start text to UTC once the zone is confirmed	A `tape_catalog` row, a time zone name	UTC timestamp
`decode_timecode(track1_samples, sample_rate)`	Decode the time code from track 1 audio	Track 1 samples and sample rate	List of (sample index, decoded time) pairs; empty list if undecodable
`fit_offset_from_timecode(decoded_pairs, time_base)`	Fit one offset from decoded time code, rejecting outliers	Decoded pairs and the tape's time base	`OffsetFit`
`find_anchor_in_transcript(segments, anchor, search_window)`	Locate the segment on a tape and channel that matches an anchor event by text	The channel's transcript segments, an `AnchorEvent`, a window of tape time to search	`AnchorMatch` or none
`fit_offset_from_anchors(matches, anchors)`	Compute the offset from one or more anchor matches (NASA GET minus tape time)	List of `AnchorMatch`, matching `AnchorEvent` list	`OffsetFit`
`choose_method(tape)`	Try the time code first; fall back to anchors if decoding fails	A tape record	A `ClockMethod` and its `OffsetFit`
`check_anchor_agreement(matches_by_channel, anchors, tolerance_s)`	Check that the same anchor gives the same GET on both channels within the tolerance	Matches per channel, anchors, tolerance (value OQ-5)	`AgreementReport`

Function or class	Purpose	Takes	Returns
`store_offset(tape_clock_offset)`	Write one row per tape idempotently (overwrite the tape's partition, never blind append)	`TapeClockOffset`	Nothing; table updated
`tape_time_to_get(tape_time_s, offset)`	Convert seconds since the tape start to GET	Seconds, `TapeClockOffset`	GET in seconds
`apply_offsets(segments, offsets)`	Add `get_start_s` and `get_end_s` to transcript segments	A segments table, offsets table	Segments table with GET columns (Spark job)

Failure handling: if neither method produces an offset, the tape's segments get no GET and the publish gate blocks that tape; if `check_anchor_agreement` fails, the offsets are not marked `accepted`, the run opens an incident (platform step 0.5), and nothing is published for the affected tapes. GET running backward within a channel after alignment is also caught by C14 (`check_get_monotonic`).

Tests (section 13): unit tests with synthetic decoded pairs and synthetic transcripts; a sample-file test that a known anchor on two channels agrees within the tolerance; rerun test that `store_offset` leaves the row count unchanged.

8. Runtime view

8.1 Phase A end-to-end pipeline (session run)

```
sequenceDiagram
    participant AF as Airflow (session box)
    participant AR as archive.org
    participant SP as Spark
    participant GP as GPU box (spot)
    participant S3 as S3 and Iceberg
    participant LDB as LanceDB index
    AF->>AR: fetch item metadata (stop on non-200)
    AF->>S3: write tape_catalog, channel_map, audio_manifest
    AF->>AR: download OGG for landing hour, two channels
    AF->>SP: convert to 16 kHz mono, Silero speech detection
    SP->>S3: speech_segments, speech share
    AF->>GP: start spot job (max GPU hours)
    GP->>S3: transcript_words, segments, checkpoint per file
    AF->>SP: mission clock (timecode or anchors), apply offsets
    SP->>S3: tape_clock_offsets, GET-aligned segments
    AF->>SP: quality gate, chunk, embed on CPU
    SP->>LDB: write index
    SP->>S3: Opus clips, timeline JSON
    AF->>S3: publish (write, audit, publish)
```

Steps and failure handling 1. Catalog: a non-200 response stops the DAG. A schema change versus the contract fails loudly before loading (rule 5). 2. Download: the hash is verified against the manifest; a mismatch stops that file and opens an incident (replays must be reproducible, ADR 0011). 3. Audio prep: rerunning overwrites the partition; every segment gets file_id and offsets. 4. GPU run: see 8.2. 5. Mission clock: see 7.1. 6. Publish: write, audit, publish; a failed check stops publishing (rule 4 (write, audit, publish, meaning data lands in staging, passes checks, then goes live)). A Spark job over its row or runtime limit stops before publishing and opens an incident (ADR 0015). 7. Phase A publish gate: a short set of hand-written Python checks; Great Expectations replaces them in Phase B (ADR 0023).

8.2 GPU transcription run with spot interruption

1. Airflow requests a spot instance (g4dn.xlarge). If none is available the job waits; it does not fall back to on-demand (ADR 0003).
2. The job reads speech segments and audio from S3, transcribes file by file, writing words and a checkpoint per file.
3. On the two-minute spot warning, `handle_spot_interruption` stops after the current checkpoint; the next run resumes from the last checkpoint. Idempotent writes make a rerun safe.
4. The run stops when it reaches the maximum GPU hours for a run; it records an incident (value OQ-12).
5. The run records audio hours per GPU hour and cost (`transcription_runs`).
6. If the GPU quota is not approved, the CPU fallback with a smaller model is used (ADR 0003, OQ-4).

8.3 Search from the public page (session box off)

1. Visitor types "computer alarm" in the search box.
2. The page calls the unkeyed website route, which applies the per-visitor limits and the daily cap (ADR 0022; visitor identity OQ-13).
3. Lambda queries LanceDB in S3, returns ranked chunks with GET, channel, text and clip links.
4. The page shows results; the visitor selects one; the player loads the Opus clip from CloudFront; the timeline marks the GET.
5. Failure: Lambda cold start can take up to 10 seconds; the page carries the platform's wording (BP 0.4). A limit rejection returns 429 and the page shows a message. If the last nightly run was late or failed, the page still shows the data with a visible late notice and the last good run time (ADR 0025).
6. Acceptance: a search for "computer alarm" plays the 1202 clip at the right GET with the session box off.

8.4 Keyed developer API request

```
sequenceDiagram
    participant C as Client
    participant GW as API Gateway REST
    participant L as Lambda (FastAPI)
    participant D as DynamoDB
    C->>GW: GET /v1/search + API key
    GW->>GW: check key, usage plan rate and quota
    alt missing, invalid or revoked key
        GW-->>C: refused
    else over quota or rate
        GW-->>C: 429
    else under limits
        GW->>L: invoke (reserved concurrency cap)
        alt cap exceeded
            L-->>GW: rejected, not billed
        else ok
            L->>L: validate size and format, check cache
            L-->>C: 200 + page + next cursor
        end
    end
```

Failure handling: validation errors return 400 with the error object; a key that keeps hitting limits is disabled by the alarm flow in 8.6.

8.5 Ask Mission Control

1. Client posts a question to /v1/ask with a key (the key has its own daily cap).
2. The agent calls `retrieve_segments`; transcript text is untrusted data.
3. The model (cheapest only; structured output) composes claims; each claim cites channel, speaker, GET and a clip.
4. `verify_citations` checks each cited time lies inside its cited clip; failing claims are removed, and if none remain the agent refuses.
5. If retrieval returns no segment that bears on the question, the agent refuses ("the tapes do not cover this").
6. The call is logged (model, prompt version, tokens, latency, cost) to S3; the trace goes to X-Ray; the daily cap and per-answer token limit apply (values OQ-12).
7. Failure: when the daily cap is reached the endpoint refuses with a limit error rather than calling the model.

8.6 Key lifecycle and automatic disabling

1. Phase A: a visitor submits the key request form, which stores the request in DynamoDB (ADR 0020; fields OQ-14); the owner issues it: a DynamoDB row and an API Gateway key attached to a usage plan with a rate limit and daily quota.
2. Phase B: self-service issue by the same service.
3. Revoke: set status revoked in DynamoDB and disable the key in API Gateway; effective immediately.
4. An alarm on spikes in rejected requests triggers `disable_abusive_key` for a key that keeps hitting its limits.

8.7 Phase B model selection and tuning

1. Run Parakeet 1-minute first-run check (fp16). On failure, compare Whisper with a smaller Whisper model.

2. Transcribe the landing hour with both models on the same T4; compute WER on air-to-ground and back-room; record audio hours per GPU hour and cost per audio hour; log MLflow runs.
3. Write the model decision record (update ADR 0004).
4. For the chosen model, run fp16 and int8 at several batch sizes with GPU samples; find and fix the largest bottleneck; measure again.
5. Only then run 5B.1 on all 30 channels one tape at a time under the 40 GB disk limit.

8.8 Nightly run

Decided by John, Oct 9, 2026 (ADR 0024): the platform nightly run (10:00 UTC, spot only, 2-hour maximum) does publish steps and quality checks only for Project 5. Transcription and other large batches, including backfills, stay in session-time batches. If a nightly run is late or failed, pages show the old data with a visible late notice (ADR 0025). The freshness target that defines "late" is open (OQ-21).

9. Data model

Format: Iceberg tables in the Glue catalog via Spark unless noted; contracts in contracts/ (ODCS YAML). Each table carries event time and a source key (rule 2 (carry event time and a source key on every record)). Table, column and contract names are Design (proposed).

Table	Layer	Phase	Contract	Notes
archive_item_metadata	Source	A	contract	archive.org item metadata
apollo_audio_files	Source	A	contract	OGG, FLAC; track 1 time code
nasa_air_to_ground_transcript	Source	A	contract	Reference transcript with GET
retrieval_eval_questions	Source (authored)	B	contract	200 questions
ask_eval_cases	Source (authored)	B	contract	Answerable, unanswerable, red-team
tape_catalog	Published	A	contract	One row per archive item per pull
channel_map	Published	A	contract	Tape track to console position
audio_manifest	Published	A	contract	URL, size, hash, fetch time
speech_segments	Published	A	contract	Silero output
transcript_words	Published	A	contract	Word timestamps and confidence
transcript_segments	Published	A	contract	Lines with GET
transcription_runs	Published	A	contract	Audio hours per GPU hour, cost
tape_clock_offsets	Published	A	contract	One row per tape
anchor_events	Published	A	contract	Anchor matches per channel
search_chunks	Published	A	contract	Chunks for search
search_index	Published (LanceDB)	A	contract	Vectors in S3
clips	Published	A	contract	Opus clips
api_key_requests	DynamoDB	A	contract	Key lifecycle
speaker_turns	Published	B	contract	pyannote output
transcript_corrections	Published	B	contract	Original kept
events	Published	B	contract	Extracted events with span
quality_flags	Published	B	contract	Great Expectations findings
wer_results	Published	B	contract	WER per channel type

Platform-owned tables used here (linked, not repeated): the pipeline metrics table (freshness and row counts, step 0.5) and llm_calls (step 1B.5): ../Platform/contracts/.

Relationships

```
erDiagram
    tape_catalog ||--o{ audio_manifest : "item_id"
    tape_catalog }o--|| channel_map : "tape_id, track"
    audio_manifest ||--o{ speech_segments : "file_id"
    speech_segments ||--o{ transcript_words : "segment_id"
    speech_segments ||--|| transcript_segments : "speech_segment_id"
    tape_catalog }o--|| tape_clock_offsets : "tape_id"
    anchor_events }o--|| tape_clock_offsets : "tape_id"
    transcript_segments ||--o{ search_chunks : "source_segment_ids"
    search_chunks ||--o{ clips : "chunk_id"
    transcript_segments ||--o{ transcript_corrections : "transcript_segment_id"
    transcript_segments ||--o{ speaker_turns : "overlap"
    transcript_segments ||--o{ events : "span"
    transcript_segments ||--o{ quality_flags : "subject"
```

Data rules from the plan applied here: history by adding one partition per pull (rule 3); business history in rows, not snapshots (rule 8 (keep business history in rows, not in snapshots)); raw archive audio kept as a manifest (rule 1); Iceberg maintenance is the last task of the nightly run (rule 7 (maintain Iceberg tables as the last task of the nightly run)).

10. API

Spec: openapi.yaml (OpenAPI 3). Base: /v1. Two doors (ADR 0022): the keyed developer API (API key header) and the unkeyed website route (per-visitor limits and a daily cap; ask off until OQ-13 decides its setting). The spec marks each operation with x-doors. All endpoints below are Design (proposed) in shape; the endpoint names come from the plan.

Endpoint	Method	Purpose	Phase first available	Notes
/v1/search	GET	Ranked chunks for a text query with filters	B (Phase A subset OQ-24)	Cursor pagination; cached repeats
/v1/segments	GET	Transcript segments by channel and GET range	B (OQ-24)	Cursor pagination; low-confidence flag
/v1/timeline	GET	Who was talking when, per channel and GET range	B (OQ-24)	Lanes
/v1/clips	GET	Clip metadata and URL by id or GET range	B (OQ-24)	CloudFront URLs
/v1/events	GET	Extracted events by type and range	B	Only after 5B.7
/v1/ask	POST	Question in, grounded answer with clips or refusal	B	Developer door: requires key, own daily cap. Website door: setting open (OQ-13), off until decided

Common behavior: 400 validation error; refused for missing, invalid or revoked key; 429 over rate, per-visitor limit or daily quota; capacity rejection when above the concurrency cap (429 with Retry-After, ADR 0028; the load test confirms it); cursor pagination; OpenAPI docs page live (hosting location OQ-13).

Numeric limits are not recorded decisions and appear in the spec as references to OQ-12 rather than numbers.

11. Deployment view

What	Runs where	When	Notes
Airflow DAGs, Spark (catalog, audio prep, clock, indexing, evals, checks)	Session box on k3s (t4g.2xlarge or m7g.2xlarge)	Work sessions; the nightly run does publish and quality checks only (ADR 0024)	Spot first, safe to rerun; 4-hour session limit
GPU transcription	g4dn.xlarge spot, T4	Work sessions; waits for spot	Per-run GPU hour limit (OQ-12); disk (OQ-11)
Developer API (keyed)	Lambda (FastAPI) behind API Gateway REST type	Always on	Reserved concurrency cap shared with the website route (OQ-12)
Website route (unkeyed)	Same Lambda code behind API Gateway (HTTP type for public pages, platform)	Always on	Per-visitor limits and daily cap (values OQ-12)

What	Runs where	When	Notes
Search index	LanceDB files in S3	Always on	Written by the session box, read by Lambda
Site, clips	CloudFront and S3	Always on	Free up to 1 TB and 10 million requests a month
Visitor data (key requests from the Phase A form)	DynamoDB with point-in-time recovery	Always on	Provisioned within free tier
Secrets	Parameter Store (SecureString)	Always on	Hugging Face token, model API key, keyed-hash key (platform pattern)
Experiment tracking	Databricks Free Edition, MLflow	During sessions	Not on the serving path; results leave by manual export to GitHub (ADR 0017)
Traces	AWS X-Ray, call log in S3	Always on	Agents run on Lambda

Environments: dev and live are fully separate (storage, catalogs, permissions); dev cannot write live data; only merges to main reach live, after the changed pipelines pass on sample data in CI; website changes get a preview link (platform, ../Platform/design-doc.md). Raw-file Object Lock (30 days) applies to live raw files only; downloaded Project 5 audio and 16 kHz intermediates live in the temporary 7-day area with no Object Lock (Decided by John, Oct 9, 2026, ADR 0011).

Schedules: platform nightly run at 10:00 UTC with 2-hour maximum (usually about 45 minutes), spot only. Project 5 runs publish and quality checks only (ADR 0024).

Infrastructure as code: Terraform. The platform rule is to use arm64 images and check each image's manifest before first use; it is stated for the session box. The image architecture for the GPU box is not stated in the plan and is settled with OQ-11 (check each image's manifest before first use).

12. Cross-cutting concerns

12.1 Security

- Keys only in Parameter Store; no key in code or GitHub; GitHub Actions through OIDC (platform).
- Least privilege per pipeline role; read-only roles for agents; ask agent has one retrieval tool and no write tool (rule 13).
- Source text (transcripts) is treated as data, never as instructions; red-team set plants instructions in transcript text (ADR 0012).
- Website route: no key; per-visitor limits and a daily cap; ask off until OQ-13 decides its setting (ADR 0022).
- Developer API: keys, per-key rate limit and daily quota, instant revocation, reserved-concurrency cap, request size and format validation, caching, spike alarm that disables a key; WAF only if abuse appears (ADR 0007, 0008).
- Scanned images (platform CI: gitleaks and Trivy).
- Threat model: written while building (security and privacy document).

12.2 Privacy

- Contact details in key requests stay in DynamoDB; the lake copy carries a keyed-hash code (ADR 0020). What a request collects: OQ-14. Deletion periods: security and privacy document.
- No visitor accounts.

12.3 Observability

- Every DAG writes freshness and row counts to the metrics table; status.json for the public status page; failed run or missed freshness target sends an email; real incidents get a record (platform step 0.5).
- Speech share per channel, audio hours per GPU hour, cost per audio hour, WER, recall at 10, event precision and recall are published (rule 14 (measure and publish the numbers)).
- Ask traces go to X-Ray with the call log in S3 (model, prompt version, tokens, latency, cost), loaded into llm_calls by the nightly run.
- GPU utilization and memory sampled with nvidia-smi or DCGM every few seconds in tuning runs.

12.4 Cost limits

See ADR 0015 for the hard-limit table. Plan estimates: Phase B month about $32 to $48; Phase C about $35 to $48.

12.5 Reliability

- Every job safe to rerun; per-file checkpoint on the GPU job; overwrite one partition, never blind-append (rule 3).
- Spot interruption two-minute warning handled; GPU waits for spot.
- Everything the public uses stays up while the session box is off (rule 16).
- Cold start wording on the page; cold-start measurement from platform step 1B.9.
- Late or failed run: the page keeps showing the old data with a visible late notice and the last good run time; nothing is hidden (ADR 0025).

12.6 Licensing and attribution

- Credit NASA; do not imply endorsement; terms confirmed before clips are published; register kept in the security and privacy document.
- Databricks Free Edition non-commercial; check licensing before charging for any project (BP assumptions).

12.7 Time

- Wall-clock times in UTC; GET separate; local conversion on the page only (ADR 0010).

13. Testing strategy

Principle: pipelines and checks are tested with small sample files in CI like ordinary unit tests; no staged production failures; real incidents are recorded (BP). Spark tests run on GitHub's free CI machines before deploy to dev and then live.

Layer	What	Examples
Unit	Pure functions with sample inputs	`parse_get`, `fit_offset_from_anchors`, `compute_wer`, `recall_at_k`, `validate_span`, cursor encode and decode, glossary rules
Sample-file pipeline	Small OGG and transcript files through each step	Audio prep on a short clip; mission clock on a synthetic two-channel sample; rerun leaves row counts unchanged
Data tests	Great Expectations on staging; schema diff against contracts	Segment has file and offset; GET not backward; clip exists; original text kept
Evals	Quantitative, in MLflow and CI	WER per channel type; retrieval (200 questions, recall at 10 threshold gate); event precision and recall; Ask faithfulness (judge calibrated on 50 hand-graded answers), citation-inside-clip exact check, refusal tests, red-team prompts planted in transcripts
API	Contract tests against openapi.yaml; k6 load test; over-quota 429; revoked key refused; beyond-cap rejected	
First-run checks	The seven checks in requirements.md	download speed, time code, OGG vs FLAC, Whisper speed, GPU quota, Parakeet 1-minute, pyannote terms
Hand checks	100 corrections; 30 hand-corrected back-room minutes; one hand-labeled hour for events; 50 hand-graded answers	

Threshold values: OQ-16.

14. Alternatives considered

Topic	Options weighed	Chosen	ADR
Project	PDF search API, Mission Control	Mission Control	0001
Start scope	All channels, clearest channels	Clearest channels	0002
GPU	On-demand, CPU only, spot	Spot g4dn.xlarge	0003
Model	Whisper large-v3, Parakeet	Decided by measurement in 5B.4	0004
Audio format	FLAC, MP3, OGG	OGG, conditional	0005

Topic	Options weighed	Chosen	ADR
Search	pgvector, LanceDB	LanceDB	0006
API type	HTTP, REST	REST (keys and quotas)	0007
Abuse controls	Alerts only, hard caps, WAF	Hard caps; WAF if abuse	0008
Clock	File-name times, time code, anchors	Time code, anchors fallback	0009
Time storage	Local, UTC	UTC plus separate GET	0010
Raw audio	Full copy, manifest	Manifest	0011
Ask	Free-form, grounded	Grounded, cited, refuse	0012
Corrections	Overwrite, keep original	Keep original, grey low confidence	0013
Events	Summary only, span required	Span required	0014
Cost control	Alerts, hard limits	Hard limits	0015
Disk	All at once, one tape at a time	One tape at a time	0016
Tracking	Self-hosted MLflow, Databricks	Databricks MLflow	0017
Clips	Full files, Opus clips	Opus clips	0018
Terms	Publish now, confirm first	Confirm first	0019
Key requests	PostgreSQL, email only, DynamoDB	DynamoDB	0020
Audio prep	Transcribe everything, speech only	Speech only	0021
API doors	One keyed door, one unkeyed door, two doors	Two doors	0022
Phase A publish gate	None, Great Expectations, a short set of Python checks	Python checks, then Great Expectations	0023
Nightly run	Transcription nightly, publish and checks only	Publish and checks only	0024
Late data	Hide, show with late notice	Show with late notice	0025

Minor decisions folded from ADRs

These were separate ADRs and are recorded here in full because each is a setting, a wording rule, a format or a repeat of a platform rule. The old numbers are kept so references still resolve.

Old ADR	Decision	Detail
ADR-0005	Download the OGG version of the audio instead of FLAC	5A.2 downloads the OGG files for the landing hour on the two Phase A channels and converts them to 16 kHz mono with ffmpeg. FLAC is used only in the accuracy comparison check.
ADR-0010	Store every wall-clock timestamp in UTC and keep GET as a separate value	Wall-clock times (tape start and end, fetch time, run time) are stored in UTC. Tape times published without a zone are stored exactly as published (as text) until the zone is confirmed (OQ-1), then converted to UTC. GET is stored in its own numeric columns and is never mixed with UTC columns. The offset per tape (ADR 0009) links the two. The page shows GET as the primary time axis; any local-time display is converted on the page only.
ADR-0011	Keep a manifest of URL, size and hash for the Apollo audio instead of a full copy	5A.1 stores item metadata (tape, channel, start time, duration, format, checksum) in Iceberg. A manifest covers every in-scope audio file with URL, size, hash and fetch time. Audio is downloaded into working storage for processing; the archive remains the source of truth. Derived audio (Opus clips) is published through CloudFront and S3 (ADR 0018). Downloaded audio and the 16 kHz mono intermediates live in the temporary 7-day area (no Object Lock). The GPU box's disk size and how Airflow hands audio to the GPU box stay open until the spike (OQ-11).
ADR-0015	Hard limits on every paid item of this project	/ Paid item / Hard limit (recorded) / /---/---/ / Each Spark job / Maximum rows read and written and maximum runtime; over either limit it stops before publishing and opens an incident / / GPU / A maximum number of GPU hours per run; spot only; if spot is unavailable the job waits / / Servers / 4-hour session limit, idle shutdown, 2-hour nightly maximum / / Public API / Cap on concurrent copies of the server plus per-key quotas / / AI model / $15 a month spend limit set in the API account before the first call; cheapest model only; a daily question cap and a token limit per answer; Phase B month capped at $10, Phase C month capped at $5 / / S3 / Clean-up rules (platform step 1B.9) plus an alarm if storage passes a set size / / DynamoDB / Provisioned within the free tier, so heavy traffic is throttled rather than billed / / Everything on AWS / The $40 budget, as a backup / Plan estimates (not limits): Phase B month about $32 to $48 all in; Phase C month about $35 to $48 all in.

Old ADR	Decision	Detail
ADR-0016	Process all 30 channels one tape at a time so the box's disk stays under 40 GB	Process one tape at a time so the box's disk stays under 40 GB. Run 5B.4 and 5B.5 on the landing hour first, then transcribe with the chosen model and settings. Throughput and cost per audio hour go in the design note.
ADR-0018	Serve speech as Opus clips through CloudFront and S3	5A.5 cuts speech into Opus clips and serves them through CloudFront. Each clip records its source file and offset, so it traces back to the original audio.
ADR-0019	Confirm source terms before publishing clips; credit NASA and do not imply endorsement	No clip is published until the terms are confirmed. The terms note goes in the README (5A.1 acceptance check). Pages and API responses credit NASA and do not imply endorsement. A licenses and attribution register is kept in the security and privacy document (written while building).
ADR-0021	Convert audio to 16 kHz mono with ffmpeg in Spark and transcribe detected speech only	5A.2 converts downloaded OGG files to 16 kHz mono with ffmpeg in Spark. Silero speech detection selects speech; only speech is transcribed (also stated for 5B.1). The speech share per channel is recorded. Every segment traces back to a file and an offset.
ADR-0023	Phase A publish gate is a short set of hand-written Python checks; Great Expectations replaces them in Phase B	Phase A: a short set of hand-written Python checks gate publishing: row counts, nulls, schema (diffed against the contract), and time ranges. A failed check stops publishing. Phase B (5B.10): Great Expectations replaces these checks and adds the transcription-error, GET-backward and missing-clip checks.
ADR-0024	The nightly run does publish and quality checks only; transcription stays in session-time batches	The nightly run for Project 5 does publish steps and quality checks only. Transcription (GPU), audio prep and other large batches run in session-time batches, never in the nightly run.
ADR-0025	When a nightly run is late or failed, show the old data with a visible late notice	The page keeps showing the old data and adds a visible late notice with the time of the last good run (converted to local time on the page only; stored in UTC). Data is never hidden because it is late. For Project 5 this applies to the published index, clips, timeline and tables the page reads. Because the archive itself does not change, the notice concerns the published outputs and checks, not the 1969 content.
ADR-0027	Store GET as whole seconds from liftoff, keep the printed text, and use the pinned ODCS and OpenAPI versions	GET is stored as an integer, the number of whole seconds from liftoff. The printed text is kept beside it in the source table, so any parse can be checked against the original. Ingestion parses "DD HH MM SS" to seconds. A row that does not parse is rejected and listed, as the source contract already requires. Fractional seconds are used only where a source gives them. Tape time keeps its own precision (ADR 0009 and ADR 0010). ODCS v3.0.2 and OpenAPI 3.0.3 are the versions for project 5, as in project 1 (ADR 0032). The "to be aligned" comments are replaced with that reference.
ADR-0028	Status codes for missing or revoked keys and for capacity rejection	A missing, invalid or revoked key returns 403 Forbidden. The body says the key is not accepted, with no detail about why. Traffic beyond the concurrency cap returns 429 Too Many Requests with a Retry-After header. This matches the rate-limit family in ADR 0007 and ADR 0008. The response shape stays the same for both (the error schema in openapi.yaml). The contract tests and the load test check these codes. If API Gateway or Lambda returns a different code, the test fails and the decision is reopened, not quietly changed.

15. Risks

Risk	Handling
GPU quota not approved	Smaller model on CPU (ADR 0003); request made in step 0.1
Spot unavailable	Job waits; no on-demand fallback for GPU
Parakeet does not run on the T4	Compare Whisper with a smaller Whisper model (ADR 0004)
Time zone or time code unreadable	Anchor events (ADR 0009); agreement check across channels

Risk	Handling
OGG less accurate than FLAC	Revisit ADR 0005
Terms prevent publishing clips	Do not publish until confirmed (ADR 0019)
Transcription cost overrun	Per-run GPU hour limit, speech-only transcription, tuning study (ADR 0015, 0021)
Missed speech by speech detection	Speech share recorded; checks flag text without speech but not missed speech; review samples when measuring
Hallucinated or ungrounded answers	Grounded retrieval, citation check, refusal, faithfulness evals (ADR 0012)
Prompt injection in transcript text	Treated as data; red-team set (rule 13)
API abuse or cost spike	Two doors: per-visitor limits and daily cap on the website route, keys and quotas on the developer API, shared concurrency cap, spike alarm, WAF if needed (ADR 0007, 0008, 0022)
Databricks Free Edition limits block automation	Manual export of results and model to GitHub (ADR 0017); spike tests automation (OQ-15)
Cold start of Lambda with LanceDB too slow	Page wording; measure in step 1B.9 (OQ-25)
Disk exhaustion in 5B.1	One tape at a time under 40 GB (ADR 0016)
Wrong GET labels	Cross-channel anchor agreement; GET monotonic check
Source change or removal at archive.org	Manifest hash check; weekly smoke test (platform 1B.9)
Data size estimates not measured	Measure the catalog first; no hours figure used until then (OQ-22)
Phase C done-when checks missing	By decision, written when Phase C is designed

16. Open questions

Answered on Oct 9, 2026 and moved into the sections above: Phase A chunking (ADR 0006), Phase A publish gate (ADR 0023), Phase A key requests (ADR 0020), where downloaded audio lives (ADR 0011), nightly run scope (ADR 0024), website versus keyed API (ADR 0022), manual export from Databricks (ADR 0017), late data (ADR 0025). Phase C done-when checks are recorded as missing until Phase C is designed (Decided by John, Oct 9, 2026, because the checks depend on the Phase C design). The earlier questions on Phase C checks and on the Phase A publish gate are removed. Answered later on Oct 9, 2026: OQ-19 and OQ-20 (ADRs 0027 and 0028), and the NASA parts of OQ-10 and OQ-17 (ADR 0026). The rows below keep their original wording and carry the answer.

Each remaining question has a matching GitHub issue (to be created before the part it blocks is built). Removed from this list once answered; the answer moves to the right section.

Questions from the Documentation Plan's "Questions still open" table for Project 5

ID	Question	What answering needs	Blocks	GitHub issue
OQ-1	Which time zone do the tape times use, and can track 1's time code be decoded?	The audio files analyzed	5A.4	issue: to be created
OQ-2	Is OGG accurate enough compared with FLAC, and how fast is the download?	The audio files	5A.2	issue: to be created
OQ-3	How fast is Whisper on the T4, and does Parakeet run on it?	The GPU server	5A.3, 5B.4	issue: to be created

Additional questions found while writing this design

ID	Question	What answering needs	Blocks	GitHub issue
OQ-4	Is the spot quota of 4 vCPUs for G instances approved?	AWS support response (platform step 0.1)	5A.3	issue: to be created
OQ-5	What is the tolerance for anchor events to line up across the two channels?	Measurements from the first aligned hour	5A.4	issue: to be created
OQ-6	Which exact hour is "the landing hour", which archive items are the air-to-ground and flight director loop channels, and does the channel map differ by tape?	Catalog pull and NASA transcript	5A.1, 5A.2	issue: to be created
OQ-7	Which embedding model on CPU, what vector dimension, and how fast?	A spike on CPU	5A.5	issue: to be created
OQ-8	What pause length ends a "run of continuous speech" for Phase A chunks (Phase A chunks and blank speaker are decided), and which chunk sizes are compared in 5B.6?	A look at the landing hour plus 5B.6	5A.5, 5B.6	issue: to be created
OQ-9	Silero speech detection parameters and the resulting speech share	Run on the landing hour	5A.2	issue: to be created
OQ-10	What do the archive.org, NASA, UT Dallas and Explore Apollo terms allow for publishing clips, and what credit wording is required?	Reading the terms	5A.5 (public release of clips)	issue: to be created. **Partly answered (Oct 9, 2026, ADR 0026):** NASA's terms are verified from NASA's own page, and NASA credit wording is set. The archive.org terms page could not be read (its address returned a title only and then 404), and the Explore Apollo and UT Dallas terms were not found. Those three stay open.
OQ-11	What is the GPU box's disk size, and how does Airflow start the GPU box and hand it audio? (Downloads and 16 kHz intermediates in the temporary 7-day area are decided.)	A spike	5A.2, 5A.3	issue: to be created
OQ-12	Values: maximum GPU hours per run, Lambda reserved-concurrency cap, per-visitor limits and daily cap on the website route, per-key rate limit and daily quota, ask daily cap and token limit per answer, request size limits, Spark job row and runtime limits (set from this project's measured size estimate)	Size estimate and review	5A.3, 5A.5, 5B.8, 5B.9	issue: to be created
OQ-13	What identifies a visitor for the website route's per-visitor limits (Project 1 uses a random ID on the phone, no internet addresses), is the ask feature capped or off on the website door, where is the docs page hosted and where does the MCP server run? (The two-doors design is decided.)	Decision in review	5A.5	issue: to be created
OQ-14	What fields does the key request form collect and how long are they kept? (Form storing requests in DynamoDB from Phase A is decided.)	Decision in review	5A.5	issue: to be created
OQ-15	Can export from Databricks Free Edition be automated, how do runs from the AWS GPU box enter Databricks MLflow given its limits (serverless only, limited outbound internet), and what is the monthly keep-alive for this project? (Manual export is decided.)	A spike with John's Databricks account	5B.3	issue: to be created

ID	Question	What answering needs	Blocks	GitHub issue
OQ-16	Thresholds: recall at 10, faithfulness, citation, refusal and red-team pass rates, event precision and recall targets, WER targets, low-confidence threshold, k6 request rate, judge agreement	First measured results	5B.3, 5B.6, 5B.9, 5B.10	issue: to be created
OQ-17	What is "the mission timeline" used in 5B.6 and 5B.7, and what are the NASA transcript's source URL, format, license and GET text format?	Locating the sources	5A.4, 5B.3, 5B.6, 5B.7	issue: to be created. **Partly answered (Oct 9, 2026, ADR 0026):** the source is NASA's Technical Air-to-Ground Voice Transcription PDF, the format is GET as DD HH MM SS from liftoff, and NASA's licence rule is recorded. Still open: the mission timeline used in 5B.6 and 5B.7.
OQ-18	Opus clip length, padding, bitrate and the source file clips are cut from	Spike	5A.5	issue: to be created
OQ-19	What status code results from capacity rejection and from a missing or revoked key?	Contract tests and load test	5B.8	issue: to be created. **Answered (Oct 9, 2026, ADR 0028, default confirmed by John):** 403 for a missing, invalid or revoked key; 429 with Retry-After for capacity rejection. The contract and load tests confirm it.
OQ-20	GET representation in stored data (seconds versus text) and the ODCS version the platform contract template uses	Platform documents	5A.1, 5A.4	issue: to be created. **Answered (Oct 9, 2026, ADR 0027):** GET is stored as whole seconds from liftoff with the printed text kept beside it. The ODCS version is v3.0.2 and the OpenAPI version is 3.0.3, as pinned in project 1 (ADR 0032).
OQ-21	What freshness target defines "late" for Project 5's published outputs, which triggers the late notice?	Decision in review	5A.5	issue: to be created
OQ-22	What are the real counts, durations and sizes of the catalog? No hours figure is used in these documents until this is measured.	The catalog pull (BP: "measure first")	5A.1	issue: to be created
OQ-23	What mechanism and lifetime caches repeated searches?	Design review	5B.8	issue: to be created
OQ-24	Which endpoints exist in Phase A for keyed access?	Decision in review	5A.5	issue: to be created
OQ-25	Lambda packaging, ASGI adapter, and cold start with LanceDB	Spike	5A.5	issue: to be created
OQ-26	Glossary rule content and the identity of "the cheapest model" for batch and live calls	Spike and pricing check	5B.2	issue: to be created
OQ-30	What condition counts as abuse appearing, which triggers adding the AWS WAF firewall rule?	A first period of real traffic	5B.8	issue: to be created
OQ-28	pyannote model terms accepted and Hugging Face token stored in Parameter Store	John's Hugging Face account	5B.2	issue: to be created

17. Glossary

Term	Meaning
Apollo 11	The mission whose Mission Control audio this project archives
MOCR	Part of the archive collection's name ("Apollo 11 MOCR ACR Collection"); the plan does not expand it
ACR	Part of the collection's name on archive.org; its expansion is not given in the plan
Channel	One console position's audio loop on a tape; also one archive item
Tape	A 30-channel recording (about 170 exist)

Term	Meaning
Track 1	The tape track that carries the time code
GET	Ground Elapsed Time, the mission clock
Time code	A signal recorded with the audio from which the tape's clock can be read
Anchor event	A known event with a GET in NASA's transcript (the 1202 alarm, the landing call) used to align a tape
Air-to-ground	The channel between the crew and ground (CAPCOM and the crew)
CAPCOM	A speaker on the air-to-ground channel, alongside the crew (named in the plan's 5B.2)
Flight director loop	One of the two Phase A channels (the plan's name for it)
1202 alarm	A program alarm during the landing, used as an anchor event
WER	Word error rate: the share of words that differ from a reference transcript
Silero	The speech detection model used to find speech
faster-whisper	The Whisper speech-to-text implementation used for transcription
Parakeet	NVIDIA's open speech model (parakeet-tdt-0.6b-v2), compared in 5B.4
NeMo	NVIDIA's toolkit that runs Parakeet
pyannote	Speaker separation (diarization) model
fp16, int8, bf16	Number formats; the T4 cannot use bf16
T4	The NVIDIA GPU in g4dn.xlarge
Spot	Cheaper AWS capacity that can be taken back with a two-minute warning
LanceDB	The file-based text and vector search store kept in S3
Embedding	A vector representing the meaning of text, used for vector search
Recall at 10	Share of questions whose expected result appears in the top 10
MRR	Mean reciprocal rank of the first correct result
Opus	The audio format of the clips
Iceberg, Glue	Lake table format and its catalog
Usage plan	API Gateway object that sets rate limit and quota for keys
Reserved concurrency	A cap on how many copies of a Lambda function run at once
MCP	Model Context Protocol, a standard for exposing tools to AI agents
ODCS	Open Data Contract Standard
ADR	Architecture decision record
C4	A model for drawing architecture at context, container and component levels
Write, audit, publish	Data lands in staging, passes checks, then goes live
Session box	The AWS server (spot, run with k3s) that is started for a work session or the nightly run and destroyed afterward; Airflow and Spark run on it. "With the session box off" means only always-on parts (S3, CloudFront, Lambda, DynamoDB) are serving.
GPU box	The separate g4dn.xlarge spot server that runs transcription
Nightly run	The platform's scheduled run at 10:00 UTC on the session box, hard maximum 2 hours; for Project 5 it does publish and quality checks only
Work session	A period when John starts the session box to build or run large batches
k3s	Lightweight Kubernetes used on the session box
Airflow, DAG	Airflow is the scheduler; a DAG is one scheduled workflow of tasks
Spark	The batch processing engine used for audio prep and table jobs
CI	Continuous integration: the automatic tests GitHub runs on every change

Term	Meaning
Lambda, API Gateway, CloudFront, S3, DynamoDB	AWS services: serverless functions, the API front door, the content delivery network, object storage and the always-on key-value database
Parameter Store	AWS store for keys and secrets; the only place keys live
X-Ray	AWS tracing service used for agent traces
MLflow, Databricks Free Edition	Experiment tracking, and the free managed service that hosts it
Great Expectations	A data checking library used from Phase B (Phase A uses a short set of Python checks)
k6	A load testing tool
nvidia-smi, DCGM	NVIDIA tools that sample GPU utilization and memory
ffmpeg	A tool that converts audio formats
Tape time	Seconds since the start of the tape (before conversion to GET)
Landing hour, landing window	The hour of the lunar landing that Phase A covers on two channels, and the wider window that Phase B covers on all 30 channels. Exact limits are open (OQ-6).
Speech segment	A span of speech found by Silero in one file (table speech_segments)
Transcript segment	The transcribed text of one speech segment (table transcript_segments)
Chunk	A unit of transcript text that is embedded and searched. In Phase A a chunk is a run of continuous speech with the speaker blank; after 5B.2 it is a speaker turn.
Diarization	Separating a recording by speaker (done by pyannote)
Website route	The unkeyed door of the API used by the public page, with per-visitor limits and a daily cap
Developer API	The keyed door of the API (REST type, per-key limits), under /v1. The plan calls it the public API.
Request form	A web form where a visitor asks for an API key; requests are stored in DynamoDB
Late notice	A visible message on a page, with the time of the last good run, shown when a nightly run is late or failed; old data stays visible
Manifest	A table of URL, size and hash for each audio file, kept instead of a full copy
Temporary 7-day area	The platform clean-up rule under which temporary files are deleted after 7 days; downloaded audio and 16 kHz files live here (no Object Lock)
Dev and live	The two fully separate environments
Terraform	The tool that defines infrastructure as code
Hugging Face	The site that hosts the pyannote model, whose terms must be accepted
Event span	The exact transcript text that supports an extracted event
Parquet	A columnar file format used for published tables
DuckDB	A small SQL engine that runs inside Lambda or on the box to query published files
dbt	A SQL transformation tool; not used in Project 5
Kafka	An event streaming system; not used in Project 5
Snowflake	A cloud data warehouse; not used in Project 5
OIDC	OpenID Connect, the login method that lets GitHub Actions reach AWS without stored keys (platform)
FastAPI	The Python web framework used for the API
FLAC	A lossless audio format; about 5 times larger than OGG here
OGG	A compressed audio format; the version downloaded for processing (ADR 0005)
Precision	The number format a model computes in (fp16, int8); in event extraction the word also means the share of extracted events that are correct
Faithfulness	Whether every claim in an answer is supported by the cited segments

Term	Meaning
Keyed hash	A one-way scrambled code computed with a secret key (kept in Parameter Store), used instead of an email address or contact detail outside DynamoDB
Partition	One slice of a table (for example one pull or one tape) that a job overwrites as a unit
Red-team	Deliberately hostile test prompts, here including instructions planted in transcript text
LLM judge	A language model that scores answers, calibrated against hand-graded answers
Object Lock	An S3 setting that blocks deletion or overwrite of live raw files for 30 days
OpenAPI	The standard format for describing the HTTP API (openapi.yaml)
Run of continuous speech	A stretch of speech with no break; the Phase A chunk. The pause length that ends a run is open (OQ-8).
Speaker turn	One speaker talking until another speaker takes over; known only after speaker separation (5B.2). Chunks follow speaker turns from then on.

Part 4: Operations

Mission Control/operations.md

Project 5: Mission Control. Operations

Project 5, Mission Control | Document: operations | Set: rev 4, Oct 9, 2026

Status: Draft for review by John. Written while building: the parts that need running data are marked open. Owner: John. Last updated: Oct 9, 2026. Sources: design-doc.md sections 11 (deployment), 12 (cross-cutting concerns) and 15 (risks); ADR 0015 (hard limits), 0016 (one tape at a time), 0024 (nightly run), 0025 (late data); platform documents in ../Platform/: ADR 0006 (session box), 0007 (Airflow database), 0008 (always-on storage), 0010 (storage clean-up and Object Lock).

1. What runs, and when

Part	Runs on	When	Stopped by
Airflow DAGs and Spark jobs (catalog, audio prep, clock, indexing, evals, checks)	Session box (k3s; t4g.2xlarge or m7g.2xlarge, spot)	Work sessions. The nightly run does publish and quality checks only (ADR 0024).	4-hour session limit; idle shutdown; 2-hour nightly maximum (platform ADR 0006)
GPU transcription	g4dn.xlarge spot (NVIDIA T4)	Work sessions; waits for spot capacity	Per-run GPU-hour limit (OQ-12)
Developer API (keyed) and website route (unkeyed)	Lambda (FastAPI) behind API Gateway	Always on	Concurrency cap shared by both doors (OQ-12)
Search index	LanceDB files in S3	Always on for reads; written by the session box	Clean-up rules (platform ADR 0010)
Site and clips	CloudFront and S3	Always on	Free tier limits
Key requests	DynamoDB, point-in-time recovery on	Always on	Free tier; throttled rather than billed
Secrets	Parameter Store (SecureString)	Always on	Access is logged
Experiment tracking	Databricks Free Edition, MLflow	During sessions only; not on the serving path	Free-tier limits; results copied to GitHub by hand (ADR 0017)
Traces and call log	AWS X-Ray; call log in S3	Always on	X-Ray sampling; log retention in security-and-privacy.md

Environments: dev and live are fully separate. Dev cannot write live data. Only merges to main reach live, after the changed pipelines pass on sample data in CI. Website changes get a preview link (platform ADR 0001).

2. Freshness and the late rule

- **Target per output:** open. The freshness target that defines "late" for each published output is not decided (OQ-21). It is needed before the late notice can be switched on.
- **Until then:** when a nightly run is late or fails, the page keeps showing the last good data with a visible late notice and the time of the last good run. Nothing is hidden (ADR 0025).
- **Nightly schedule:** 10:00 UTC, two-hour maximum, usually about 45 minutes (platform ADR 0006). For Project 5 the nightly run publishes and checks only (ADR 0024).

3. Monitoring and alerts

Signal	Where it goes	Who acts
Rows and freshness for every task run	Metrics table (an Iceberg table in the lake, the platform default pending OQ-P01)	Status page; email on a missed target or failure
Public status file	status.json on the site	Read by the status page

Signal	Where it goes	Who acts
Real incidents	One incident record each (platform step 0.5)	John, with a runbook entry afterwards
GPU use and memory (tuning runs)	nvidia-smi or DCGM sampled every few seconds	John, during tuning
API spike	Spike alarm; abusive keys disabled automatically (ADR 0008)	John; WAF added only if abuse appears (ADR 0022)
Storage size	Alarm when storage passes a set size (ADR 0015)	John; size set after measuring (OQ-22)
Session limit and idle shutdown	Platform alarms	Automatic
Model API spend	Spend limit in the API account (ADR 0015)	Automatic at the limit

Ask traces go to AWS X-Ray. The call log (model, prompt version, tokens, latency, cost) is written to S3 and loaded into the `llm_calls` table by the nightly run.

4. Limits

The hard limits are recorded in ADR 0015. Numbers that are not yet set are open in OQ-12.

Paid item or resource	Limit	Status
Each Spark job	Maximum rows read and written; maximum runtime. Over either limit, the job stops before publishing and opens an incident.	Rule recorded; numbers open (OQ-12, OQ-22)
GPU	Maximum GPU hours per run; spot only; waits when spot is unavailable	Rule recorded; hours open (OQ-12)
Servers	4-hour session limit; idle shutdown; 2-hour nightly maximum	Recorded (platform ADR 0006)
Public API	Cap on concurrent copies of the server, shared with the website route; per-key quota	Cap open (OQ-12)
Website route	Per-visitor limits and a daily cap	Values open (OQ-12)
Ask agent	Daily question cap; token limit per answer; cheapest model only	Values open (OQ-12); the spend limit is $15 a month in ADR 0015, a conflict with $10 in project 1 (see the decisions list in the changelog)
S3	Clean-up rules; storage alarm	Rules in platform ADR 0010; alarm size open
DynamoDB	Provisioned within the free tier	Recorded
Everything on AWS	$40 AWS budget, as a backup	Recorded

Plan estimates, not limits: Phase B month about $32 to $48 all in; Phase C month about $35 to $48 all in (ADR 0015). Measured cost per run is recorded in the first runs and compared with these estimates.

5. Runbooks

A runbook is written after a real failure, not before. Each real failure gets an incident record (platform step 0.5), and the runbook entry is written from that record. The table below lists the failures the design expects, with the first check and the first response. These responses are design (proposed) and are confirmed by the first real failure.

Failure	First check	First response	Source
GPU quota not approved	Quota status in AWS support	Use a smaller model on CPU (ADR 0003)	Risk table
Spot capacity unavailable	Spot request status	Job waits; no on-demand fallback for the GPU	Risk table
Parakeet does not run on the T4	Parakeet one-minute first-run check (fp16)	Compare Whisper with a smaller Whisper model (ADR 0004)	Design doc 16

Failure	First check	First response	Source
Time code unreadable	Anchor event agreement across channels	Use anchor events (ADR 0009)	Risk table
Source changed or removed at archive.org	Manifest hash check; weekly smoke test (platform 1B.9)	Stop the affected tape; keep the old manifest entry	Risk table
Disk exhaustion during transcription	Disk use on the box	One tape at a time under 40 GB (ADR 0016)	Risk table
Wrong GET labels	Cross-channel anchor agreement; monotonic GET check	Mark the tape for review; do not publish its labels	Risk table
Lambda cold start too slow	Measured cold start	Keep the page wording; measure in 1B.9 (OQ-25)	Risk table
API abuse or cost spike	Spike alarm; key usage	Disable the key (automatic); tighten per-visitor limits; WAF only if abuse persists (ADR 0008)	ADR 0008
Model API spend near the limit	Spend and call log	Daily question cap refuses questions; no new calls past the limit	ADR 0015
Databricks export fails	Export log	Repeat the manual export by hand (ADR 0017)	Risk table
Nightly run late or failed	Metrics table; run log	Late notice on the page; email; incident record	ADR 0025

6. Recovery and backups

- Pipelines are safe to rerun. Each job checkpoints per file (GPU) and overwrites one partition, never a blind append (design rule 3).
- Live raw files are protected by S3 Object Lock for 30 days (platform ADR 0010). Downloaded audio and 16 kHz intermediates are in the temporary 7-day area with no Object Lock.
- Published data and raw data are kept. Other storage follows the clean-up rules in platform ADR 0010.
- The Airflow database is saved to S3 after each run and restored at startup (platform ADR 0007).

7. Postmortems

After each real incident that affects the public site or the published numbers, a blameless postmortem is written in the incident record: what happened, the timeline, the cause, the fix, and what changes so it does not happen again. Postmortems describe systems, not people.

8. Open items

Item	Needed by
Freshness target per output and the late rule (OQ-21)	Before the late notice is switched on
Numeric limits (OQ-12)	Before the first paid run
Measured cost per run and per month	After the first full run
Storage alarm size (OQ-22)	After the catalog is measured
Approval of the metrics table default (platform OQ-P01)	Before the status page is built

Part 5: Model card

Mission Control/model-card.md

Project 5: Mission Control. Model card

Project 5, Mission Control | Document: model card | Set: rev 4, Oct 9, 2026

Status: Draft for review by John. Owner: John. Last updated: Oct 9, 2026. Sources: design-doc.md sections 5, 7 (components C5 to C9), 12 and 15; ADR 0004 (transcription model), 0012 (answers only from retrieved segments), 0013 (low-confidence lines), 0014 (event spans), 0021 (speech-only transcription); security-and-privacy.md for licences.

This card covers the machine-learning models and the AI features of Mission Control. The project is a portfolio demo, so no model is used for a decision about a person.

1. Models and components

Component	Model	Used for	Licence	Status
Speech detection	Silero VAD	Finds speech in each audio file (C2, 5A.2)	MIT (verified: Silero Team, 2020-present)	Chosen; parameters open (OQ-9)
Transcription	faster-whisper with Whisper large-v3	Turns speech into text on the spot GPU (C3, 5A.3)	Code MIT (verified: openai/whisper, copyright OpenAI 2022). Model weights and faster-whisper licence not yet checked	Chosen (ADR 0004)
Transcription (comparison)	NVIDIA Parakeet, parakeet-tdt-0.6b-v2	Compared with Whisper on the same audio (Phase B)	CC-BY-4.0 (verified on the model card). Attribution required	Compared in 5B.4; the T4 test is open (OQ-3)
Speaker separation	pyannote speaker-diarization-3.1	Who is speaking, on mixed channels (C9, 5B.2)	MIT (verified). Gated: the user must accept the conditions and use an access token	Phase B; acceptance of terms and token open (OQ-28)
Text embeddings	To be chosen	Vector search over chunks (C5, 5A.5)	Open	Open (OQ-7)
Ask agent and LLM judge	The cheapest model of the provider's current range	Answers from retrieved segments with citations; scores answers (C13, 5B.9)	Provider terms; not yet checked	Identity open (OQ-26); spend limit open (see section 6)

Transcripts are the input to every text model. They are treated as data and never as instructions (ADR 0012).

2. Intended use

- Search and listen to archive audio of Apollo 11 Mission Control: find a spoken line, play the exact clip, see who was speaking on the timeline.
- Ask questions about the mission. The Ask agent answers only from segments it retrieves, cites each one, and refuses when the transcript does not support an answer (ADR 0012).
- Read published numbers: word error rate, retrieval scores, event accuracy and cost, all recorded in the design doc and published with the demo.

3. Not intended for

- Official records, legal, medical or safety decisions, or history research where a single misheard word matters. The transcript is a machine transcript; the audio is the record.
- Claims about what a crew member felt or intended beyond what is said on the recording.
- Identifying a person from a voice. Speaker labels are channel and role labels from the transcript, not voice identities.

- Commercial use. The inputs carry licences that limit reuse, and Databricks Free Edition is non-commercial (see security-and-privacy.md).

4. Evaluations

Eval	What it measures	Threshold	Status
Retrieval	Share of 200 questions whose expected segment is in the top 10 (recall at 10) (C11, 5B.6)	Open (OQ-16)	200 questions are in the authored contract; threshold set from the first measured result
Transcription	Word error rate per channel type, against the NASA transcript for the air-to-ground channel (C10, 5B.3)	Open (OQ-16)	Reference source recorded in ADR 0026
Events	Precision and recall of extracted events (each needs a valid span, ADR 0014) (C12, 5B.7)	Open (OQ-16)	Planned in Phase B
Ask faithfulness	Share of answer claims supported by the cited segments; LLM judge calibrated on 50 hand-graded answers (C13, 5B.9)	Open (OQ-16)	Judge agreement with the hand grades is measured before the judge is trusted
Citation check	Each citation is an exact match to a segment inside its clip	Pass rate open (OQ-16)	Exact check, not a judgement
Refusal	Unanswerable questions are refused	Open (OQ-16)	Eval cases in the authored contract
Red team	Hostile prompts planted in transcript text do not change behaviour (C13, 5B.9)	Open (OQ-16)	Planned in Phase B

Baseline without AI (default, chosen because John had no preference): the same search without the Ask agent, returning ranked segments only. The Ask agent is compared against this baseline on the retrieval set so that its value is shown, not assumed.

5. Known failure cases

Failure	Where it shows	What limits it
Speech missed by the speech detector	Lines that are never transcribed	Speech share is recorded per channel; spot checks on samples (risk table)
Wrong words on noisy or garbled audio	Transcript lines; search misses	Original text kept beside the corrected text; low-confidence lines are shown in grey (ADR 0013)
Wrong GET time label	Timeline and clip positions	Cross-channel anchor agreement; monotonic GET check; anchor events (ADR 0009)
Speakers mixed up on overlapping speech	Speaker labels on the timeline	Labels are shown as channel and role; diarization is measured in 5B.2
Ungrounded answer	Ask agent output	Answers only from retrieved segments; citation check; refusal; faithfulness eval (ADR 0012)
Prompt injection in transcript text	Ask agent output	Transcript text is data; red-team set (rule 13)
Parakeet does not run on the T4	Model comparison	Compared with a smaller Whisper model (ADR 0004)
Events without a valid span	Event list	Not published; a valid span is required (ADR 0014)

6. Safety, privacy and cost

- **Questions sent to the model provider.** A visitor's question is sent to the model provider to get an answer. No visitor identity is sent with it (ADR 0022 and the per-visitor identity question, OQ-13). Whether question text is kept after the answer is in security-and-privacy.md, section 3.
- **Spend.** The model API has a monthly spend limit set in the account before the first call, a daily question cap and a token limit per answer (ADR 0015). The monthly figure is open: ADR 0015 records $15, and project 1 records $10 (see the decisions list in the changelog).

- **Cheapest model.** Only the cheapest model is used for now. A stronger model is compared in Phase C only when the evals justify it (build plan decisions, AI).
- **Tracing.** Every Ask call is traced in AWS X-Ray and written to the call log in S3 with the model, prompt version, tokens, latency and cost.

7. Open items

Item	ID
Embedding model, vector dimension and speed	OQ-7
Model identity for the Ask agent and the judge	OQ-26
Thresholds for each eval	OQ-16
Whether Whisper weights and faster-whisper licences are verified	Before the first transcription run
Terms accepted for the pyannote model, and the access token in Parameter Store	OQ-28
Monthly spend limit for the model API	Decision for John (see the changelog)

Part 6: Security and privacy

Mission Control/security-and-privacy.md

Project 5: Mission Control. Security and privacy

Project 5, Mission Control | Document: security and privacy | Set: rev 4, Oct 9, 2026

Status: Draft for review by John. Owner: John. Last updated: Oct 9, 2026. Sources: design-doc.md sections 12.1, 12.2 and 12.6; ADR 0008 (abuse controls), 0012 (grounded answers), 0019 (terms and credit), 0020 (key requests); platform documents in ../Platform/ for keys, roles, retention and CI scans.

Scope: the Mission Control site, its public API and its keyed developer API, the pipelines that build them, and the models they use. The platform's own security rules are in ../Platform/design-doc.md and apply here too.

1. Controls in place

Control	How	Source
Keys	Every key in AWS Parameter Store (encrypted). None in code or GitHub.	Platform ADR 0004
CI access to AWS	GitHub Actions uses OIDC; no stored AWS keys	Platform ADR 0004
Least privilege	One role per pipeline. The Ask agent has one retrieval tool and no write tool.	Design doc 12.1; rule 13
Dev and live	Separate storage, catalogs and roles. Dev cannot write live data.	Platform ADR 0005
Scans	gitleaks and Trivy in CI on every push	Platform ADR 0001 and step 0.1
Public API abuse	Keys with per-key limits and daily quota; instant revocation; automatic disabling of a key on a spike; shared concurrency cap; request validation; caching	ADR 0007, ADR 0008
Website route	No key; per-visitor limits and a daily cap. Ask is off on this route until OQ-13 decides.	ADR 0022
Raw files	S3 Object Lock for 30 days on live raw files	Platform ADR 0010
Terraform state	Own encrypted bucket; only the Terraform role reads it	Platform ADR 0002

2. Threat model

Asset	Threat	Mitigation	Residual risk and status
Transcript text	Instructions planted in transcript text change how the Ask agent behaves (prompt injection)	Transcript text is data, never instructions; the agent has one retrieval tool and no write tool; a red-team set plants instructions in transcript text	Eval pass rate open (OQ-16)
Model API budget	Heavy use of the Ask agent runs up model cost	Spend limit in the API account; daily question cap; token limit per answer; cheapest model only (ADR 0015)	Monthly figure open: $15 in ADR 0015 and $10 in project 1 (see the changelog)
Public API	Floods, scraping or fake keys	Keys with quotas; instant revocation; spike alarm disables a key; shared concurrency cap; WAF only if abuse appears (ADR 0008, ADR 0022)	Numeric limits open (OQ-12); abuse trigger open (OQ-30 in the design doc)
API keys	A key leaks or is shared	Keys issued on request; revocation; keys stored only as hashes (default, section 4); no key in logs	Status codes for a missing or revoked key are set by default (OQ-19); the contract tests confirm them

Asset	Threat	Mitigation	Residual risk and status
Key request data	A requester's contact details are exposed	Contact details stay in DynamoDB; the lake copy carries only a keyed-hash code (ADR 0020)	Fields collected and retention open (OQ-14)
Visitor identity	Per-visitor limits need an identifier	No internet addresses are used for limits in project 5 until OQ-13 decides; identity is open	Open (OQ-13)
Source audio	The archive removes or changes an item	Manifest with size and hash; a change is detected and the affected tape stops (manifest check in the risk table)	Low; monitored by the weekly smoke test (platform 1B.9)
Published clips	Clips break a rights or terms condition	No clip is published until the terms are confirmed (ADR 0019)	Open for archive.org, Explore Apollo and UT Dallas (OQ-10)
Supply chain	A malicious or vulnerable package or container image	Trivy and gitleaks in CI; pinned images; the Marquez image is one pinned build (platform ADR 0013)	Low
Cost spike from pipelines	A job runs away	Hard limits on rows, runtime and GPU hours (ADR 0015); session limit; idle shutdown	Numeric limits open (OQ-12)
Logs	Logs carry personal data or secrets	Logs hold no keys or contact details; retention below	Confirm when logging is built

3. Personal data inventory

Data	Where it is	Why it is kept	Who can read it	Retention	Status
Key request fields (contact details and the fields the form collects)	DynamoDB, point-in-time recovery on	To issue and manage a key	John, through the key service	Open (OQ-14)	Open
Keyed-hash code for a requester	Lake (as a published or internal table)	To count unique requesters without storing contact details	Pipelines and John	Kept with the lake tables (platform clean-up rules)	Decided (ADR 0020)
Keyed-hash key	Parameter Store	To compute the code	Pipelines through their roles	Kept while the key is in use	Decided
Visitor identifier for per-visitor limits	Not decided	To apply per-visitor limits	Not decided	Not decided	Open (OQ-13)
Visitor questions (Ask)	Sent to the model provider to get an answer	To answer	The model provider, under its terms	Not stored by the project (default, see section 4)	Default
Call log (model, prompt version, tokens, latency, cost)	S3, loaded into `llm_calls`	Cost and quality checks	John	90 days (default, see section 4)	Default
Pipeline logs	S3	Debugging	John	90 days (platform ADR 0010)	Decided
Security logs	S3	Security review	John	1 year (platform ADR 0010)	Decided
Dev data	Dev storage	Testing	Dev roles	14 days (platform ADR 0010)	Decided
Temporary files and failed uploads	Temporary 7-day area	Processing	Pipelines	7 days (platform ADR 0010)	Decided

The mission audio, transcripts and clips are historical public archive material. The pages do not collect visitor data beyond what the key request collects and what OQ-13 decides for the website route.

4. Defaults chosen in this document

These are minor decisions, made where John had no preference. Each can be changed.

- **Visitor questions are not stored by the project.** The Ask agent sends the question to the model provider to get an answer, and the call log does not hold the question text. This keeps the question out of storage. It can be revisited when OQ-13 is decided.
- **The call log is kept for 90 days.** This matches the pipeline logs (platform ADR 0010).
- **API keys are stored only as hashes.** The service keeps a hash of each key and never the key itself, so a leak of the store does not reveal a working key. This is the standard practice for issued keys.

5. Licences and attribution register

Each entry says what was checked and where. "Verified" means the publisher's own page was read on Oct 9, 2026. "Unverified" means it was not read, and the item stays blocked where it matters.

Item	Publisher	Licence or terms	Status	Attribution	Used for
Apollo 11 MOCR audio (archive.org item "Apollo 11 MOCR ACR Collection")	Uploaded by NASA; digitized by UT Dallas and NASA	Uploaded with no licence field. archive.org terms page could not be read (its address returned a title only, then 404).	Unverified. Publishing clips is blocked until confirmed (ADR 0019).	Credit NASA (see below); credit line for UT Dallas and Explore Apollo open	Source audio, clips
Explore Apollo and collectSPACE (digitization)	UT Dallas and NASA	Terms not found in search	Unverified	Open (OQ-10)	Source audio
NASA air-to-ground transcript (ALSJ PDF)	NASA	NASA Images and Media page (updated Aug 13, 2026): NASA content "generally are not subject to copyright in the United States"; NASA should be acknowledged as the source; no endorsement	Verified for NASA content. The transcript page itself carries no separate notice.	"Source: NASA" (ADR 0026)	Reference transcript, WER, anchors
NASA material on pages and in API responses	NASA	As above	Verified	Credit NASA as the source; do not imply endorsement (ADR 0019)	Pages and API responses
Whisper (code)	OpenAI	MIT (openai/whisper LICENSE, copyright OpenAI 2022)	Verified for the code	MIT notice kept with any copy of the code	Transcription
Whisper large-v3 model weights	OpenAI	Not checked on the model page	Unverified	To check before first use	Transcription
faster-whisper	Its authors	Not checked	Unverified	To check before first use	Transcription
Parakeet tdt 0.6b v2 model	NVIDIA	CC-BY-4.0 (model card; "ready for commercial/non-commercial use")	Verified	Credit NVIDIA and the model; link the licence; note changes (CC BY 4.0 terms). Exact line set when used.	Model comparison (5B.4)
NeMo toolkit	NVIDIA	Not checked	Unverified	To check before first use	Parakeet runs
Silero VAD	Silero Team	MIT (copyright Silero Team, 2020-present)	Verified	MIT notice kept	Speech detection
pyannote speaker-diarization-3.1	pyannote	MIT. Gated: the user accepts the conditions and uses an access token.	Verified for the licence; the conditions text was not read	Open (OQ-28)	Speaker separation

Item	Publisher	Licence or terms	Status	Attribution	Used for
Databricks Free Edition	Databricks	Non-commercial (build plan assumption)	Taken from the plan; not re-checked	None	Experiment tracking
Open-source libraries (LanceDB, DuckDB, FastAPI, Spark, Airflow and others)	Their authors	Not checked in this document	Unverified; check each when first used	Per licence	Platform
Model API (Ask and judge)	The provider	Provider terms; the model identity is open (OQ-26)	Unverified	Per provider terms	Ask agent
Opus clips and manifest	This project, from the source audio	Same terms as the source audio	Blocked by OQ-10	Credit as for the source	Clips

6. Takedown and rights handling

If an archive item is removed, changed, or challenged:

1. Stop publishing the affected tape or clip the same day. Keep the raw file under its Object Lock.
2. Record the event in the changelog and in an incident record (platform step 0.5).
3. Check the terms again for the item before it is published again.

This process is a default. It is a minor decision, made because John had no preference.

6a. Open items

Item	ID
Terms and credit for archive.org, Explore Apollo and UT Dallas	OQ-10
Fields collected by the key request form and how long they are kept	OQ-14
Visitor identity for per-visitor limits; whether Ask is on for the website route	OQ-13
Numeric limits for the public API and the Ask agent	OQ-12
Monthly spend limit for the model API ($15 or $10)	Decision for John
pyannote terms accepted and access token stored	OQ-28
Whisper weights, faster-whisper and NeMo licences checked	Before first use
Abuse trigger for the WAF	OQ-30

Part 7: Architecture decision records

`Mission Control/adrs/0001-project-scope-mission-control.md`

ADR 0001: Project 5 is Mission Control, a searchable and playable Apollo 11 audio archive

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

The earlier Project 5 idea, a PDF search API, was judged boring. John wanted a project that adds unstructured audio at scale with AI and gives a memorable demo. Source material is Apollo 11 Mission Control audio from roughly 170 thirty-channel tapes, digitized by UT Dallas and NASA and published on archive.org and the Explore Apollo site.

Decision

Project 5 is "Mission Control": every spoken line is transcribed and tagged with its channel (console position) and Ground Elapsed Time (GET). A timeline shows who was talking at any moment, search finds and plays the exact clips, "Ask Mission Control" answers questions with clips, and a public API exposes the same data. It is built in three phases (A live, B depth, C scale) and every phase ends deployed.

Alternatives considered

- The PDF search API: rejected by John as boring (Build Plan Decisions, "Project 5").
- Removing the project: rejected; John's rule was to keep every project.

Consequences

- The project needs GPU transcription, which is its largest cost (see ADR 0003 and ADR 0015).
- The project uses unstructured data, search and AI agents, so it needs a model card and eval suites (see the design doc, sections 12 and 13).
- Source terms must be confirmed before clips are published (ADR 0019).

Sources: Build Plan Decisions, "Projects and scope"; Build Plans, "Project 5: Mission Control".

Amendment (Decided by John, Oct 9, 2026)

No archive hours figure is used in these documents until the catalog has been measured (OQ-22). The roughly 170 thirty-channel tapes figure is from the plan.

`Mission Control/adrs/0002-clearest-channels-first.md`

ADR 0002: Start with the clearest channels; noise reduction and low-confidence marking come later

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

The archive holds 30 channels per tape. Some channels are clear (air-to-ground and the flight director loop); others are noisier back-room channels. Transcription quality differs by channel.

Decision

- Phase A covers the landing hour on the two clearest channels only: air-to-ground and the flight director loop.
- Noisier back-room channels come in Phase B (5B.1, all 30 channels for the landing window).
- Noise reduction is tested in Phase C and kept only where it lowers word error rate.
- Published accuracy per channel and grey "unverified" low-confidence lines arrive in Phase B (5B.3, 5B.10).

Alternatives considered

- Start with all 30 channels: rejected; the choice records "start where transcription works best".
- Apply noise reduction from the start: deferred to Phase C and judged by measured word error rate.

Consequences

- A first version limited to two clear channels can go public early.
- Phase B must handle mixed-speaker and noisy audio (pyannote, glossary rules, quality checks).
- The channel map (tape track to console position) must identify the two Phase A channels (open question OQ-6 in the design doc).

Sources: Build Plan Decisions, "Projects 3, 5, 2 and 4", row "Audio quality"; Build Plans, Project 5 Phase A, Phase B and Phase C.

`Mission Control/adrs/0003-gpu-transcription-on-spot-g4dn.md`

ADR 0003: Transcribe on a g4dn.xlarge spot GPU box with faster-whisper and NeMo

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026). The specific model is decided separately in ADR 0004.

Context

Transcription is the project's largest cost. The platform's GPU choice is stated as: g4dn.xlarge on spot (NVIDIA T4), with faster-whisper and NVIDIA NeMo, chosen as cheapest per audio hour transcribed. Spot price is about $0.27 to $0.30 an hour. New AWS accounts often start with a GPU quota of zero.

Decision

- GPU work runs only on g4dn.xlarge spot instances (one NVIDIA T4). If spot is unavailable the job waits.
- Phase A uses faster-whisper large-v3, with a checkpoint per file. If the GPU quota is not yet approved, a smaller model on CPU is used.
- A spot quota of 4 vCPUs for G instances is requested in platform step 0.1.
- NVIDIA NeMo (Parakeet) is used in Phase B for the model comparison (ADR 0004).
- Every job is safe to rerun because spot instances can be taken back with a two-minute warning.
- NVIDIA's GPU profiling (nvidia-smi or DCGM) is used in Phase B (5B.5).

Alternatives considered

- On-demand GPU: not chosen; the plan states GPU work is spot only.
- CPU-only transcription: kept only as the fallback when the GPU quota is not granted.
- RAPIDS: dropped by John because it did not fit.

Consequences

- The GPU job needs per-file checkpoints and idempotent writes.

- Cost figures: Phase B month 8 to 12 GPU hours ($2.20 to $3.60); Phase C month 15 to 25 GPU hours ($4 to $7.50). These are plan estimates, not limits.
- A maximum number of GPU hours per run is required by the hard-limits rule; its value is not yet set (OQ-12).
- Whisper speed on the T4 is unmeasured (OQ-3).

Sources: Build Plans, "One tool per job" (GPU work), "How the box is run", "Hard limits on everything that costs money"; Build Plan Decisions, "Tools and platform" (NVIDIA work).

`Mission Control/adrs/0004-transcription-model-whisper-vs-parakeet.md`

ADR 0004: Choose the transcription model by measured accuracy and cost (Whisper large-v3 versus Parakeet)

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Proposed. The decision is made in step 5B.4 and this record is then updated to Accepted with the result. Until then faster-whisper large-v3 is used (ADR 0003).

Context

John chose "Whisper vs NVIDIA Parakeet on the same audio; cost tuning on the GPU. Choose by measured accuracy and cost." Parakeet (parakeet-tdt-0.6b-v2 through NeMo, English, word timestamps) appears in NVIDIA's test hardware list for the T4, but the T4's architecture is not on the supported list, and the T4 cannot use bf16. It is not known whether Parakeet runs on the T4 (OQ-3).

Decision

In 5B.4, transcribe the same landing hour with Parakeet and with faster-whisper large-v3 on the same T4 and compare: - word error rate on air-to-ground and on back-room audio, - audio hours per GPU hour, - cost per audio hour. The model used in 5B.1 is chosen from the comparison table in the README, and the result is written into this record. Run Parakeet in fp16.

Fallback: if Parakeet fails on the T4 (1-minute first-run check), 5B.4 compares Whisper large-v3 with a smaller Whisper model instead.

Alternatives considered

- Always use Whisper large-v3: not chosen in advance; John wants the choice by measurement.
- Always use Parakeet: not chosen in advance, and it may not run on the T4.

Consequences

- 5B.1 must wait for 5B.4 and 5B.5, which run on the landing hour first.
- The comparison table and tuning results are tracked in MLflow (ADR 0017).
- Fine-tuning a small Whisper model in Phase C compares word error rate with base and large models.

Sources: Build Plan Decisions, row "Model choice"; Build Plans, 5B.4, 5B.5, Phase B first-run checks, "Risks and open items" (Transcription model).

`Mission Control/adrs/0006-lancedb-search-from-lambda.md`

ADR 0006: Text and vector search with LanceDB files in S3, queried from Lambda

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

The search must keep working when the session box is off. The platform's "one tool per job" table assigns text and vector search to LanceDB files in S3, searchable from Lambda with the session box off. pgvector was removed as a duplicate. Embeddings for Project 5 are computed on CPU.

Decision

- Chunk transcripts by speaker turn, embed on CPU and write a LanceDB index to S3 (5A.5).
- Query the index from Lambda behind API Gateway with the session box off.
- 5B.6 compares chunk sizes and hybrid versus vector-only search, so the search mode is a measured choice, not fixed in advance.

Alternatives considered

- pgvector: removed by John as a duplicate engine.
- PostgreSQL on the box: off most of the time, so it cannot serve search when the session box is off.

Consequences

- The index is a published artifact that must be rebuilt when transcripts, chunks or corrections change.
- Lambda package size and cold start for LanceDB are unmeasured (OQ-25).
- The embedding model and vector dimension are not yet chosen (OQ-7).
- The page must carry the cold-start wording from the platform serving template.

Sources: Build Plans, "One tool per job" (Text and vector search), 5A.5, 5B.6; Build Plan Decisions, "Tools and platform" (Duplicate engines).

Amendment (Decided by John, Oct 9, 2026)

In Phase A a chunk is a run of continuous speech with the speaker left blank (null). The index is rebuilt by speaker turn after 5B.2 (pyannote). The pause length that ends a run, and the chunk sizes compared in 5B.6, stay open (OQ-8). Phase A therefore does not use real speaker turns, although the plan's 5A.5 says "speaker turn".

`Mission Control/adrs/0007-public-api-rest-type-keys-and-quotas.md`

ADR 0007: Public API on API Gateway's REST type with keys and quotas; keys on request in Phase A, self-service from Phase B

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

The API must be secure and cannot risk being abused. The platform decision is: FastAPI on Lambda behind API Gateway, using the HTTP type for public pages and the REST type wherever API keys and per-key limits are needed, because only the REST type supports API keys and usage plans.

Decision

- The public API is FastAPI on Lambda behind API Gateway's REST type, under /v1, with endpoints: search, segments, timeline, clips, events and ask (5B.8).
- Phase A: any API access uses keys issued on request, each with a rate limit and a daily quota (5A.5).
- From Phase B: keys are self-service, with per-key rate limits and daily quotas, instant revocation, a reserved-concurrency cap on the Lambda function, request size and format validation, caching of repeated searches, cursor pagination, OpenAPI docs, contract tests, a k6 load test and an MCP server.

- The ask endpoint requires a key and has its own daily cap, under the model spend limit.
- AWS WAF (about $6 a month or more) is added only if abuse appears.

Alternatives considered

- HTTP API type for the public API: rejected because it does not support API keys or usage plans.
- Open access without keys: rejected; the API must not risk abuse.
- WAF from day one: deferred until abuse appears, to save at least about $6 a month.

Consequences

- The numeric limits (rate, daily quota, concurrency cap, request size, ask daily cap, token limit per answer) are not recorded decisions and are listed under open question OQ-12.
- The public web page uses the unkeyed website route (ADR 0022); what identifies a visitor there is open (OQ-13).
- Phase A keys are requested through a simple form stored in DynamoDB (ADR 0020); what the form collects is open (OQ-14).
- Abuse handling is in ADR 0008.

Sources: Build Plan Decisions, "Projects 3, 5, 2 and 4" (Public API); Build Plans, "One tool per job" (Apps and APIs), 5A.5, 5B.8, "Hard limits on everything that costs money".

Amendment (Decided by John, Oct 9, 2026)

The keyed REST API is the developer door only. The public page uses a second, unkeyed website route with per-visitor limits and a daily cap. See ADR 0022. Keys are issued on request in Phase A and opened fully in Phase B; numbers stay open (OQ-12).

`Mission Control/adrs/0008-api-abuse-controls-and-concurrency-cap.md`

ADR 0008: Cap Lambda concurrency and disable abusive keys automatically

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026; numeric values still to be set)

Context

"Nothing in the project should surprise with cost." The public API is open to self-service keys from Phase B, so traffic is the main cost and abuse risk. The plan's hard-limits table lists for the public API: a cap on how many copies of the server can run at once, plus per-key quotas.

Decision

- A reserved-concurrency cap on the API Lambda function limits the number of copies that can run at once; traffic beyond the cap is rejected rather than billed.
- Each key has a rate limit and daily quota; an over-quota key gets a 429.
- Revocation is instant; a revoked key is refused.
- An alarm on spikes in rejected requests disables a key that keeps hitting its limits.
- The ask endpoint has its own daily cap and a token limit per answer, within the monthly model spend limit ($15 overall; the model API is capped at $10 in a Phase B month and $5 in a Phase C month).
- Request size and format validation, and caching of repeated searches, reduce load.
- WAF only if abuse appears.

Alternatives considered

- No concurrency cap, rely on budget alerts: rejected; AWS budget alerts lag by hours.

- WAF from the start: deferred (ADR 0007).

Consequences

- The 429 and revoked-key behaviors are acceptance checks of 5B.8, verified by contract tests and the k6 load test.
- Status codes for capacity rejection and revoked keys are set by ADR 0028 and confirmed by those tests.
- Values are open (OQ-12).

Sources: Build Plan Decisions, "Cost, limits and operations" (AI spending, Surprise costs); Build Plans, 5B.8, "Hard limits on everything that costs money", "Monthly cost" table.

Amendment (Decided by John, Oct 9, 2026)

The website route (ADR 0022) is also protected: per-visitor limits and a daily cap, sharing the reserved-concurrency cap with the keyed API. Its numbers are open (OQ-12, OQ-13).

`Mission Control/adrs/0009-mission-clock-timecode-or-anchor-events.md`

ADR 0009: Map tape time to GET by decoding the track 1 time code, with anchor events as the fallback

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026). Which method applies is settled by the first-run check (OQ-1).

Context

Search results and the timeline must show Ground Elapsed Time (GET, the mission clock). The tape database lists start and end clock times without a time zone. Landing-day tapes cover both readings of the time zone, so the time zone needs the audio's time code or anchor events. NASA's public air-to-ground transcript carries GET timestamps and is the reference for timing and accuracy.

Decision

5A.4 maps tape time to GET by: 1. decoding the time code on track 1 of the tape; or, if it cannot be decoded, 2. anchoring known events from NASA's air-to-ground transcript (the 1202 alarm and the landing call). One offset per tape is stored. The anchor events must line up across the two channels within a stated tolerance (the tolerance value is not yet set, OQ-5).

Alternatives considered

- Trust the file-name times: rejected; the time zone is unknown.
- Align every segment to the NASA transcript by text: not chosen; the plan stores an offset per tape.

Consequences

- The function-level design is in design doc section 7 (component C4).
- A wrong time zone shifts all GET values of a tape; anchor agreement across channels is the check.
- Clock method and fit quality are stored with each offset so a reader can see how it was derived.

Sources: Build Plans, 5A.4, Phase A first-run checks, "Risks and open items" (Apollo tape times); Documentation Plan, "Questions still open".

Mission Control/adrs/0012-ask-answers-only-from-retrieved-segments.md

ADR 0012: Ask Mission Control answers only from retrieved segments, cites every claim, and refuses when uncovered

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

"Ask Mission Control" is an agent. Platform rules require structured output on every call, a call log, tracing to AWS X-Ray, the cheapest model only for now, source text treated as data and never as instructions, and a hard limit on AI spending. Transcript text may contain planted instructions.

Decision

- The agent answers only from retrieved segments.
- Every claim cites channel, speaker and GET and links a playable clip.
- The agent refuses when the tapes do not cover a question.
- Transcript text is untrusted data; red-team prompts planted in transcript text are part of the eval suite.
- Evals: faithfulness judged by an LLM calibrated on 50 hand-graded answers; exact checks that cited times fall inside the cited clips; refusal tests; red-team prompts. CI fails below thresholds (values open, OQ-16).
- The cheapest model only; a stronger model is compared only in Phase C where the plan allows it.

Alternatives considered

- Free-form chat answers from model knowledge: rejected; every claim must be grounded in a clip.
- Long-context stuffing without retrieval: not chosen; the plan builds retrieval with measured recall.

Consequences

- Citation checks are mechanical (cited time inside the cited clip), so they run in CI without a judge.
- The agent needs a retrieval tool, a refusal path, and a call log in S3 loaded into the llm_calls table by the nightly run (platform; see ../Platform/).
- Daily question cap and token limit per answer are required (values open, OQ-12).

Sources: Build Plans, 5B.9, "Engineering standards" rules 13 and 14, "Model API"; Build Plan Decisions, "AI" table.

Amendment (Decided by John, Oct 9, 2026)

On the unkeyed website route the ask feature has an open setting (OQ-13) and is off until it is decided (ADR 0022; which one is open in OQ-13). The keyed developer API keeps its own daily cap.

Mission Control/adrs/0013-keep-original-text-and-show-low-confidence.md

ADR 0013: Never overwrite the original transcript; show low-confidence lines in grey as unverified

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

Transcripts of noisy 1969 audio contain errors. Phase B adds jargon fixes by glossary rules and a batch model call for low-confidence segments only. John chose "grey low-confidence lines" and published accuracy as the honest treatment of uncertainty.

Decision

- Jargon fixes use glossary rules first. A batch model call then handles low-confidence segments only, in structured output listing each change.
- The original text is always kept; a corrected segment keeps its original.
- A hand check of 100 corrections reports how many were right.
- Low-confidence lines are shown in grey and marked unverified on the page.
- Word error rate per channel type is published (5B.3), and per channel in the README (5B.10).

Alternatives considered

- Overwrite text with the corrected version: rejected; the original must always remain.
- Hide low-confidence lines: rejected; the choice is to show them marked.
- Send every segment to the model: rejected; cost; only low-confidence segments are sent.

Consequences

- The data model stores original and corrected text side by side (transcript_corrections).
- The confidence threshold that makes a segment "low-confidence" is not yet set (OQ-16).
- The batch model call is part of the $1 to $3 of batch model calls in a Phase B month.

Sources: Build Plans, 5B.2, 5B.3, 5B.10, "Monthly cost" table; Build Plan Decisions, "Projects 3, 5, 2 and 4" (Audio quality).

`Mission Control/adrs/0014-event-extraction-requires-valid-span.md`

ADR 0014: Extracted events must point to an exact transcript span or be rejected

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026)

Context

5B.7 uses a batch model call to extract events (alarms, go/no-go polls, system problems, decisions, handovers) from transcripts. Model output can be wrong or ungrounded. Platform rules require structured output and treating source text as data.

Decision

- Each event has type, GET, channel, speakers, confidence and the exact transcript span.
- Events without a valid span are rejected before publishing.
- The timeline shows events as markers.
- Precision and recall per event type are published against the mission timeline and one hand-labeled hour.

Alternatives considered

- Accept events with a text summary only: rejected; a span ties every event to evidence.
- Rule-based extraction only: not chosen; the plan uses a batch model call (structured extraction from unstructured data).

Consequences

- "Valid span" needs a mechanical definition (the span's text exists verbatim in the cited transcript segments); the design uses that definition, to be confirmed in review.
- The mission timeline source is not named in the plan (OQ-17).
- Cost is part of the $1 to $3 of batch model calls in a Phase B month (plan estimate).

Sources: Build Plans, 5B.7.

`Mission Control/adrs/0017-mlflow-on-databricks-for-experiment-tracking.md`

ADR 0017: Track accuracy, speed and cost experiments as MLflow runs on Databricks

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (decided by John, Oct 2026). Export is manual (see amendment); whether it can be automated, and how GPU-box runs are logged, is open (OQ-15).

Context

The platform assigns ML training and experiment tracking to Databricks Free Edition with MLflow (free managed MLflow and registry); self-hosted MLflow was removed as a duplicate. Databricks Free Edition is serverless only, has a daily fair-use quota, limited outbound internet and is non-commercial. Project 1 pushes data in through the Files API and copies results to the GitHub repo after each run, so the site never depends on Databricks.

Decision

Project 5 tracks as MLflow runs on Databricks: every model and setting for word error rate (5B.3), the Parakeet versus Whisper comparison (5B.4), the tuning runs at fp16 and int8 and several batch sizes (5B.5), and retrieval evaluation scores (5B.6). Published results are also copied into the repo so nothing public depends on Databricks being up.

Alternatives considered

- Self-hosted MLflow: removed by John as a duplicate.
- Tracking in the README only: not chosen; the plan requires MLflow runs.

Consequences

- CI that fails when recall at 10 drops below a threshold (5B.6) must read a stored copy, not call Databricks.
- Non-commercial limits apply (a licensing assumption noted in the plan).
- The method for logging runs from an AWS GPU job into Databricks Free Edition is unconfirmed; this may require a design change (OQ-15).

Sources: Build Plans, "One tool per job" (ML training), "Where each job runs", 5B.3 to 5B.6; Build Plan Decisions, "Tools and platform" (ML training).

Amendment (Decided by John, Oct 9, 2026)

Results and models leave Databricks Free Edition by manual export: download them and commit them to GitHub. A spike tests whether automation is possible (OQ-15). How runs from the AWS GPU box enter Databricks MLflow is part of that spike.

Mission Control/adrs/0020-api-key-requests-in-dynamodb.md

ADR 0020: Store API key requests and key status in DynamoDB; personal data stays there

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted as a consequence of platform decisions. Phase A request form Decided by John, Oct 9, 2026. What the form collects is not decided (OQ-14).

Context

The platform's "one tool per job" table assigns visitor data, explicitly including API key requests, to DynamoDB: always on and writable 24/7 while the session box is off, free tier covers the traffic. Platform decisions also state that email addresses stay in DynamoDB and appear everywhere else only as a scrambled code (keyed hash, key in Parameter Store), that DynamoDB has point-in-time recovery on, and that keys are stored only in Parameter Store (API Gateway holds the issued API keys themselves). Project 5 is the first project to issue keys to the public.

Decision

- Key requests (Phase A) and key lifecycle records (request, issued, rejected, revoked, disabled) are stored in a DynamoDB table (contract: api_key_requests).
- Any email address or contact detail is replaced with a keyed-hash code in the nightly copy into the lake; the raw value never leaves DynamoDB.
- The secret key value is never stored in this table.

Alternatives considered

- PostgreSQL on the box: off most of the time; rejected by the platform.
- A form that emails the owner with no table: not chosen; revocation and automatic disabling need a record.

Consequences

- DynamoDB is used from Phase A (see amendment).
- A deletion period for this personal data is required by the security and privacy document (written while building).

Sources: Build Plans, "One tool per job" (Visitor data), "Risks and open items" (Personal data), 5A.5, 5B.8; Build Plan Decisions, "Data storage, security and privacy".

Amendment (Decided by John, Oct 9, 2026)

From Phase A, key requests come through a simple request form that stores them in DynamoDB. This corrects the platform technology table, which lists DynamoDB for Project 5 from Phase B. The fields the form collects, and their retention, stay open (OQ-14).

Mission Control/adrs/0022-two-doors-for-the-api.md

ADR 0022: Two doors for the API: an unkeyed website route and a keyed developer API

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted (Decided by John, Oct 9, 2026). Numbers are still open (OQ-12, OQ-13).

Context

Phase A requires the public page to search with the session box off (5A.5), while the public API issues keys on request (5A.5) and needs API Gateway's REST type for keys and usage plans (ADR 0007). A page cannot hold a secret key. The platform uses API Gateway's HTTP type for public pages and the REST type wherever API keys and per-key limits are needed.

Decision

- Door 1, the website route: unkeyed, used by the public page. It has per-visitor limits and a daily cap. The AI "ask" feature is either setting open (OQ-13), off until decided on this door (which of the two is open, OQ-13).
- Door 2, the developer API: keyed, API Gateway REST type, per-key limits, under /v1. Keys are issued on request in Phase A (simple request form, ADR 0020) and opened fully (self-service) in Phase B.
- Both doors share the same Lambda code, the same LanceDB index and the same reserved-concurrency cap (ADR 0008).
- All numbers (per-visitor limits, daily cap, per-key limits, concurrency cap) stay open (OQ-12).

Alternatives considered

- One keyed door for everything: rejected; the page cannot hold a key.
- One unkeyed door for everything: rejected; the developer API needs per-key limits and instant revocation.

Consequences

- ADR 0007 is amended to point here.
- What identifies a visitor for per-visitor limits on the website door is not decided (Project 1 uses a random ID on the phone with no internet addresses; whether Project 5 does the same is OQ-13).
- The OpenAPI spec lists which operations are on which door (x-doors).
- The shared concurrency cap means heavy website traffic can crowd out keyed traffic and the reverse; this is accepted until measured by the load test.

Sources: Decided by John, Oct 9, 2026, for the reason given in the Context section; Build Plans, "One tool per job" (Apps and APIs), 5A.5, 5B.8.

`Mission Control/adrs/0026-nasa-air-to-ground-transcript-source-and-credit.md`

ADR 0026: Use NASA's Technical Air-to-Ground Voice Transcription as the reference transcript

Project 5, Mission Control | Document: architecture decision record (ADR) | Set: rev 4, Oct 9, 2026

Status: Accepted. Confirmed by John, Oct 9, 2026. Previously recorded as: Default (chosen from primary sources on Oct 9, 2026; John can change it). Closes the source, format and NASA-licence parts of OQ-17. The mission timeline part of OQ-17 stays open, and the archive.org, Explore Apollo and UT Dallas terms stay open under OQ-10.

Context

The design needs a reference transcript for the air-to-ground channel (anchor events in 5A.4, word error rate in 5B.3). OQ-17 asked for its source URL, format, licence and GET text format. Checked on Oct 9, 2026:

- NASA's Apollo Lunar Surface Journal hosts the "Technical Air-to-Ground Voice Transcription (GOSS NET 1)" as a PDF covering tapes 1 to 20: https://www.nasa.gov/wp-content/uploads/static/history/alsj/a11/a11transcript_tec.pdf
- A USGS report that uses this transcript says the lines are listed in chronological order, with the time given as days, hours, minutes and seconds from liftoff (Ground Elapsed Time). The leading "00 00 00" is therefore day, hour, minute, second.

- NASA's Images and Media page (last updated Aug 13, 2026, https://www.nasa.gov/nasa-brand-center/images-and-media/) says NASA content "generally are not subject to copyright in the United States", that NASA "should be acknowledged as the source of the material", and that NASA does not endorse commercial products or activities.
- The NASA transcript page that links the PDF (https://www.nasa.gov/history/alsj/a11/a11transcript_tec.html, last updated Dec 19, 2025) has no copyright notice of its own. It links to the NASA Brand and Usage Guidelines.

Decision

- Source: the NASA PDF above. The contract src_nasa_air_to_ground_transcript records it.
- Format: PDF. Each line starts with GET as "DD HH MM SS" (days, hours, minutes, seconds from liftoff), then the speaker code, then the text. The ingestion step parses GET to whole seconds from liftoff (ADR 0027) and keeps the printed text.
- Licence and credit: NASA's stated rule applies. Pages and API responses credit NASA as the source and do not imply endorsement. The credit line is set to that wording, pending the other three sources (OQ-10).
- The legend (three dots for garbled passages, dashes for pauses or interruptions) is reported in a search summary. It is to be checked against the PDF when the first file is parsed.

Alternatives considered

- The USGS geology transcript (Bailey and Ulrich, 1974): rejected as the reference. It removes housekeeping and engineering talk, so it is not complete.
- The Command Module onboard recorder transcript: rejected for this purpose. It records the crew's communication on the spacecraft recorder, not the air-to-ground channel.
- A copy of the PDF in the project repository: deferred. The source is a public NASA file, so the project keeps the URL, the download date and a hash (manifest approach, ADR 0011).

Consequences

- Accuracy caveat: a 2011 independent correction effort (Heiko Kueffen, reported in search results) questioned the GET timing of some lines during the moonwalk. Anchor alignment therefore has to be checked against the audio, not taken from the transcript alone (OQ-5 and ADR 0009).
- The mission timeline used in 5B.6 and 5B.7 is still not named, so OQ-17 stays open for that part.
- The archive.org audio, Explore Apollo and UT Dallas terms are not yet verified. ADR 0019 keeps the rule that no clip is published until they are.

Sources: NASA Apollo Lunar Surface Journal, "Technical Air-to-Ground Voice Transcription" (https://www.nasa.gov/wp-content/uploads/static/history/alsj/a11/a11transcript_tec.pdf); NASA Images and Media (https://www.nasa.gov/nasa-brand-center/images-and-media/); USGS, "Apollo 11 voice transcript pertaining to the geology of the landing site" (https://pubs.usgs.gov/publication/70043037); Build Plans, "Project 5: Mission Control", 5A.1 and 5B.3.

Part 8: Data contracts

Mission Control/contracts/anchor_events.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): anchor_events. Written by step 5A.4 (component C4).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-anchor-events
name: anchor_events
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Where known events from NASA's air-to-ground transcript (the 1202 alarm, the landing call) were found on
each channel's tape. Used to derive and check offsets.
  limitations: Anchors are matched by text; a poor match score must not be accepted silently.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
  - {property: source_contract, value: src_nasa_air_to_ground_transcript.yaml}
schema:
  - name: anchor_events
    physicalType: table
    properties:
      - {name: anchor_id, logicalType: string, required: true, primaryKey: true, primaryKeyPosition: 1, description:
"Anchor id from the anchor definitions"}
      - {name: channel_id, logicalType: string, required: true, primaryKey: true, primaryKeyPosition: 2, description:
"Channel where it was matched"}
      - {name: label, logicalType: string, required: true, description: "For example the 1202 alarm or the landing
call"}
      - {name: nasa_get_s, logicalType: number, required: true, description: "GET from NASA's transcript, seconds"}
      - {name: tape_id, logicalType: string, required: true, description: "Tape"}
      - {name: file_id, logicalType: string, required: true, description: "File"}
      - {name: tape_time_s, logicalType: number, required: true, description: "Seconds since the tape start where it
was found"}
      - {name: matched_segment_id, logicalType: string, required: true, description: "Transcript segment matched"}
      - {name: match_score, logicalType: number, required: true, description: "Text match score"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: library, rule: unique, description: "(anchor_id, channel_id) unique"}
```

Mission Control/contracts/api_key_requests.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE, DynamoDB): api_key_requests. Written by component C8 (ADR 0020).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-api-key-requests
name: api_key_requests
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Key request and key lifecycle records for the public API (request, issued, rejected, revoked, disabled).
  limitations: Requests come through a simple request form from Phase A (Decided by John, Oct 9, 2026). What the form
collects is not decided (OQ-14). The secret key value is never stored here. Contact details stay in DynamoDB; the
nightly lake copy carries only a keyed-hash code. Deletion period to be set in the security and privacy document.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "DynamoDB, provisioned within the free tier, point-in-time recovery on"}
  - {property: freshness_target, value: "Not applicable: operational store."}
  - {property: personal_data, value: "requester_contact is personal data"}
schema:
  - name: api_key_requests
```

```
    physicalType: table
    properties:
      - {name: request_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Partition
key"}
      - {name: requested_at, logicalType: timestamp, required: true, description: "UTC"}
      - {name: requester_contact, logicalType: string, required: false, description: "Contact detail if collected (OQ-
14); personal data", classification: personal}
      - {name: requester_contact_hash, logicalType: string, required: false, description: "Keyed hash used outside
DynamoDB (key in Parameter Store)"}
      - {name: status, logicalType: string, required: true, description: "requested, issued, rejected, revoked or
disabled"}
      - {name: api_gateway_key_id, logicalType: string, required: false, description: "API Gateway key id (not the
secret)"}
      - {name: usage_plan, logicalType: string, required: false, description: "Usage plan holding the rate limit and
daily quota"}
      - {name: issued_at, logicalType: timestamp, required: false, description: "UTC"}
      - {name: revoked_at, logicalType: timestamp, required: false, description: "UTC"}
      - {name: revoke_reason, logicalType: string, required: false, description: "manual or repeated_limit_hits"}
quality:
  - {type: text, description: "No secret key value in any attribute"}
  - {type: text, description: "Lake copy contains requester_contact_hash only, never requester_contact"}
```

Mission Control/contracts/audio_manifest.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): audio_manifest. Built by step 5A.1 (component C1).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-audio-manifest
name: audio_manifest
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "Manifest of URL, size and hash for in-scope audio files instead of a full copy (platform rule 1 (keep raw
files immutable with source URL, fetch time and hash), ADR 0011). Acceptance: the manifest covers the in-scope files
with hashes."
  limitations: Hash algorithm is chosen when the manifest is first built.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Not applicable: static archive."}
  - {property: source_contract, value: src_apollo_audio_files.yaml}
schema:
  - name: audio_manifest
    physicalType: table
    properties:
      - {name: file_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
for the file (source key)"}
      - {name: item_id, logicalType: string, required: true, description: "archive.org identifier"}
      - {name: format, logicalType: string, required: true, description: "OGG, FLAC or MP3"}
      - {name: url, logicalType: string, required: true, description: "Source URL"}
      - {name: size_bytes, logicalType: integer, required: true, description: "Size in bytes"}
      - {name: hash_algorithm, logicalType: string, required: true, description: "Algorithm of hash"}
      - {name: hash, logicalType: string, required: true, description: "Content hash"}
      - {name: fetched_at, logicalType: timestamp, required: true, description: "Fetch time, UTC"}
      - {name: in_scope, logicalType: boolean, required: true, description: "True when the file is processed in the
current phase"}
      - {name: scope_phase, logicalType: string, required: true, description: "A, B or C"}
quality:
  - {type: library, rule: notNull, description: "url, size_bytes, hash not null for every in-scope row"}
```

Mission Control/contracts/channel_map.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): channel_map. Built by step 5A.1 (component C1).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-channel-map
name: channel_map
```

```
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Maps tape track to console position (for example air-to-ground, flight director loop).
  limitations: Whether the mapping differs by tape is not known (OQ-6), so tape_id is part of the key. The authored
source of the mapping is not yet identified.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Not applicable: reference data changed by pull request."}
schema:
  - name: channel_map
    physicalType: table
    properties:
      - {name: tape_id, logicalType: string, required: true, primaryKey: true, primaryKeyPosition: 1, description:
"Tape"}
      - {name: track_number, logicalType: integer, required: true, primaryKey: true, primaryKeyPosition: 2,
description: "Track on the tape"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel key used in other tables"}
      - {name: console_position, logicalType: string, required: true, description: "Console position name"}
      - {name: channel_type, logicalType: string, required: false, description: "Grouping used when publishing WER per
channel type (for example air-to-ground versus back-room)"}
      - {name: in_phase_a_scope, logicalType: boolean, required: true, description: "True for the two Phase A
channels"}
      - {name: map_source, logicalType: string, required: true, description: "Where the mapping came from"}
quality:
  - {type: library, rule: unique, description: "(tape_id, track_number) unique"}
```

Mission Control/contracts/clips.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): clips. Built by step 5A.5 (component C5).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-clips
name: clips
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Metadata for Opus clips served through CloudFront (ADR 0018). Used by the page, the API clips endpoint and
the Ask citation check (cited time inside cited clip).
  limitations: Clip length, padding and bitrate are open (OQ-18). Publication depends on the terms confirmation (ADR
0019).
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3 for metadata; Opus files in S3 behind CloudFront"}
  - {property: freshness_target, value: "Built with search_chunks."}
schema:
  - name: clips
    physicalType: table
    properties:
      - {name: clip_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: chunk_id, logicalType: string, required: true, description: "Chunk the clip plays"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: get_start_s, logicalType: number, required: true, description: "GET at the first sample of the clip"}
      - {name: get_end_s, logicalType: number, required: true, description: "GET at the last sample of the clip"}
      - {name: source_file_id, logicalType: string, required: true, description: "File the clip was cut from"}
      - {name: source_start_offset_s, logicalType: number, required: true, description: "Offset in the source file"}
      - {name: duration_s, logicalType: number, required: true, description: "Clip duration"}
      - {name: format, logicalType: string, required: true, description: "opus"}
      - {name: path, logicalType: string, required: true, description: "Path under the CloudFront distribution"}
      - {name: size_bytes, logicalType: integer, required: true, description: "File size"}
```

```
      - {name: created_at, logicalType: timestamp, required: true, description: "UTC"}
quality:
  - {type: text, description: "The file at path exists for every row (missing clips are a Great Expectations check,
5B.10)"}
```

Mission Control/contracts/events.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): events. Built by step 5B.7 (component C12). ADR 0014.
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-events
name: events
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Events extracted by a batch model call (alarms, go/no-go polls, system problems, decisions, handovers), each
tied to an exact transcript span. Events without a valid span are rejected before publishing.
  limitations: Precision and recall per event type are published (5B.7). Extraction thresholds are open (OQ-16).
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
schema:
  - name: events
    physicalType: table
    properties:
      - {name: event_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: event_type, logicalType: string, required: true, description: "alarm, go_no_go_poll, system_problem,
decision or handover"}
      - {name: get_s, logicalType: number, required: true, description: "GET seconds"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: speakers, logicalType: array, required: false, description: "Speaker labels involved"}
      - {name: confidence, logicalType: number, required: true, description: "Model confidence"}
      - {name: span_segment_ids, logicalType: array, required: true, description: "Transcript segments holding the
span"}
      - {name: span_char_start, logicalType: integer, required: true, description: "Character offset of the span start
in the first segment"}
      - {name: span_char_end, logicalType: integer, required: true, description: "Character offset of the span end in
the last segment"}
      - {name: span_text, logicalType: string, required: true, description: "Exact text of the span"}
      - {name: summary, logicalType: string, required: false, description: "Short label returned by the model, if any"}
      - {name: model_name, logicalType: string, required: true, description: "Model"}
      - {name: prompt_version, logicalType: string, required: true, description: "Prompt version in git"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "span_text appears verbatim in the cited segments, else the row is rejected (ADR 0014)"}
  - {type: text, description: "event_type is one of the five listed types"}
```

Mission Control/contracts/quality_flags.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): quality_flags. Built by step 5B.10 (component C14).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-quality-flags
name: quality_flags
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Findings from Great Expectations checks (Phase B; Phase A uses a short set of Python checks, ADR 0023) that
flag likely transcription errors (repeated phrases, text where no speech was detected), GET running backward within a
channel, and missing clips.
  limitations: Does not detect speech that was never detected.
team:
  - username: John
```

```
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
schema:
  - name: quality_flags
    physicalType: table
    properties:
      - {name: flag_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: flag_type, logicalType: string, required: true, description: "repeated_phrase, text_without_speech,
get_backward or missing_clip"}
      - {name: subject_table, logicalType: string, required: true, description: "Table the flag refers to"}
      - {name: subject_id, logicalType: string, required: true, description: "Row id in that table"}
      - {name: channel_id, logicalType: string, required: false, description: "Channel"}
      - {name: detail, logicalType: string, required: false, description: "Evidence for the flag"}
      - {name: detected_at, logicalType: timestamp, required: true, description: "UTC"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "flag_type is one of the four listed types"}
```

Mission Control/contracts/search_chunks.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): search_chunks. Built by step 5A.5 (component C5).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-search-chunks
name: search_chunks
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Transcript chunks (runs of continuous speech in Phase A, speaker turns after 5B.2; ADR 0006) that are
embedded and searched. The vectors are stored in the LanceDB index, see search_index.
  limitations: "Decided by John, Oct 9, 2026: in Phase A a chunk is a run of continuous speech with speaker null;
rebuilt by speaker turn after 5B.2. Pause length that ends a run and chunk sizes (compared in 5B.6) are open (OQ-8)."
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Rebuilt whenever transcripts, chunk policy or corrections change."}
schema:
  - name: search_chunks
    physicalType: table
    properties:
      - {name: chunk_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: tape_id, logicalType: string, required: true, description: "Tape"}
      - {name: speaker, logicalType: string, required: false, description: "Speaker label; null in Phase A (decided)"}
      - {name: text, logicalType: string, required: true, description: "Text searched (corrected text when a correction
exists; original always kept elsewhere)"}
      - {name: get_start_s, logicalType: number, required: true, description: "GET seconds"}
      - {name: get_end_s, logicalType: number, required: true, description: "GET seconds"}
      - {name: source_segment_ids, logicalType: array, required: true, description: "Transcript segments in the chunk"}
      - {name: chunk_policy, logicalType: string, required: true, description: "Policy name and parameters (compared in
5B.6)"}
      - {name: low_confidence, logicalType: boolean, required: false, description: "Phase B grey and unverified"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "get_end_s not before get_start_s; get_start_s not null (published chunks only for tapes
with an accepted offset)"}
```

Mission Control/contracts/search_index.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): search_index, the LanceDB table stored as files in S3 and queried from Lambda.
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-search-index
```

```
name: search_index
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Text and vector search over search_chunks, with the session box off (ADR 0006).
  limitations: Embedding model and vector dimension are not chosen (OQ-7). Hybrid versus vector-only is chosen from
5B.6 results.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "LanceDB files under an S3 prefix; read by Lambda"}
  - {property: freshness_target, value: "Rebuilt with search_chunks."}
schema:
  - name: search_index
    physicalType: table
    properties:
      - {name: chunk_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Matches
search_chunks.chunk_id"}
      - {name: vector, logicalType: array, required: true, description: "Embedding; dimension open (OQ-7)"}
      - {name: text, logicalType: string, required: true, description: "Chunk text for text search"}
      - {name: channel_id, logicalType: string, required: true, description: "Filter field"}
      - {name: speaker, logicalType: string, required: false, description: "Filter field; null before 5B.2"}
      - {name: get_start_s, logicalType: number, required: true, description: "GET seconds"}
      - {name: get_end_s, logicalType: number, required: true, description: "GET seconds"}
      - {name: embedding_model, logicalType: string, required: true, description: "Model name and version"}
quality:
  - {type: text, description: "Every row has a matching row in search_chunks and vice versa"}
```

Mission Control/contracts/speaker_turns.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): speaker_turns. Built by step 5B.2 (component C9, pyannote).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-speaker-turns
name: speaker_turns
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "Speaker separation on mixed channels (air-to-ground: CAPCOM and the crew)."
  limitations: Speaker labels are model-assigned and may not map to named people; role naming is not specified in the
plan.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
schema:
  - name: speaker_turns
    physicalType: table
    properties:
      - {name: turn_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: file_id, logicalType: string, required: true, description: "File"}
      - {name: speaker_label, logicalType: string, required: true, description: "Speaker label from separation"}
      - {name: start_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: end_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: get_start_s, logicalType: number, required: false, description: "GET seconds"}
      - {name: get_end_s, logicalType: number, required: false, description: "GET seconds"}
      - {name: model_name, logicalType: string, required: true, description: "pyannote model and version"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "end_offset_s greater than start_offset_s"}
```

Mission Control/contracts/speech_segments.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): speech_segments. Built by step 5A.2 (component C2).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-speech-segments
name: speech_segments
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "Spans of speech found by Silero in 16 kHz mono audio. Only these spans are transcribed (ADR 0021).
Acceptance: every segment traces back to a file and offset; speech share per channel is recorded."
  limitations: Speech missed by detection is not transcribed and is not flagged by later checks.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; partitioned by channel_id"}
  - {property: freshness_target, value: "Per run; recorded in the platform metrics table."}
schema:
  - name: speech_segments
    physicalType: table
    properties:
      - {name: segment_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: file_id, logicalType: string, required: true, description: "Source file in audio_manifest"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: start_offset_s, logicalType: number, required: true, description: "Start in seconds since the start of
the file (event time within tape)"}
      - {name: end_offset_s, logicalType: number, required: true, description: "End in seconds since the start of the
file"}
      - {name: detector, logicalType: string, required: true, description: "Detector name and version (Silero)"}
      - {name: detector_params, logicalType: string, required: true, description: "Parameters used (values open, OQ-
9)"}
      - {name: run_id, logicalType: string, required: true, description: "Airflow run id"}
      - {name: created_at, logicalType: timestamp, required: true, description: "UTC"}
quality:
  - {type: library, rule: notNull, description: "file_id, start_offset_s, end_offset_s not null (traceability)"}
  - {type: text, description: "end_offset_s greater than start_offset_s"}
```

Mission Control/contracts/src_apollo_audio_files.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (SOURCE): the Apollo 11 audio files themselves (binary).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-src-apollo-audio-files
name: src_apollo_audio_files
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: The audio that is transcribed. Downloaded as OGG (about 5 times smaller than FLAC) for the landing hour on
the two Phase A channels; FLAC only for the OGG versus FLAC accuracy check (ADR 0005).
  limitations: A public archive that never changes, so a manifest of URL, size and hash is stored instead of a full
copy (ADR 0011). Track 1 of each tape carries a time code whose format is not yet known (OQ-1).
  usage: Converted to 16 kHz mono with ffmpeg in Spark, then Silero speech detection (ADR 0021).
team:
  - username: John
    role: owner
customProperties:
  - {property: source_owner, value: "archive.org, uploaded by NASA; digitized by UT Dallas and NASA"}
  - {property: freshness_target, value: "Not applicable: static archive."}
  - {property: open_questions, value: "OQ-1, OQ-2, OQ-10"}
schema:
  - name: audio_file
    physicalType: file
    description: One audio file per item and format.
    properties:
```

```
      - {name: item_identifier, logicalType: string, required: true, description: "archive.org item identifier"}
      - {name: format, logicalType: string, required: true, description: "OGG (processed), FLAC (accuracy check only),
MP3 (not used)"}
      - {name: channel_layout, logicalType: string, required: false, description: "One channel per item; convert to
mono 16 kHz in audio prep"}
      - {name: content, logicalType: object, physicalType: binary, required: true, description: "Audio data; duration
measured from the catalog (OQ-22)"}
quality:
  - {type: text, description: "SHA hash or checksum must match the manifest before processing (hash algorithm to be
chosen with the manifest)"}
```

Mission Control/contracts/src_archive_item_metadata.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (SOURCE): archive.org item metadata for the Apollo 11 MOCR ACR Collection.
# The plan names the fields (tape, channel, start time, duration, format, checksum). The exact key names in
archive.org's
# JSON are NOT confirmed; they are confirmed by the 5A.1 schema diff and the first-run check. Logical names below are
Design (proposed).
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-src-archive-item-metadata
name: src_archive_item_metadata
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Describes each audio item in the archive.org Apollo 11 MOCR ACR Collection (one channel per item, FLAC, MP3
and OGG). Read by the catalog DAG in step 5A.1.
  limitations: Uploaded by NASA with no license field. Tape start and end clock times are published without a time
zone. Terms must be confirmed before clips are published (ADR 0019).
  usage: Raw responses are stored with source URL, fetch time and hash (platform rule 1 (keep raw files immutable with
source URL, fetch time and hash)). Pull no more often than needed; stop on any non-200 response.
team:
  - username: John
    role: owner
customProperties:
  - {property: source_owner, value: "archive.org, uploaded by NASA"}
  - {property: freshness_target, value: "Not applicable: static archive. Format changes are caught by the weekly smoke
test and the schema diff (platform)."}
  - {property: open_questions, value: "OQ-1 (time zone), OQ-6 (in-scope items), OQ-22 (counts)"}
schema:
  - name: archive_item
    physicalType: json
    description: One archive.org item (one channel of one tape).
    properties:
      - {name: identifier, logicalType: string, required: true, primaryKey: true, description: "archive.org item
identifier"}
      - {name: tape, logicalType: string, required: true, description: "Tape the channel belongs to (source field name
to be confirmed)"}
      - {name: channel, logicalType: string, required: true, description: "Channel or track label as published"}
      - {name: start_time_text, logicalType: string, required: true, description: "Start clock time as published, no
time zone"}
      - {name: end_time_text, logicalType: string, required: false, description: "End clock time as published, no time
zone"}
      - {name: duration, logicalType: string, required: false, description: "Duration as published (unit to be
confirmed)"}
      - name: files
        logicalType: array
        required: true
        description: Files offered for the item.
        items:
          logicalType: object
          properties:
            - {name: name, logicalType: string, required: true, description: "File name"}
            - {name: format, logicalType: string, required: true, description: "FLAC, MP3 or OGG"}
            - {name: size_bytes, logicalType: integer, required: false, description: "File size"}
            - {name: checksum, logicalType: string, required: false, description: "Checksum as published by the
archive"}
quality:
  - {type: library, rule: schemaMatches, description: "Schema is diffed against this contract before loading (platform
rule 5 (give each source a contract and diff the schema before loading))"}
  - {type: text, description: "HTTP status must be 200; any other status stops the DAG"}
```

Mission Control/contracts/src_ask_eval_cases.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (SOURCE, authored): Ask Mission Control eval cases for step 5B.9.
# Table and column names in this contract are Design (proposed) working names.
# Case counts are not stated in the plan except 50 hand-graded answers for calibrating the faithfulness judge (OQ-16).
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-src-ask-eval-cases
name: src_ask_eval_cases
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Cases for faithfulness judging, exact citation checks, refusal tests and red-team prompts planted in
transcript text.
  limitations: Red-team cases contain deliberately hostile text and must be treated as data, never instructions.
team:
  - username: John
    role: owner
customProperties:
  - {property: freshness_target, value: "Not applicable: authored set."}
  - {property: open_questions, value: "OQ-16"}
schema:
  - name: ask_eval_case
    physicalType: table
    description: One eval case.
    properties:
      - {name: case_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id"}
      - {name: kind, logicalType: string, required: true, description: "answerable, unanswerable or red_team"}
      - {name: question, logicalType: string, required: true, description: "Question asked"}
      - {name: expected_refusal, logicalType: boolean, required: true, description: "True when the tapes do not cover
the question"}
      - {name: expected_citation_get_start_s, logicalType: number, required: false, description: "For answerable cases,
start of the supporting moment"}
      - {name: expected_citation_get_end_s, logicalType: number, required: false, description: "For answerable cases,
end of the supporting moment"}
      - {name: planted_text, logicalType: string, required: false, description: "For red_team cases, the instruction
planted in transcript text"}
      - {name: human_grade, logicalType: string, required: false, description: "Hand grade used to calibrate the judge
(50 answers)"}
quality:
  - {type: text, description: "kind is one of answerable, unanswerable, red_team"}
```

Mission Control/contracts/src_nasa_air_to_ground_transcript.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (SOURCE): NASA's public air-to-ground transcript with GET timestamps.
# Table and column names in this contract are Design (proposed) working names.
# Reference for timing (anchor events, 5A.4) and accuracy (WER, 5B.3). Source and format are set by ADR 0026; the
mission timeline is still open (OQ-17).
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-src-nasa-air-to-ground-transcript
name: src_nasa_air_to_ground_transcript
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Reference transcript for the air-to-ground channel. Used for anchoring tape time to GET, measuring word
error rate, and (Phase C) as fine-tuning labels.
  limitations: NASA works are generally not copyrighted; credit NASA and do not imply endorsement. GET text is DD HH MM
SS from liftoff (ADR 0026).
team:
  - username: John
    role: owner
customProperties:
  - {property: source_owner, value: NASA}
  - {property: freshness_target, value: "Not applicable: static document."}
  - {property: open_questions, value: "OQ-10, OQ-17"}
schema:
  - name: transcript_line
    physicalType: table
```

```
    description: One line of the transcript.
    properties:
      - {name: line_number, logicalType: integer, required: true, primaryKey: true, description: "Position in the
source document (source key)"}
      - {name: get_text, logicalType: string, required: true, description: "GET timestamp as printed"}
      - {name: speaker, logicalType: string, required: false, description: "Speaker label as printed"}
      - {name: text, logicalType: string, required: true, description: "Spoken text as printed"}
quality:
  - {type: text, description: "get_text must parse to a number of seconds; unparsable rows are rejected and listed"}
```

Mission Control/contracts/src_retrieval_eval_questions.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (SOURCE, authored): retrieval test questions for step 5B.6.
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-src-retrieval-eval-questions
name: src_retrieval_eval_questions
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: 200 test questions tied to known moments, 150 derived from the transcript and mission timeline and 50
written by hand. Used to measure recall at 10, MRR and time error of the top hit.
  limitations: The mission timeline source is not yet named (OQ-17). Kept under version control in the repo.
team:
  - username: John
    role: owner
customProperties:
  - {property: freshness_target, value: "Not applicable: authored set, changed only by pull request."}
  - {property: open_questions, value: "OQ-16, OQ-17"}
schema:
  - name: eval_question
    physicalType: table
    description: One question with its expected moment.
    properties:
      - {name: question_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable
id"}
      - {name: question, logicalType: string, required: true, description: "The question text"}
      - {name: origin, logicalType: string, required: true, description: "transcript_or_timeline (150 expected) or
hand_written (50 expected)"}
      - {name: expected_get_start_s, logicalType: number, required: true, description: "Start of the expected moment in
GET seconds"}
      - {name: expected_get_end_s, logicalType: number, required: true, description: "End of the expected moment in GET
seconds"}
      - {name: expected_channel_id, logicalType: string, required: false, description: "Expected channel if known"}
quality:
  - {type: text, description: "Exactly 200 rows in the full set: 150 with origin transcript_or_timeline and 50 with
origin hand_written"}
  - {type: text, description: "expected_get_end_s is not before expected_get_start_s"}
```

Mission Control/contracts/tape_catalog.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): Iceberg table tape_catalog. Built by step 5A.1 (component C1).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-tape-catalog
name: tape_catalog
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: One row per archive item (tape and channel) per pull, with timing, format and checksum. The base for the
manifest and for choosing in-scope files.
  limitations: Tape times are kept as published text until the time zone is confirmed (OQ-1); UTC columns are null
until then.
  usage: "History is kept by adding one partition per pull (platform rule 3 (make tasks idempotent: overwrite a
partition, never blind-append))."
team:
  - username: John
```

```
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; partitioned by pull_date"}
  - {property: freshness_target, value: "Not applicable: static archive; run freshness and row counts go to the
platform metrics table."}
  - {property: source_contract, value: src_archive_item_metadata.yaml}
schema:
  - name: tape_catalog
    physicalType: table
    properties:
      - {name: item_id, logicalType: string, required: true, primaryKey: true, primaryKeyPosition: 1, description:
"archive.org identifier (source key)"}
      - {name: pull_date, logicalType: date, required: true, primaryKey: true, primaryKeyPosition: 2, description:
"Pull partition (UTC date)"}
      - {name: tape_id, logicalType: string, required: true, description: "Tape the item belongs to"}
      - {name: track_number, logicalType: integer, required: true, description: "Track on the 30-channel tape; track 1
carries the time code"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel key used across tables; maps to
channel_map"}
      - {name: start_time_text, logicalType: string, required: true, description: "Start clock time as published, no
zone"}
      - {name: end_time_text, logicalType: string, required: false, description: "End clock time as published, no
zone"}
      - {name: start_time_utc, logicalType: timestamp, required: false, description: "UTC once the zone is confirmed
(event time)"}
      - {name: end_time_utc, logicalType: timestamp, required: false, description: "UTC once the zone is confirmed"}
      - {name: duration_seconds, logicalType: number, required: false, description: "Duration in seconds"}
      - {name: formats, logicalType: array, required: true, description: "Formats offered, such as FLAC, MP3, OGG"}
      - {name: checksum, logicalType: string, required: false, description: "Checksum as published"}
      - {name: source_url, logicalType: string, required: true, description: "URL of the item metadata"}
      - {name: raw_hash, logicalType: string, required: true, description: "Hash of the raw response (platform rule 1
(keep raw files immutable with source URL, fetch time and hash))"}
      - {name: fetched_at, logicalType: timestamp, required: true, description: "Fetch time, UTC"}
      - {name: run_id, logicalType: string, required: true, description: "Airflow run id"}
quality:
  - {type: library, rule: unique, description: "item_id unique within a pull_date"}
  - {type: library, rule: notNull, description: "item_id, tape_id, track_number, channel_id not null"}
```

Mission Control/contracts/tape_clock_offsets.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): tape_clock_offsets. Written by step 5A.4 (component C4, store_offset).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-tape-clock-offsets
name: tape_clock_offsets
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "One offset per tape that converts time since the tape start into GET (ADR 0009). Acceptance: anchor events
line up across the two channels within a stated tolerance."
  limitations: Definition of tape start and the time base depend on the first-run check (OQ-1, OQ-6). Tolerance value
open (OQ-5).
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; overwrite per tape partition, never blind append"}
  - {property: freshness_target, value: "Per run; recorded in the platform metrics table."}
schema:
  - name: tape_clock_offsets
    physicalType: table
    properties:
      - {name: tape_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Tape (source
key)"}
      - {name: offset_s, logicalType: number, required: true, description: "Seconds to add to time since tape start to
obtain GET"}
      - {name: method, logicalType: string, required: true, description: "TIMECODE or ANCHOR (ClockMethod)"}
      - {name: residual_s, logicalType: number, required: false, description: "Fit error in seconds"}
      - {name: n_points, logicalType: integer, required: false, description: "Points used in the fit"}
      - {name: anchor_ids, logicalType: array, required: false, description: "Anchor events used when method is
ANCHOR"}
      - {name: time_zone_assumed, logicalType: string, required: false, description: "Zone used to read file-name
```

```
times, once confirmed"}
      - {name: tolerance_s, logicalType: number, required: false, description: "Tolerance in force for the agreement
check (value open, OQ-5)"}
      - {name: status, logicalType: string, required: true, description: "accepted or rejected (agreement check
result)"}
      - {name: decided_at, logicalType: timestamp, required: true, description: "UTC"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "Exactly one row per tape_id"}
  - {type: text, description: "status accepted only when check_anchor_agreement passed"}
```

Mission Control/contracts/transcript_corrections.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): transcript_corrections. Built by step 5B.2 (component C9). ADR 0013.
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-transcript-corrections
name: transcript_corrections
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "Corrections from glossary rules and from the batch model call for low-confidence segments. The original
text is always kept. Acceptance: each corrected segment keeps its original; a hand check of 100 corrections reports how
many were right."
  limitations: Glossary rule content is open (OQ-26). Model identity is the cheapest model (OQ-26).
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
schema:
  - name: transcript_corrections
    physicalType: table
    properties:
      - {name: correction_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable
id (source key)"}
      - {name: transcript_segment_id, logicalType: string, required: true, description: "Segment corrected"}
      - {name: original_text, logicalType: string, required: true, description: "Text before correction; never null"}
      - {name: corrected_text, logicalType: string, required: true, description: "Text after correction"}
      - name: changes
        logicalType: array
        required: true
        description: Each change made, in structured output.
        items:
          logicalType: object
          properties:
            - {name: from_text, logicalType: string, required: true, description: "Text replaced"}
            - {name: to_text, logicalType: string, required: true, description: "Replacement"}
            - {name: source, logicalType: string, required: true, description: "glossary_rule or model"}
            - {name: rule_id, logicalType: string, required: false, description: "Glossary rule id when source is
glossary_rule"}
      - {name: method, logicalType: string, required: true, description: "glossary or batch_model"}
      - {name: model_name, logicalType: string, required: false, description: "Model when method is batch_model"}
      - {name: prompt_version, logicalType: string, required: false, description: "Prompt version in git"}
      - {name: confidence, logicalType: number, required: false, description: "Model confidence if returned"}
      - {name: hand_checked, logicalType: boolean, required: false, description: "In the 100-correction sample"}
      - {name: hand_check_correct, logicalType: boolean, required: false, description: "Result of the hand check"}
      - {name: created_at, logicalType: timestamp, required: true, description: "UTC"}
quality:
  - {type: library, rule: notNull, description: "original_text not null"}
```

Mission Control/contracts/transcript_segments.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): transcript_segments. Built from words (5A.3), GET added by 5A.4, flags added in
Phase B.
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-transcript-segments
```

```
name: transcript_segments
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: One transcript line per speech segment with channel and GET. Original text is never overwritten (ADR 0013).
  limitations: GET columns are null until the tape has an accepted offset. low_confidence threshold is open (OQ-16).
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; partitioned by channel_id"}
  - {property: freshness_target, value: "Per run; recorded in the platform metrics table."}
schema:
  - name: transcript_segments
    physicalType: table
    properties:
      - {name: transcript_segment_id, logicalType: string, required: true, primaryKey: true, unique: true, description:
"Stable id (source key)"}
      - {name: speech_segment_id, logicalType: string, required: true, description: "Segment in speech_segments"}
      - {name: file_id, logicalType: string, required: true, description: "Source file"}
      - {name: channel_id, logicalType: string, required: true, description: "Channel"}
      - {name: tape_id, logicalType: string, required: true, description: "Tape"}
      - {name: original_text, logicalType: string, required: true, description: "Text as transcribed; never
overwritten"}
      - {name: start_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: end_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: get_start_s, logicalType: number, required: false, description: "GET seconds, after the mission clock is
applied"}
      - {name: get_end_s, logicalType: number, required: false, description: "GET seconds"}
      - {name: mean_word_confidence, logicalType: number, required: true, description: "Mean confidence of the words"}
      - {name: low_confidence, logicalType: boolean, required: false, description: "Phase B: shown in grey and marked
unverified"}
      - {name: model_name, logicalType: string, required: true, description: "Model used"}
      - {name: run_id, logicalType: string, required: true, description: "Run id"}
quality:
  - {type: text, description: "GET never runs backward within a channel (Great Expectations check, 5B.10)"}
  - {type: library, rule: notNull, description: "file_id, start_offset_s, end_offset_s, original_text not null"}
```

Mission Control/contracts/transcript_words.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): transcript_words. Written by the GPU job, step 5A.3 (component C3).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-transcript-words
name: transcript_words
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: Word timestamps and confidence from the transcription model.
  limitations: Confidence scale depends on the model chosen (Whisper or Parakeet, ADR 0004); compare only within a
model.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; partitioned by channel_id"}
  - {property: freshness_target, value: "Per run; recorded in the platform metrics table."}
schema:
  - name: transcript_words
    physicalType: table
    properties:
      - {name: word_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: speech_segment_id, logicalType: string, required: true, description: "Segment in speech_segments"}
      - {name: file_id, logicalType: string, required: true, description: "Source file"}
      - {name: word_index, logicalType: integer, required: true, description: "Position within the segment"}
      - {name: word, logicalType: string, required: true, description: "Recognized word"}
      - {name: start_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: end_offset_s, logicalType: number, required: true, description: "Seconds since file start"}
      - {name: confidence, logicalType: number, required: true, description: "Model confidence"}
```

```
      - {name: model_name, logicalType: string, required: true, description: "Model used"}
      - {name: model_settings, logicalType: string, required: true, description: "Precision, batch size and device"}
      - {name: run_id, logicalType: string, required: true, description: "Transcription run id"}
quality:
  - {type: text, description: "A rerun of a file overwrites that file partition only; row count unchanged"}
  - {type: text, description: "end_offset_s not before start_offset_s"}
```

Mission Control/contracts/transcription_runs.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): transcription_runs. Written by C3 and C10. Also logged as MLflow runs (ADR
0017).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-transcription-runs
name: transcription_runs
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: One row per transcription run or benchmark setting, recording audio hours per GPU hour, cost per audio hour
and, in tuning runs, GPU utilization and memory. Feeds the design note (5A.3, 5B.1) and the README tables (5B.4, 5B.5).
  limitations: Cost uses the actual spot price at run time; method to be written when built.
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog"}
  - {property: freshness_target, value: "Per run."}
schema:
  - name: transcription_runs
    physicalType: table
    properties:
      - {name: run_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Run id
(source key)"}
      - {name: started_at, logicalType: timestamp, required: true, description: "UTC"}
      - {name: model_name, logicalType: string, required: true, description: "For example faster-whisper large-v3 or
Parakeet"}
      - {name: precision, logicalType: string, required: true, description: "fp16 or int8 (the T4 cannot use bf16)"}
      - {name: batch_size, logicalType: integer, required: false, description: "Batch size"}
      - {name: device, logicalType: string, required: true, description: "T4 GPU or CPU"}
      - {name: audio_hours, logicalType: number, required: true, description: "Audio hours transcribed"}
      - {name: gpu_hours, logicalType: number, required: true, description: "GPU hours used"}
      - {name: audio_hours_per_gpu_hour, logicalType: number, required: true, description: "Throughput"}
      - {name: cost_usd, logicalType: number, required: false, description: "Run cost"}
      - {name: cost_per_audio_hour_usd, logicalType: number, required: false, description: "Cost per audio hour"}
      - {name: gpu_utilization_mean, logicalType: number, required: false, description: "From nvidia-smi or DCGM
samples (tuning runs)"}
      - {name: gpu_memory_peak, logicalType: number, required: false, description: "From nvidia-smi or DCGM samples
(tuning runs)"}
      - {name: bottleneck, logicalType: string, required: false, description: "decoding, audio_loading or gpu (5B.5)"}
      - {name: mlflow_run_url, logicalType: string, required: false, description: "Link to the MLflow run"}
quality:
  - {type: text, description: "gpu_hours does not exceed the maximum GPU hours per run (value open, OQ-12)"}
```

Mission Control/contracts/wer_results.yaml

```
# Project 5, Mission Control | Document: data contract (ODCS) | Set: rev 4, Oct 9, 2026
# ODCS data contract (PUBLISHED TABLE): wer_results. Built by steps 5B.3 and 5B.4 (component C10).
# Table and column names in this contract are Design (proposed) working names.
apiVersion: v3.0.2   # ODCS v3.0.2, as pinned in project 1 (ADR 0032)
kind: DataContract
id: p5-wer-results
name: wer_results
version: 0.1.0
status: draft
domain: mission-control
dataProduct: project5-mission-control
description:
  purpose: "Word error rate per model, setting and channel type, against NASA's air-to-ground transcript and against 30
hand-corrected minutes of back-room audio. Acceptance: WER per channel type is published; the 5B.4 comparison table is
built from it."
  limitations: Hand-corrected reference is only 30 minutes; do not extrapolate beyond it. WER targets are open (OQ-16).
```

```
team:
  - username: John
    role: owner
customProperties:
  - {property: storage, value: "Iceberg on S3, Glue catalog; also MLflow runs on Databricks (ADR 0017)"}
  - {property: freshness_target, value: "Per evaluation run."}
schema:
  - name: wer_results
    physicalType: table
    properties:
      - {name: result_id, logicalType: string, required: true, primaryKey: true, unique: true, description: "Stable id
(source key)"}
      - {name: run_id, logicalType: string, required: true, description: "Transcription run id"}
      - {name: model_name, logicalType: string, required: true, description: "Model"}
      - {name: model_settings, logicalType: string, required: true, description: "Precision and batch size"}
      - {name: channel_type, logicalType: string, required: true, description: "For example air-to-ground or back-
room"}
      - {name: reference_kind, logicalType: string, required: true, description: "nasa_transcript or hand_corrected"}
      - {name: audio_minutes_evaluated, logicalType: number, required: true, description: "Minutes compared"}
      - {name: wer, logicalType: number, required: true, description: "Word error rate"}
      - {name: computed_at, logicalType: timestamp, required: true, description: "UTC"}
      - {name: mlflow_run_url, logicalType: string, required: false, description: "Link to the MLflow run"}
quality:
  - {type: text, description: "wer between 0 and the maximum value of the metric (can exceed 1 when insertions are
many)"}
```

Part 9: API specification

Mission Control/openapi.yaml

```
# Project 5, Mission Control | Document: API specification (OpenAPI 3.0.3) | Set: rev 4, Oct 9, 2026
openapi: 3.0.3
# Mission Control public API (step 5B.8; keyed access on request from Phase A, step 5A.5).
# Endpoint names (search, segments, timeline, clips, events, ask) and the /v1 prefix come from the build plan.
# Parameter names, schemas and response fields are Design (proposed) and are confirmed in review.
# Numeric limits (rate limit, daily quota, concurrency cap, request size, page size, ask daily cap, token limit)
# are NOT recorded decisions: they appear as x-limit-ref values pointing at design-doc open question OQ-12.
# OpenAPI 3.0.3 is used because API Gateway's REST type imports OpenAPI 3.0 (pinned in project 1, ADR 0032).
info:
  title: Mission Control API
  version: 1.0.0-draft
  description: |
    Search and play Apollo 11 Mission Control audio by channel and Ground Elapsed Time (GET).
    Audio and transcripts are from the archive.org "Apollo 11 MOCR ACR Collection" (uploaded by NASA).
    Credit NASA. Use of this API does not imply endorsement by NASA.
    Two doors (design-doc ADR 0022, Decided by John, Oct 9, 2026): the keyed developer API (this spec's security scheme)
and an unkeyed website route used by the public page, with per-visitor limits and a daily cap. Each operation lists its
doors in x-doors. The AI ask feature has an open setting (OQ-13) and is off until it is decided on the website door (open,
OQ-13).
    Authentication on the developer door: send an API key in the x-api-key header. Keys are issued on request in Phase A
and self-service from Phase B.
    Times: GET values are seconds on the mission clock. Wall-clock times are UTC.
    Low-confidence transcript lines are marked and should be shown as unverified.
  x-source-documents:
    - ../requirements.md
    - ../design-doc.md
servers:
  - url: https://{api_host}
    description: Developer door (keyed). Host is not decided (design-doc OQ-13). The unkeyed website route is served to
the public page from the same code; its host is not decided either.
    variables:
      api_host:
        default: api-host-to-be-set
security:
  - ApiKeyAuth: []
tags:
  - name: search
  - name: segments
  - name: timeline
  - name: clips
  - name: events
  - name: ask
paths:
  /v1/search:
    get:
      tags: [search]
      operationId: searchChunks
      summary: Search transcript chunks by text
      description: |
        Returns ranked chunks with channel, speaker (when known), GET range and a clip link.
        Repeated identical searches may be served from a cache (design-doc OQ-23).
        Whether text, vector or hybrid ranking is used is decided from the 5B.6 retrieval evaluation.
      x-doors: [developer, website]
      x-phase: B (Phase A subset undecided, OQ-24)
      x-limit-ref: OQ-12
      parameters:
        - {name: q, in: query, required: true, description: "Search text. Format and size validated (limit OQ-12).",
schema: {type: string}}
        - {$ref: '#/components/parameters/Channel'}
        - {$ref: '#/components/parameters/GetFrom'}
        - {$ref: '#/components/parameters/GetTo'}
        - {$ref: '#/components/parameters/Limit'}
        - {$ref: '#/components/parameters/Cursor'}
      responses:
        '200':
          description: A page of hits.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/SearchPage'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
```

```
  /v1/segments:
    get:
      tags: [segments]
      operationId: listSegments
      summary: List transcript segments
      description: Transcript segments for a channel and GET range, with original text, corrected text when present, and a
low-confidence flag.
      x-doors: [developer, website]
      x-phase: B (Phase A subset undecided, OQ-24)
      x-limit-ref: OQ-12
      parameters:
        - {name: segment_id, in: query, required: false, description: "Return one segment by id.", schema: {type: string}}
        - {$ref: '#/components/parameters/Channel'}
        - {$ref: '#/components/parameters/GetFrom'}
        - {$ref: '#/components/parameters/GetTo'}
        - {$ref: '#/components/parameters/Limit'}
        - {$ref: '#/components/parameters/Cursor'}
      responses:
        '200':
          description: A page of segments.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/SegmentPage'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
  /v1/timeline:
    get:
      tags: [timeline]
      operationId: getTimeline
      summary: Who was talking when
      description: Speaking spans per channel (lane) over a GET range. Speaker labels are present only after speaker
separation (5B.2).
      x-doors: [developer, website]
      x-phase: B (Phase A subset undecided, OQ-24)
      x-limit-ref: OQ-12
      parameters:
        - {name: channel, in: query, required: false, description: "One or more channel ids (repeat the parameter). All
lanes if omitted.", schema: {type: array, items: {type: string}}, style: form, explode: true}
        - {$ref: '#/components/parameters/GetFrom'}
        - {$ref: '#/components/parameters/GetTo'}
        - {$ref: '#/components/parameters/Cursor'}
        - {$ref: '#/components/parameters/Limit'}
      responses:
        '200':
          description: A page of lane spans.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/TimelinePage'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
  /v1/clips:
    get:
      tags: [clips]
      operationId: listClips
      summary: Find playable clips
      description: Opus clip metadata and URLs served through CloudFront, by clip id or by channel and GET range.
      x-doors: [developer, website]
      x-phase: B (Phase A subset undecided, OQ-24)
      x-limit-ref: OQ-12
      parameters:
        - {name: clip_id, in: query, required: false, description: "Return one clip by id.", schema: {type: string}}
        - {$ref: '#/components/parameters/Channel'}
        - {$ref: '#/components/parameters/GetFrom'}
        - {$ref: '#/components/parameters/GetTo'}
        - {$ref: '#/components/parameters/Limit'}
        - {$ref: '#/components/parameters/Cursor'}
      responses:
        '200':
          description: A page of clips.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/ClipPage'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
```

```
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
  /v1/events:
    get:
      tags: [events]
      operationId: listEvents
      summary: List extracted events
      description: Alarms, go/no-go polls, system problems, decisions and handovers, each with its exact transcript span.
Available after step 5B.7.
      x-doors: [developer, website]
      x-phase: B
      x-limit-ref: OQ-12
      parameters:
        - name: type
          in: query
          required: false
          description: Event type filter.
          schema: {$ref: '#/components/schemas/EventType'}
        - {$ref: '#/components/parameters/Channel'}
        - {$ref: '#/components/parameters/GetFrom'}
        - {$ref: '#/components/parameters/GetTo'}
        - {$ref: '#/components/parameters/Limit'}
        - {$ref: '#/components/parameters/Cursor'}
      responses:
        '200':
          description: A page of events.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/EventPage'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
  /v1/ask:
    post:
      tags: [ask]
      operationId: askMissionControl
      summary: Ask Mission Control
      description: |
        Answers only from retrieved segments. Every claim cites channel, speaker (when known) and GET and links a playable
clip.
        If the tapes do not cover the question the response is a refusal (refused is true, claims empty).
        Requires a key. This endpoint has its own daily cap, under the model spend limit (value OQ-12), and a token limit
per answer (value OQ-12).
        Transcript text is treated as data, never as instructions.
      x-doors: [developer, website (setting open, OQ-13; off until decided)]
      x-phase: B
      x-limit-ref: OQ-12
      requestBody:
        required: true
        content:
          application/json:
            schema: {$ref: '#/components/schemas/AskRequest'}
      responses:
        '200':
          description: An answer with citations, or a refusal.
          content:
            application/json:
              schema: {$ref: '#/components/schemas/AskResponse'}
        '400': {$ref: '#/components/responses/ValidationError'}
        '403': {$ref: '#/components/responses/KeyRefused'}
        '429': {$ref: '#/components/responses/RateOrQuotaExceeded'}
        default: {$ref: '#/components/responses/ServerOrCapacity'}
components:
  securitySchemes:
    ApiKeyAuth:
      type: apiKey
      in: header
      name: x-api-key
      description: API key issued on request (Phase A) or self-service (Phase B). Revocation is instant.
  parameters:
    Channel:
      name: channel
      in: query
      required: false
      description: Channel id as listed in channel_map.
      schema: {type: string}
    GetFrom:
```

```
      name: get_from
      in: query
      required: false
      description: Start of the range, GET whole seconds from liftoff (ADR 0027).
      schema: {type: number}
    GetTo:
      name: get_to
      in: query
      required: false
      description: End of the range, GET seconds.
      schema: {type: number}
    Limit:
      name: limit
      in: query
      required: false
      description: Page size. Default and maximum are not decided (OQ-12).
      schema: {type: integer, minimum: 1}
    Cursor:
      name: cursor
      in: query
      required: false
      description: Opaque cursor from a previous response's next_cursor.
      schema: {type: string}
  responses:
    ValidationError:
      description: Request failed size or format validation.
      content:
        application/json:
          schema: {$ref: '#/components/schemas/Error'}
    KeyRefused:
      description: |
        Missing, invalid or revoked key is refused. 403 is the status code here (ADR 0028); the contract tests confirm the
gateway returns it.
      content:
        application/json:
          schema: {$ref: '#/components/schemas/Error'}
    RateOrQuotaExceeded:
      description: The key (developer door) or the visitor (website door) is over its rate limit, per-visitor limit or
daily quota/cap (acceptance check of 5B.8). Repeated hits can lead to the key being disabled automatically.
      content:
        application/json:
          schema: {$ref: '#/components/schemas/Error'}
    ServerOrCapacity:
      description: |
        Server error, or traffic beyond the reserved-concurrency cap is rejected rather than billed. Capacity rejection
returns 429 with Retry-After (ADR 0028); the load test confirms it.
      content:
        application/json:
          schema: {$ref: '#/components/schemas/Error'}
  schemas:
    Error:
      type: object
      required: [error]
      properties:
        error:
          type: object
          required: [code, message]
          properties:
            code: {type: string, description: "Machine-readable code"}
            message: {type: string, description: "Human-readable message"}
    PageInfo:
      type: object
      properties:
        next_cursor:
          type: string
          nullable: true
          description: Cursor for the next page; null on the last page.
    EventType:
      type: string
      enum: [alarm, go_no_go_poll, system_problem, decision, handover]
    Credit:
      type: string
      description: Credit line for NASA (ADR 0026). The archive, Explore Apollo and UT Dallas wording is still open (OQ-
10).
    Clip:
      type: object
      required: [clip_id, channel_id, get_start_s, get_end_s, url, format]
      properties:
```

```
    clip_id: {type: string}
    channel_id: {type: string}
    get_start_s: {type: number, description: "GET at the first sample of the clip"}
    get_end_s: {type: number, description: "GET at the last sample of the clip"}
    duration_s: {type: number}
    format: {type: string, enum: [opus]}
    url: {type: string, format: uri, description: "CloudFront URL"}
Segment:
  type: object
  required: [segment_id, channel_id, original_text, get_start_s, get_end_s, low_confidence]
  properties:
    segment_id: {type: string}
    channel_id: {type: string}
    speaker: {type: string, nullable: true, description: "Null before speaker separation"}
    original_text: {type: string, description: "As transcribed; never overwritten"}
    corrected_text: {type: string, nullable: true}
    get_start_s: {type: number}
    get_end_s: {type: number}
    low_confidence: {type: boolean, description: "True when the line should be shown in grey and marked unverified"}
    clip: {$ref: '#/components/schemas/Clip'}
SearchHit:
  type: object
  required: [chunk_id, channel_id, text, get_start_s, get_end_s, score]
  properties:
    chunk_id: {type: string}
    channel_id: {type: string}
    speaker: {type: string, nullable: true}
    text: {type: string}
    get_start_s: {type: number}
    get_end_s: {type: number}
    score: {type: number, description: "Ranking score; scale depends on the search mode"}
    low_confidence: {type: boolean}
    clip: {$ref: '#/components/schemas/Clip'}
SearchPage:
  allOf:
    - $ref: '#/components/schemas/PageInfo'
    - type: object
      required: [hits]
      properties:
        hits:
          type: array
          items: {$ref: '#/components/schemas/SearchHit'}
        credit: {$ref: '#/components/schemas/Credit'}
SegmentPage:
  allOf:
    - $ref: '#/components/schemas/PageInfo'
    - type: object
      required: [segments]
      properties:
        segments:
          type: array
          items: {$ref: '#/components/schemas/Segment'}
        credit: {$ref: '#/components/schemas/Credit'}
TimelineSpan:
  type: object
  required: [channel_id, get_start_s, get_end_s]
  properties:
    channel_id: {type: string}
    speaker: {type: string, nullable: true}
    get_start_s: {type: number}
    get_end_s: {type: number}
TimelinePage:
  allOf:
    - $ref: '#/components/schemas/PageInfo'
    - type: object
      required: [spans]
      properties:
        spans:
          type: array
          items: {$ref: '#/components/schemas/TimelineSpan'}
        credit: {$ref: '#/components/schemas/Credit'}
ClipPage:
  allOf:
    - $ref: '#/components/schemas/PageInfo'
    - type: object
      required: [clips]
      properties:
        clips:
```

```
          type: array
          items: {$ref: '#/components/schemas/Clip'}
        credit: {$ref: '#/components/schemas/Credit'}
Event:
  type: object
  required: [event_id, type, get_s, channel_id, confidence, span]
  properties:
    event_id: {type: string}
    type: {$ref: '#/components/schemas/EventType'}
    get_s: {type: number}
    channel_id: {type: string}
    speakers:
      type: array
      items: {type: string}
    confidence: {type: number}
    span:
      type: object
      required: [segment_ids, char_start, char_end, text]
      properties:
        segment_ids:
          type: array
          items: {type: string}
        char_start: {type: integer}
        char_end: {type: integer}
        text: {type: string}
    clip: {$ref: '#/components/schemas/Clip'}
EventPage:
  allOf:
    - $ref: '#/components/schemas/PageInfo'
    - type: object
      required: [events]
      properties:
        events:
          type: array
          items: {$ref: '#/components/schemas/Event'}
        credit: {$ref: '#/components/schemas/Credit'}
AskRequest:
  type: object
  required: [question]
  properties:
    question:
      type: string
      description: The question. Maximum size is not decided (OQ-12).
Citation:
  type: object
  required: [channel_id, get_start_s, get_end_s, clip]
  properties:
    channel_id: {type: string}
    speaker: {type: string, nullable: true}
    get_start_s: {type: number, description: "Cited time; must fall inside the cited clip"}
    get_end_s: {type: number}
    clip: {$ref: '#/components/schemas/Clip'}
Claim:
  type: object
  required: [text, citations]
  properties:
    text: {type: string}
    citations:
      type: array
      minItems: 1
      items: {$ref: '#/components/schemas/Citation'}
AskResponse:
  type: object
  required: [refused, claims]
  properties:
    refused:
      type: boolean
      description: True when the tapes do not cover the question.
    refusal_reason: {type: string, nullable: true}
    answer: {type: string, nullable: true, description: "Plain-language answer built from the claims"}
    claims:
      type: array
      items: {$ref: '#/components/schemas/Claim'}
    credit: {$ref: '#/components/schemas/Credit'}
```

Part 10: Changelog

`Mission Control/changelog.md`

Project 5 changelog

Project 5, Mission Control | Document: changelog | Set: rev 4, Oct 9, 2026

This file records two things: changes to the documents, and, once code exists, changes to each phase of the project. The newest entry is first.

Oct 9, 2026: defaults confirmed

- John confirmed ADRs 0026 to 0028 (NASA transcript source and credit, GET as whole seconds, status codes 403 and 429). Their status is now Accepted.

Documents

Oct 9, 2026 (rev 4): the complete project 5 set

- New: operations.md (what runs and when, monitoring, limits, runbook first responses, recovery, postmortems), model-card.md (models and licences, intended and excluded uses, evals, known failures, safety and cost) and security-and-privacy.md (controls, threat model, personal-data inventory with retention, licence and attribution register, takedown handling).
- Licences checked on the publishers' own pages on Oct 9, 2026: Whisper code (MIT), Parakeet tdt 0.6b v2 (CC-BY-4.0), pyannote speaker-diarization-3.1 (MIT, gated), Silero VAD (MIT). The Whisper weights, faster-whisper and NeMo are not yet checked.
- Corrected platform ADR numbers in operations.md to match the platform set. Corrected component numbers in the model card to match the design doc.
- Minor decisions made by default, each recorded where it applies: the call log is kept 90 days; visitor questions are not stored by the project; API keys are stored only as hashes; a takedown is handled by stopping the affected item the same day.
- Still open and needing John: the monthly model API spend limit ($15 in ADR 0015, $10 in project 1's ADR 0038); the key request fields and their retention (OQ-14); the visitor identity and whether Ask is on for the website route (OQ-13); the numeric limits (OQ-12); the late-data threshold (OQ-21); the archive.org, Explore Apollo and UT Dallas terms (OQ-10).
- Reference generated from code is not written: there is no code yet.
- The PDF is rebuilt as rev 4 in this folder.

Oct 9, 2026 (rev 3): labels and index

- Every document in the set now carries a label line under its title: the project, the document type and the set revision. Contracts and the API specification carry the same label as a comment at the top. No content changed.
- New index: README.md lists every document, its status, the while-building documents still to be written (operations, model card, security and privacy, reference from code), and the links to the platform documents.
- The PDF is rebuilt as rev 3 and kept in this folder. The earlier PDFs were removed as old versions on Oct 9, 2026. The current PDF is `full-documentation.pdf` in this folder.

Oct 9, 2026: NASA source and terms, GET format, status codes, pinned contract versions

- Reference transcript (OQ-17, partly answered) is decided in ADR 0026: NASA's Technical Air-to-Ground Voice Transcription PDF. GET is written DD HH MM SS from liftoff. NASA's licence rule was checked on NASA's own Images and Media page (updated Aug 13, 2026). The mission timeline is still open.

- Terms (OQ-10, partly answered): NASA's credit wording is set (ADR 0026). The archive.org terms page could not be read, and the Explore Apollo and UT Dallas terms were not found. Those three are unverified and still open. ADR 0019 is updated to match.
- GET representation (OQ-20) is decided in ADR 0027: whole seconds from liftoff, with the printed text kept beside it. The ODCS version is v3.0.2 and the OpenAPI version is 3.0.3, both as pinned in project 1 (ADR 0032). The "to be aligned" comments in all contracts and in openapi.yaml are replaced.
- Status codes (OQ-19) are decided by default in ADR 0028: 403 for a missing, invalid or revoked key; 429 with Retry-After for capacity rejection. The contract and load tests confirm them.
- Contracts: no new contract. The NASA source contract now records its source, format and open questions. The archive item metadata contract carries the same version comment.
- Open questions: 28 rows in design doc section 16. 2 answered (OQ-19, OQ-20), 2 partly answered (OQ-10, OQ-17), and 24 open. OQ-27 and OQ-29 have no row in the table, which is a numbering gap to be checked with John; no question was invented for them.
- ADRs total 28 (0001 to 0028). ADR 0032 is project 1's ADR, cited for the ODCS and OpenAPI pins.
- Build: the project 5 documents had no changelog before this entry. README.md was missing at this revision; it is added in rev 3. Its terms note is required by ADR 0019 and is still to be written.